Regulatory Capture by Superstars: Lobbying, Innovation, and Industry Concentration*

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Abstract

A long-established concern about corporate lobbying is regulatory capture: incumbent firms shaping regulation in their favor. We study which regulations firms lobby for and how this affects concentration, innovation, and welfare. Using LLMs, we classify federal lobbying reports of public firms and trade associations, assigning each request to a policy tool from a new comprehensive taxonomy, with the direction and scope of the requested change. Lobbying for firm-specific benefits such as tax decreases and government contracts is one-sided: nearly every firm asks for more. Lobbying over regulatory costs is instead contested: the largest firms are the most likely to ask for higher costs, while trade associations mostly ask for lower ones. To quantify the consequences of lobbying, we build a dynamic general equilibrium model of oligopolistic competition and innovation in which firms lobby for firm-specific transfers and over industry-wide regulatory costs, set in a contest between strategic incumbents and an association of the competitive fringe. In the model estimated on US data, lobbying raises regulatory costs and lowers welfare through industry composition: each incumbent innovates more, but industries become more concentrated, lowering output and growth. Almost all the welfare loss comes from lobbying over regulation rather than for benefits.

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1 Introduction

Americans rank the role of money in politics among the country's most serious problems (Pew Research Center, 2026). One reason for concern is regulatory capture: firms can use money to shape regulation in their favor at the expense of other firms and the public interest. For example, large AI companies are asking the government to regulate their industry. Some might think the goal is to ensure AI safety, while others might argue it aims to reduce the number of smaller competitors and secure higher, more stable market share and profits, which is regulatory capture. Concerns about regulatory capture extend well beyond AI. In 2015–2016, Altria and Reynolds lobbied on a bill that would have exempted the vaping products already on sale from premarket review by the Food and Drug Administration; in 2020–2021, both lobbied in support of FDA regulation of synthetic-nicotine products. H&R Block, the largest listed tax-preparation firm, lobbied for minimum competency standards for paid tax preparers. C.H. Robinson, the largest listed freight broker, lobbied to raise the bond that brokers must post to guarantee their payments to shippers and truckers; in 2012, Congress raised it from $10,000 to $75,000, the same amount for every broker.

Firms shape government policy mainly through lobbying and political donations, and lobbying is their largest political expense (for public firms, it is 16 times their PAC contributions). The main advantage of analyzing corporate lobbying is that mandatory lobbying reports show which policies firms discussed with the government. These policies fall into groups with very different economic effects. For instance, one group of policies benefits firms without directly harming other firms: increasing subsidies, appropriations, and contracts; reducing tax rates. The other group affects industry- or economy-wide regulatory costs of operation: compliance rules, safety/certification standards, data privacy, licensing. Unlike the first group, these act as fixed rather than variable costs, and they can disproportionately hurt smaller firms, increase concentration, and raise barriers to entry (Ryan, 2012; Johnson et al., 2023; Kalmenovitz et al., 2025). These groups therefore can have very different implications for concentration and welfare.

We ask the following *research questions*. First, what policies do firms lobby for? Although a vast literature documents how much firms spend on lobbying, how persistent lobbying is, and which broad topics it covers, we do not know which specific policy tools firms lobby for, which position they take on each, or how often they lobby on them. To answer this question, we measure which policies firms lobby for by using LLM classification of the federal lobbying reports of public firms and trade associations. We assign each request a firm makes to a specific policy tool from a new, comprehensive taxonomy, together with the direction of the requested change and whether the firm supports it.

Second, how does lobbying shape concentration, innovation, and growth, and does it reduce welfare? The answer is not obvious and requires analyzing not only static concentration but also dynamic effects. For instance, when regulatory costs increase, they increase the profits of large incumbents. This can reduce welfare through an inefficient reduction in supply and misallocation. However, profits fund innovation; thus, regulations could accelerate growth and raise welfare. Additionally, higher regulation due to lobbying might bring higher welfare by correcting externalities, but we abstract from the non-consumption effects of regulation in this paper. To quantify the net effect of lobbying on consumption-equivalent welfare, we build a dynamic general equilibrium model of oligopolistic competition and innovation in which firms lobby for firm-specific transfers and over industry-wide regulatory costs, and we evaluate the counterfactual of banning lobbying.

Empirical results. To learn what firms lobby for, we construct a new dataset of the specific policy tools the firms are lobbying for using the LLM classification of federal mandatory lobbying reports from 2008 to 2022. We analyze the free text of lobbying reports and, for each lobbying entity, extract each lobbying request and assign it to a concrete policy tool (corporate taxes, subsidies, appropriations, R&D credits, privacy regulation, licensing, etc.) from a new taxonomy of 87 instruments. For each instrument requested, we record the direction of change, whether the filer supports it, and which firms and industries the policy applies to. We focus on public firms, which lobby the most and whose size, costs, and innovation we observe, and on trade associations, the nonprofits that represent firms, whose revenues and dues we observe in their tax filings. We classify the reports of public firms headquartered in the United States and of trade associations, which carry 58% of lobbying dollars. We find that large firms lobby on their own, and that firms lobby most on appropriations and on regulatory costs such as privacy, conduct, and reporting rules. Almost all firms and associations ask for more appropriations and lower taxes, but they disagree on regulatory costs: among those that take a side, 63% of the asks of the largest 5% of firms within their NAICS4 industries are for higher costs, against 47% for all public firms and 36% for associations. Lobbying participation rises steeply with firm size and is persistent.

Motivated by these findings, we focus on three lobbying instruments: regulatory costs (e.g., privacy, licensing, and compliance requirements); reductions in the corporate tax rate; and increases in procurement contracts. We chose these instruments for several reasons. First, they let us study the industry-wide contest over regulatory costs alongside the own-benefit instruments, taxes and procurement, which are the focus of the existing literature. They also let us compare the welfare consequences of the two kinds of lobbying; additionally, these instruments have measurable outcomes. From the data, we measure participation in each instrument by firm size, participation persistence, and returns to lobbying, which we

observe in tax rates and procurement contracts. For regulatory costs, we observe their total level and how firms' lobbying affects the enactment of bill provisions that raise or lower them.

Quantitative model. In the data, we see that large firms lobby on their own, while small firms rarely lobby by themselves, and associations side with the smaller firms more often than with the largest. Thus, we build a dynamic quantitative model in which each industry consists of a few large strategic firms, the superstars, and a mass of atomistic fringe firms. Superstars compete à la Cournot, and fringe firms are perfectly competitive. Superstars innovate to raise their productivity, and fringe firms can make a breakthrough innovation and join the oligopoly. Fringe firms differ in their fixed costs of operating, so higher regulatory costs make the marginal fringe firms stop producing.

Large firms lobby for lower corporate taxes and for more procurement contracts, and all firms lobby over regulatory costs. To lobby on an instrument, a firm pays a fixed participation cost, and it can make its lobbying more effective by spending more. Because fringe firms are atomistic, none of them would lobby on its own, so they lobby through a trade association that acts on behalf of its members. Regulatory costs are set in a contest that aggregates the lobbying of firms that want them higher against the lobbying of firms that want them lower, and firms take into account how regulatory costs change the number of fringe firms and how many break through into the oligopoly.

In the estimated model, large firms lobby for higher regulatory costs, while the trade association mostly lobbies against them. The largest firms favor higher regulatory costs both because their profits rise as marginal fringe firms exit and because they are protected against entrants; the smallest superstars favor them mainly as protection against displacement by innovating entrants. On net, lobbying raises regulatory costs by 3%. This reduces the number of producing fringe firms by 0.7%, raises the profits of large firms, and lowers the rate at which fringe firms break through into the oligopoly. Industries become more concentrated.

Lobbying lowers welfare through lobbying spending and an industry composition effect. With higher regulatory costs, large firms innovate more. But industries become more concentrated: fewer fringe firms break into the oligopoly, resulting in fewer innovating oligopolistic firms and lower aggregate innovation. Overall, lobbying reduces consumption-equivalent welfare by 0.2% through lobbying spending, lower output, higher compliance costs, and slower growth. The paper provides a welfare analysis of how lobbying-driven regulatory costs affect consumption and labor supply, and we abstract from any offsetting or complementary effects on the non-consumption utility effects of regulation.

Importantly, we can decompose the welfare effect by policy instrument. Lobbying over regulatory costs accounts for almost the entire welfare loss, while tax and procurement

lobbying, the focus of the literature, accounts for almost none: it mostly shifts money from the household as a taxpayer to the household as a firm owner. Thus, not all lobbying is the same, and we need to think carefully about which types of lobbying should be allowed. The most salient outcome of lobbying, large firms paying lower taxes, is not the one that reduces welfare; it is lobbying over regulatory costs.

Related literature. This paper contributes to three strands of the literature. The first is the empirical literature on lobbying. It documents that lobbying rises with firm size and is persistent over time (Huneeus and Kim, 2024). Arayavechkit et al. (2018) find that taxes are the most frequent lobbying topic, measure lobbying frequency by general issue code, and study tax lobbying by atomistic firms. Closest to our paper, Macedoni and Weinberger (2026) classify lobbying reports by topic, such as investment, trade policy, and labor policy, and find that firms with low capital intensity are more likely to lobby for increases in fixed costs rather than variable costs. We contribute a systematic classification of the specific policy instruments firms and associations lobby for, such as corporate taxes, subsidies, appropriations, R&D tax credits, privacy regulation, and licensing, along with the requested change, the filer's position, and the persistence of lobbying on each instrument.

The second strand is the theoretical literature on lobbying. In static models of lobbying contests, regulation is determined by aggregating the pressures of groups lobbying pro or against the policy (Becker, 1983; Becker and Mulligan, 2003; Grossman and Helpman, 1994; Bombardini, 2008; Kang, 2016). Dynamic models study atomistic firms that lobby for lower taxes or higher sales (Arayavechkit et al., 2018; Huneeus and Kim, 2024; Hossein Dad and Nguyen, 2025; Macedoni and Weinberger, 2026). However, because these firms are atomistic, they can not move industry-wide regulation on their own. We contribute by developing a dynamic quantitative model in which firms lobby both over industry-wide regulation, in a contest between strategic firms and an association of atomistic firms, and for firm-specific benefits such as taxes and procurement.

The third strand is the literature on oligopoly and innovation (Ericson and Pakes, 1995; Akcigit and Ates, 2023). The closest paper, and the model we build on, is Cavenaile et al. (2026), which introduces innovating oligopolists and a competitive fringe in general equilibrium. We add lobbying and adapt the model so that lobbying can operate through the right channels. First, fringe firms differ in their fixed costs, letting us study how regulation affects different types of small firms and adding a selection margin. Second, a breakthrough by fringe firms doesn't just create new firms at the lowest productivity to become strategic; the entrant can displace incumbents and even leapfrog the leader. This creates an additional lobbying motive: large firms lobby not only to raise their own profits and market shares, but also to protect themselves from innovative smaller firms. Thus, we contribute to the

oligopoly and innovation literature by enabling the analysis of how lobbying and regulation affect industry structure and innovation.

The paper is organized as follows. Section 2 describes the data and classification, Section 3 presents the empirical facts, Section 4 develops the model, and Section 5 describes the model estimation. Section 6 explores the effects of lobbying, its welfare cost, and the counterfactuals. Section 7 concludes.

2 Measuring what firms lobby for

To study the welfare consequences of lobbying over regulations, we first need to know what firms lobby for. Existing lobbying literature shows how much each firm spends, the broad issue areas it lobbies on, and its positions on bills, but not which specific policy tool it mentions, how the tool changes, or whether the firm supports that change. We construct these measures from the free text of quarterly federal lobbying reports filed under the Lobbying Disclosure Act (LDA) in 2008-2022 (see the sample report in the Appendix A). It describes the specific issues each client lobbied on, often mentioning the bills and regulations. We turn this text into a dataset of lobbying requests.

Our unit of observation is a lobbying report: one client, one registrant (a lobbying firm or the client's own lobbyists), and one quarter. We match the client to a public firm or trade association. We split the free text of the reports into individual requests and assign each to one of 87 policy instruments, along with the requested change relative to current law, who the policy covers, and the client's position. For example, the request "support a decrease in the corporate tax rate" is recorded as the corporate income tax rate, a decrease, covering all firms, and supported by the client. At the end of the analysis, for every client, quarter, and instrument, we record whether the client lobbies on the instrument and the side the client takes. The rest of this section describes how we build the dataset in more detail and how we check its correctness.

2.1 Reports and clients

Every registered lobbyist files a report (form LD-2) for each client every quarter. The report names the client and the lobbyist, states the amount spent, and, for each general issue area, describes in free text the specific issues lobbied, often citing bills, public laws, and regulations. We use every report filed from 2008 to 2022, assembled from OpenSecrets (Center for Responsive Politics, 2024), LobbyView (Kim, 2018), and the Senate Office of Public Records. Free issue text is largely missing before 2006: among reports with a positive

amount, 21% include it in 1999–2006, against 96% in 2007 and 98% in 2008–2022, the latter carrying 99% of the dollars. We start the dataset in 2008 because the Honest Leadership and Open Government Act of 2007 made the issue-reporting requirement stricter, requiring quarterly rather than semiannual reporting; thus, every year in our sample has the same reporting frequency and better issue-text coverage.

We match clients to public firms and trade associations, which account for 65% of lobbying dollars, 36% of reports, and 52% of the issues listed in 2008–2022 (Figure 1). We choose this set of clients because we can observe industry, size, costs, and innovation for public firms, and the revenues and dues for associations, which also represent smaller firms. We match public firms to the Compustat dataset by name, and we match their subsidiaries and merged companies using CRSP, their SEC filings, and EDGAR, so we attribute a subsidiary's lobbying to its parent. We select firms headquartered in the United States, which together with the associations carry 58% of lobbying dollars, and obtain 75,957 firm-years for 9,690 firms across 268 four-digit NAICS industries. Trade associations are matched to the IRS register of business leagues (501(c)(6) organizations), and industry associations are assigned to the industries they represent.

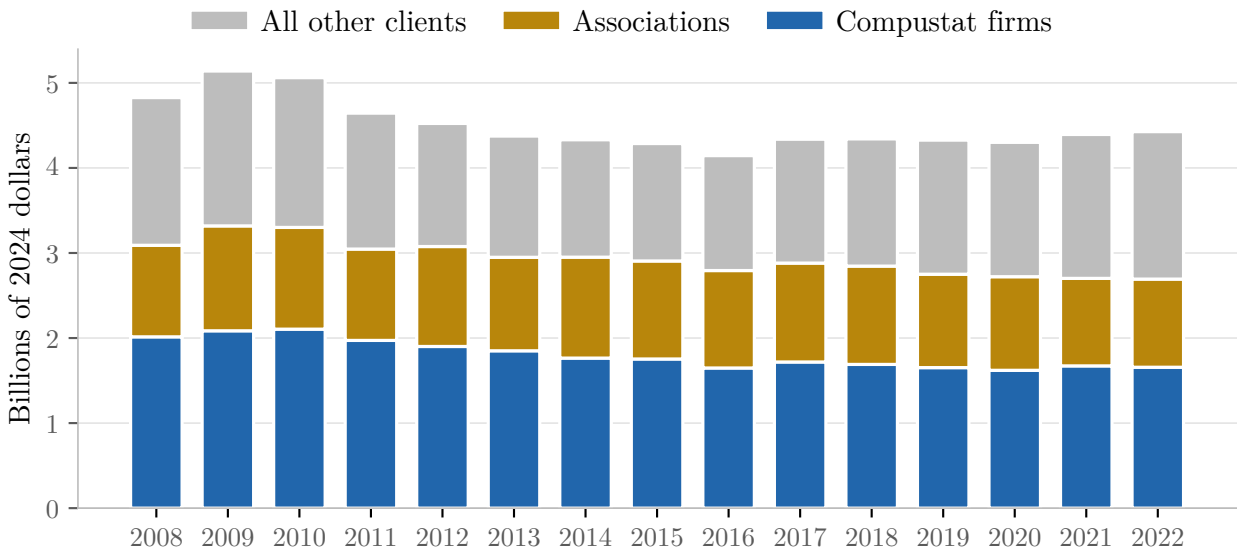


Figure 1. Who Lobbies: Compustat Firms and Associations

Notes: Federal lobbying spending by type of client, billions of 2024 dollars, 2008–2022. Associations are clients matched to an IRS 501(c)(6) organization.

2.2 From text to policy instruments

The issue text is the central object of the measurement. A single issue text block often contains several requests on different policies, including bills and regulations. We process it

in five steps.

Extracting requests. Each issue text block is split into requests that each address a single policy. We separate text blocks into requests using pattern rules for unambiguous structures, such as enumerated lists; for blocks whose boundaries require judgment, we split them with a large language model and require it to reproduce the original text verbatim (Appendix B).

Bills. Every cited bill and public law is matched to the congressional record and corrected for mistyped numbers using the titles and context of the bills; cited regulations are matched to their titles and summaries on Regulations.gov. We classify all bill provisions the same as issue text.

Instruments. Each request is assigned to an instrument in a taxonomy of 87 policy instruments. We created the taxonomy that unites the government policies that affect firms from the Treasury tax-expenditure list, the taxonomy of regulatory forms of Xie and Perez (2018), the Harmonized Tariff Schedule and the UNCTAD classification of non-tariff measures, the federal Assistance Listings and the Federal Acquisition Regulation, and the patent, exclusivity, and immigration laws, with the tools-of-government catalog of Salamon (2002) as a check on completeness. We map each instrument to the part of the firm's problem it moves, which we call its *wedge* (Table 1), such as affecting labor costs, output, innovation cost, etc. Instruments are defined by their effect on the firm rather than who they apply to: a research credit applying to one firm and one applying to every firm are the same instrument with different eligibility (Appendix C).

Direction, coverage, and side. For each request, the classifier records the change requested relative to current law (for example introduce, raise, lower, widen, narrow, extend, fund, exempt, delay, or repeal, with the magnitude when the text states it), whom the policy covers (industry, geography, product, legal form, and thresholds on firm characteristics), and the client's position on it (support, oppose, amend, monitor, or unstated). For each instrument, the codebook records its payoff sign for different types of firms (buyer vs seller of the good, domestic vs foreign firms) and its shape (proportional, quasi-fixed, or mixed). The *direct payoff* of a request for each firm is the direction of the change times the instrument's sign: increasing a credit is favorable, increasing an excise tax is not. The *side* on the instrument is the position multiplied by the direction of the change: supporting a bill that raises a compliance requirement or opposing its repeal both attribute the client to the side of supporting higher regulatory costs. However, positions are stated rarely in the lobbying reports: the client supports or opposes the policy only for 6% of firms' matched issues and 16% of associations' (Table 2).

Table 1. Wedges of the Firm's Problem and the Instruments That Affect Them

Wedge	Instruments (examples)	Number
Demand, s^d	Appropriations and earmarks, procurement contracts, buyer purchase subsidies, tariffs and quotas, antidumping duties	14
Regulatory cost, s^R	Licenses and charters, product certification, pre-market approval, facility permits, recurring reporting, labeling, data privacy and cybersecurity, liability rules	18
Output, s^y	Per-unit production credits, excise taxes, emission charges, performance standards with per-unit costs, quotas	17
Profit tax, s^T	The corporate income tax, deductions, reduced rates on classes of income, foreign income, losses	7
Innovation cost, s^i	Research credits and expensing, research grants, patents, product exclusivity	5
Capital cost, s^k	Investment credits, accelerated depreciation, tax-exempt bonds, federal lending	6
Labor cost, s^w	Hiring credits, payroll taxes, wage floors, worker protections, visas	8
Set prices, s^p	Government payment rates, mandatory discounts, price ceilings and floors, rate regulation	5
Competition, s^A	Merger control, platform regulation, antitrust exemptions, mandated access	4
Lump-sum transfers, s^t	Grants and direct payments, regulatory user fees	2
Place-based policies	Enterprise zones	1

Notes: The codebook's 87 instruments are categorized by the wedge of the firm's problem they affect. A regulatory cost is a requirement whose cost of compliance does not scale with the firm's output, or does so only in part (liability and conduct rules); requirements whose cost scales with output are output wedges. Appendix C lists the instruments.

From requests to firm measures. We record the client as lobbying on an instrument in a quarter if any of its reports names the instrument in the client's own words, at high or medium confidence, or cites a bill, a named section or title of a bill, or a regulation whose provisions that the citation refers to all concern this instrument (among the provisions classified at high or medium confidence). The bill's name plays no role. A citation whose provisions concern several instruments counts for none of them, also when the instruments belong to the same part of the firm's problem; otherwise, citing a defense authorization would make a firm active on procurement, export controls, and cybersecurity. Appendix E counts every ask instead, giving each instrument of a citation the share of its provisions on that instrument. A client's side on an instrument in a year is the share of its requests on the instrument in that year that point to a higher level, over all the reports filed for the client, counting only requests stated in its own words at high or medium confidence and citations that state support or opposition and whose provisions concern a single instrument.

Coverage and validation. Table 2 reports the coverage of the classification by Claude Opus 5.5. Among issues mentioned in the reports, 46% for firms and 47% for associations are matched to a policy instrument in the text itself or through a cited bill whose section or side the text names, and about a third are bare citations of a bill or rule, which enter through the provisions of the legislation; the rest contain no firm policy instrument or are too vague to classify, such as "climate change" or "tax reform". Low confidence issues (10% of firms' and 11% of associations' matched issues) are dropped. Classifying the same text twice gives the same instruments in 94% of cases when neither classification has low confidence, and the same position in 99%. An independent classification by the authors with a second language model, Claude Fable 5.1, assigns the same instrument to 83% of a sample of 1,000 high- and medium-confidence issues of industry associations in 2008–2018, and to 89% of a sample of 1,000 bill provisions. The positions we record on bills agree with those recorded by MapLight for 95% of the 492 client–bill pairs both cover (93% of the 55 firm–bill pairs). A hand-coded validation sample, which we will use to debias the estimates following Carlson and Dell (2025), is in progress.

Table 2. Classification: Coverage and Confidence

	Compustat firms 2008–2022	Associations 2008–2022
Reports with issue text	169,612	125,556
Issues listed in the reports	1,298,944	1,557,060
matched to a policy instrument	46%	47%
bare citation of a bill or rule	31%	35%
no policy instrument	11%	8%
unclear (e.g., "climate change", "tax reform")	13%	10%
Confidence of the matched issues:		
high	51%	45%
medium	37%	40%
low (dropped)	10%	11%
cites a bill's section or states a side, no own words	2%	4%
Position stated in the matched issues:		
support or oppose	6%	16%
monitor	5%	4%
unstated	89%	80%

3 What Firms Lobby For

We document four facts about the content and the structure of lobbying. Together they motivate the model in Section 4: lobbying is concentrated on two kinds of policy, firms agree on one and disagree on the other, and the decision to lobby is persistent across instruments.

Participation in lobbying. Lobbying is concentrated among large firms and large associations. Figure 2 reports the share of all public firm-years and association-years with a lobbying report, by the firm's sales percentile within its industry (for associations, by revenue percentile among 501(c)(6) organizations): 6.1% of the firm-years below the industry's median of sales have a report, 19% between the median and the 80th percentile, 35% between the 80th and the 90th, 48% between the 90th and the 95th, and 63% among the largest 5%. The same pattern holds for associations. The facts below describe what these clients lobby on.

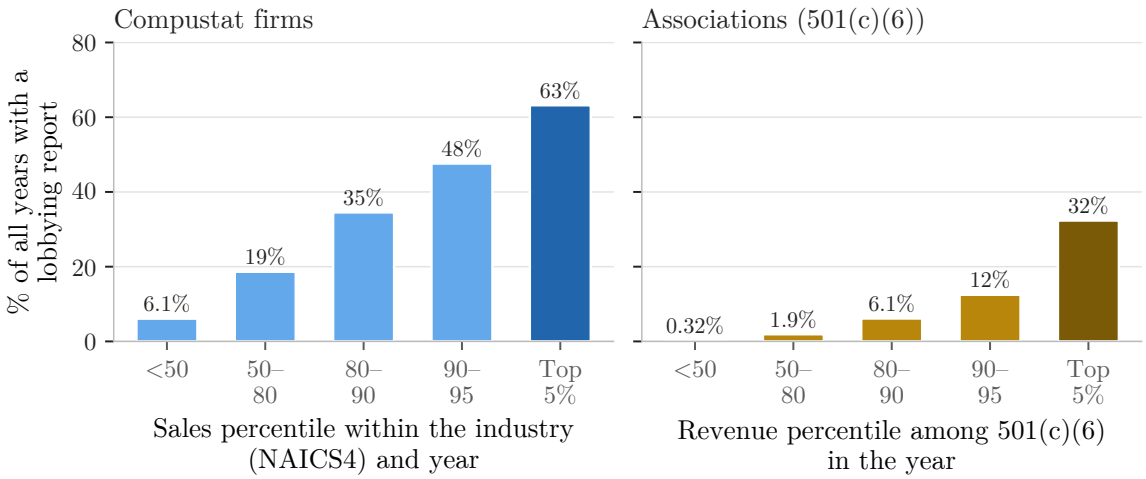


Figure 2. Lobbying by Firm Size

Notes: Share of all firm-years with a lobbying report (a report with a positive amount that names a classified instrument), by the firm's sales percentile within its industry (four-digit NAICS) and year; 75,957 firm-years of Compustat firms headquartered in the United States. Associations: share of the years in which a 501(c)(6) organization filing an IRS Form 990 or 990-EZ with positive revenue (30,112 a year on average) is a lobbying client with a positive amount, whether or not its reports name a classified instrument, by its revenue percentile among these organizations in the year. 2008–2022.

3.1 Fact 1: Firms lobby most on government demand and on regulatory costs

Figure 3 reports, for each wedge, the share of the quarters with a lobbying report in which the firm (or association) lobbies on it. Two wedges lead. Lobbying firms lobby on government demand (appropriations, procurement, buyer subsidies, public infrastructure,

and trade measures) in 63% of their quarters and on regulatory costs in 50%; for associations the shares are 66 and 49%. The profit tax, the subject of a large literature on corporate lobbying (Richter et al., 2009; Arayavechkit et al., 2018), appears in about a fifth of the quarters (20% for firms, 15% for associations). The largest 5% of firms lobby on every wedge more often, on government demand in 75% of their quarters and on regulatory costs in 66%. These shares describe what lobbying clients lobby on.

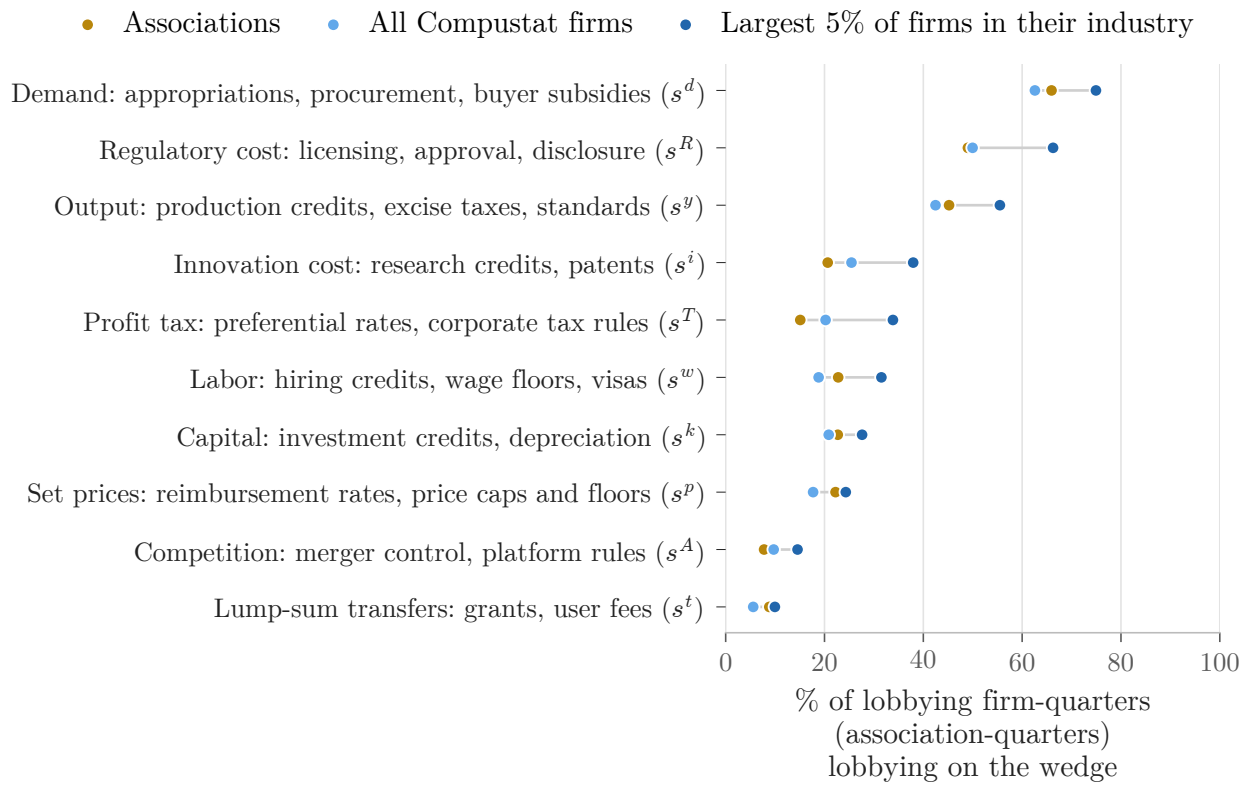


Figure 3. Lobbying by Wedge of the Firm's Problem

Notes: Share of lobbying firm-quarters (association-quarters) in which the client lobbies on the wedge. 52,079 Compustat firm-quarters, of which 14,970 of the largest 5% of firms in their industry by sales; 60,596 association-quarters; 2008–2022.

At the level of instruments, program funding and appropriations lead for every group (32% of firm-quarters, 38% of association-quarters, and 43% for the largest 5% of firms), followed for firms by data privacy and cybersecurity, the most lobbied regulatory cost (21%), technology and performance standards (20%), procurement contracts (19%), buyer subsidies and purchase mandates (19%), and public infrastructure (16%). Associations lobby more than firms on standards, reporting duties, and payment rates, and far less on procurement (8 versus 19%). In the six sectors of Table 3, firms lobby on what they sell to the government and on what their regulator controls, and a regulatory cost is among the three most lobbied instruments in five of them: conduct rules on transactions in health care services and in

finance, pre-market approval in pharmaceuticals and devices, privacy in technology, and facility permits in energy and utilities (Table 3).

Table 3. What Each Sector Lobbies on Most

	Most common	Second	Third
Aerospace and defense	Procurement (s^d)	Appropriations (s^d)	R&D grants (s^i)
Health care services	Payment rates (s^p)	Appropriations (s^d)	Conduct rules (s^R)
Pharma and devices	Appropriations (s^d)	Payment rates (s^p)	Pre-market approval (s^R)
Technology	Appropriations (s^d)	Procurement (s^d)	Privacy (s^R)
Finance and insurance	Conduct rules (s^R)	Buyer subsidies (s^d)	Privacy (s^R)
Energy and utilities	Technology standards (s^y)	Facility permits (s^R)	Buyer subsidies (s^d)

Notes: The three instruments lobbied in the largest share of the sector's lobbying firm-years, the profit-tax instruments counted as one; Compustat firms, 2008–2022. Conduct rules: rules on how a transaction is made.

3.2 Fact 2: Firms agree on transfers and disagree on regulatory costs

Figure 4 reports, for each instrument, the mean share of asks for more of it among the firm-years (association-years) taking a side. On instruments that benefit the firm directly, firms speak with one voice: 96 to 100% of asks are for more funding, contracts, grants, and credits (53% on patents and copyright, an instrument that favors firms holding intellectual property and burdens those using it), 97% of asks on the profit tax are for a lower one, and 99% of asks on contracts are for more. On regulatory costs, firms split. Depending on the instrument, between 6% (facility permits) and 66% (labeling) of the asks of firms taking a side are for *higher* regulatory costs; pooling licenses, reporting duties, labeling, and privacy, the mean share is 47% for firms and 36% for associations.

The firms that ask for higher regulatory costs are disproportionately large. Counting each firm-year once, the share of asks for higher regulatory costs is 27% among firms below the 80th percentile of their industry's sales, 43% between the 80th and 95th percentiles, and 63% among the largest 5%; counting every ask, so that firms with more asks weigh more, these shares are 21, 41, and 72% (Figure 5). Associations ask for higher regulatory costs less often than the largest firms, at 36% (37% counting every ask). Within the same industry and year, the associations' share is 27 percentage points below that of the largest 5% of firms (standard error 10) and close to that of firms below the 80th percentile (10 points above, standard error 10), comparing the industry associations of the 53 industries in which firms take a side with those firms, with industry and year fixed effects and standard errors clustered by filer. On privacy, the most lobbied regulatory cost, the share is 81% among the

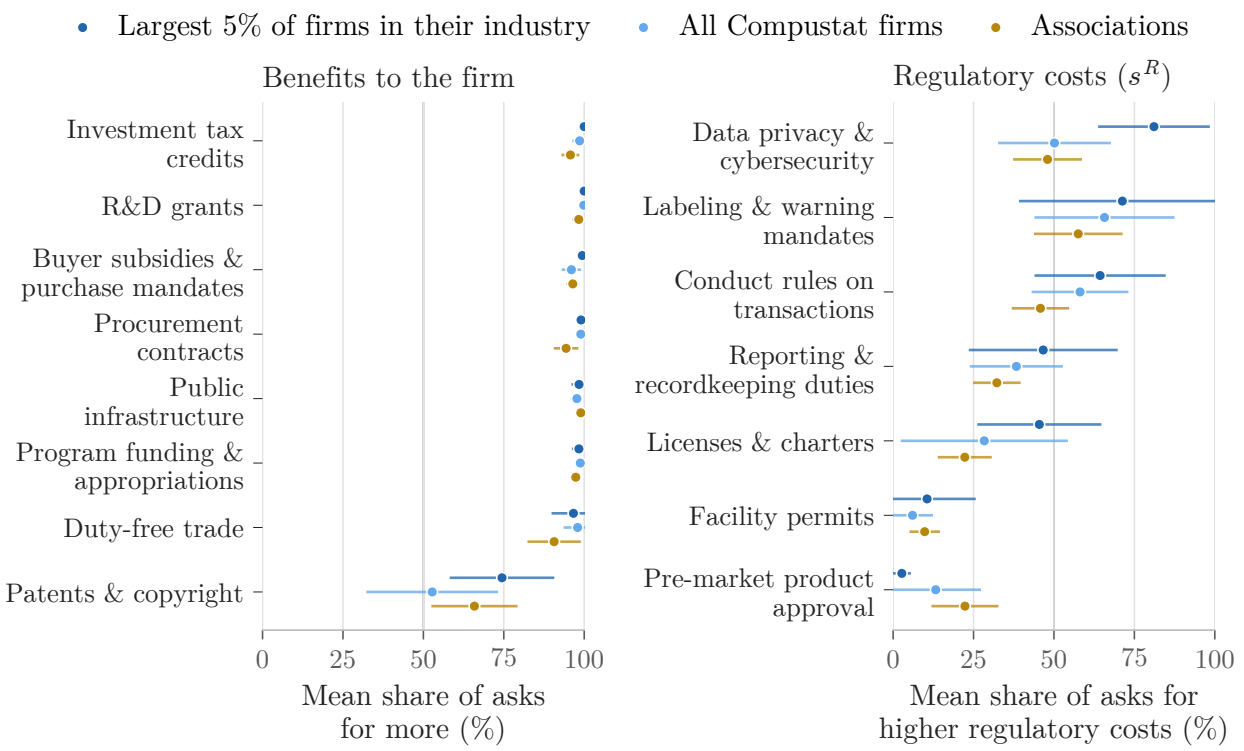


Figure 4. The Side Firms Take: Benefits and Regulatory Costs

Notes: Left: mean share of asks for more of the benefit among the firm-years (association-years) taking a side. Right: mean share of asks for higher regulatory costs. A filer-year's share is the share of its requests on the instrument in the year that point to a higher level (Section 2.2), and each filer-year counts once. 95% confidence intervals clustered by filer. 2008–2022.

largest firms and 50% among all firms. However, this pattern rests on few observations: only 278 firm-years take a side on regulatory costs. Counting each firm-year once, the largest 5% exceed firms below the 80th percentile by 36 percentage points across all firms (standard error 10) and by 24 points within an industry and year (standard error 13); counting every ask, by 50 and 26 points (standard errors 12 and 12). Only 11 of 53 industries have firms stating a side both above and below the industry's 80th percentile of sales, and counting each firm once rather than each firm-year lowers the first two gaps to 18 points (standard error 11) and 10 points (standard error 16). We therefore target the overall shares of sides in estimation and treat the size gradient as an untargeted pattern.

A natural concern is that the sides we observe are selected. Sides are written down rarely: for 1.9% of requests on regulatory costs and 4.2% of requests on other instruments, and 5% of the firm-years that lobby on regulatory costs state a side. Three checks show that the stated sides are representative, with one exception. First, large firms state a side in more years only because they file more requests: across all instruments, the fifth of firms largest relative to their industry and year file a median of 101 requests a year, against 8 for

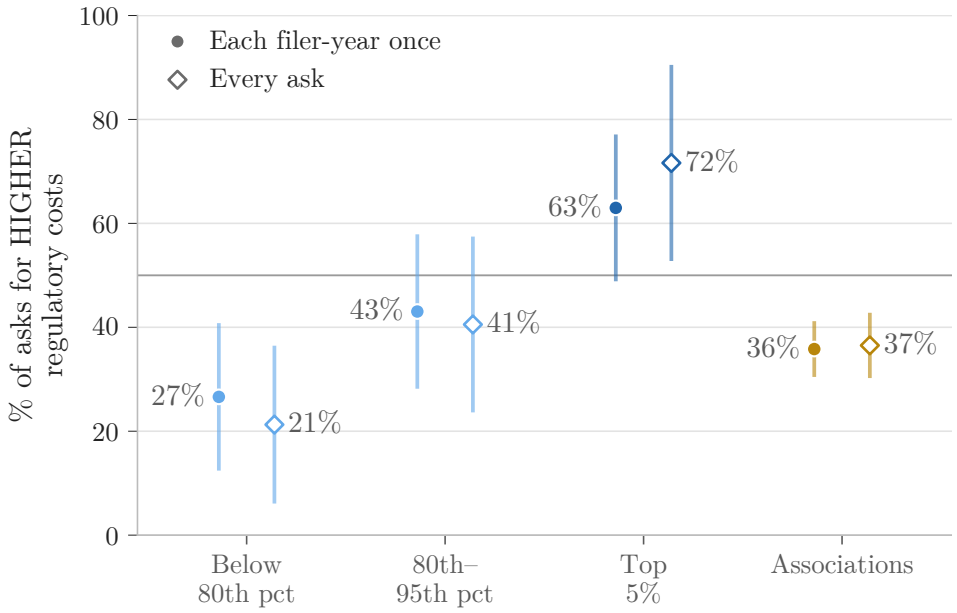


Figure 5. Share Asking for Higher Regulatory Costs, by Firm Size

Notes: Share of asks on regulatory costs (licenses, reporting duties, labeling, privacy) that are for higher regulatory costs, by the firm's sales percentile within its industry and year. Circles: the mean over firm-years (association-years) of each year's share of asks, each year counted once. Diamonds: asks for higher costs over all asks of the group, so that filers with more asks weigh more. 278 firm-years of 112 firms (1,284 asks) and 945 association-years of 264 associations (4,858 asks) take a side; 2008–2022. 95% confidence intervals clustered by filer.

the smallest fifth, and state a side 17.8 percentage points more often (standard error 2.2), but no more often at the same number of requests (−1.2, standard error 2.3); on regulatory costs, they are no more likely to state a side either way (0.0 percentage points, standard error 1.5, and −4, standard error 2.0, at the same number of requests). Second, on most instruments, whether a side is written down depends mostly on the lobbying firm that writes the report rather than on the client: a client's reports state a side in 11.0% of cases when its lobbying firm states sides for its other clients and in 2.6% otherwise, and clients that state sides often do not choose lobbying firms that do (correlation 0.02). On regulatory costs, it is the client's own habit (5.2 versus 0.2%). Third, and this is the exception, the 27 firms that state a side on regulatory costs in more than a third of their other years lean against them: within an industry and year, their share of asks for higher regulatory costs is 37 percentage points below that of firms that state a side only once (standard error 14). Counting each firm once rather than each firm-year would raise the mean share from 47 to 50%; we keep the firm-year share.

3.3 Fact 3: Lobbying rises steeply with firm size on every instrument

We now focus on the three instruments that the model will carry: the firm's own profit tax, its own federal procurement contracts, and *regulatory costs*, the requirements whose cost grows less than proportionally with the firm's size (licenses, recurring reporting duties, labeling, and privacy and cybersecurity rules). Reports that include these instruments carry 61% of firms' lobbying dollars, and the instruments span both kinds of policy: tax and procurement are one-sided, regulatory costs two-sided. Participation on each instrument rises steeply with size (Figure 6): the share that lobbies on regulatory costs in a year rises from 0.8% of firm-years below the industry's median of sales to 35% among the largest 5%, on the profit tax from 0.4 to 25%, and on procurement from 1.1 to 17%. Most of this gradient is whether a firm lobbies at all: 6% of the firm-years below the median and 63% of those of the largest 5% have a lobbying report (Figure 2). Among the years in which a firm lobbies, the shares still rise with size, from 13 to 55% on regulatory costs, from 6 to 39% on the profit tax, and from 18 to 27% on procurement (Appendix Figure 14). Within an industry and year, each log point of sales raises the probability that a lobbying firm lobbies on regulatory costs by 9.1 percentage points, on the profit tax by 7.8, and on procurement by 2.7 (standard errors 0.4; 13,669 lobbying firm-years, 2008–2022), against averages of 36, 23, and 22%. Among the four largest firms of each industry, the gradient continues by rank: in a given quarter, 24% of industry leaders lobby on regulatory costs, against 19, 15, and 12% of the second through fourth firms (Table 4).

Table 4. Participation and Persistence of the Four Largest Firms

	Lobbying in a quarter, by rank				Next quarter	
Instrument	Leader	#2	#3	#4	Stay	Join
Regulatory costs	0.236	0.192	0.153	0.122	0.924	0.018
Profit tax	0.181	0.131	0.100	0.082	0.904	0.015
Procurement	0.133	0.090	0.059	0.058	0.897	0.010

Notes: The four largest Compustat firms headquartered in the United States in each four-digit NAICS industry and year, all 268 industries, 1,658 firms. Regulatory costs (licenses, reporting duties, labeling, privacy): a firm lobbies in a quarter if any of its reports in the quarter lobbies on them, 2008–2022, 49,832 firm-quarters. Profit tax (all corporate income tax instruments) and procurement contracts: lobbying in the last quarter of the fiscal year, fiscal years 2008–2022, 12,325 firm-years. Monitoring counts. Stay: probability of lobbying on the instrument next quarter given lobbying this quarter; join: given not lobbying.

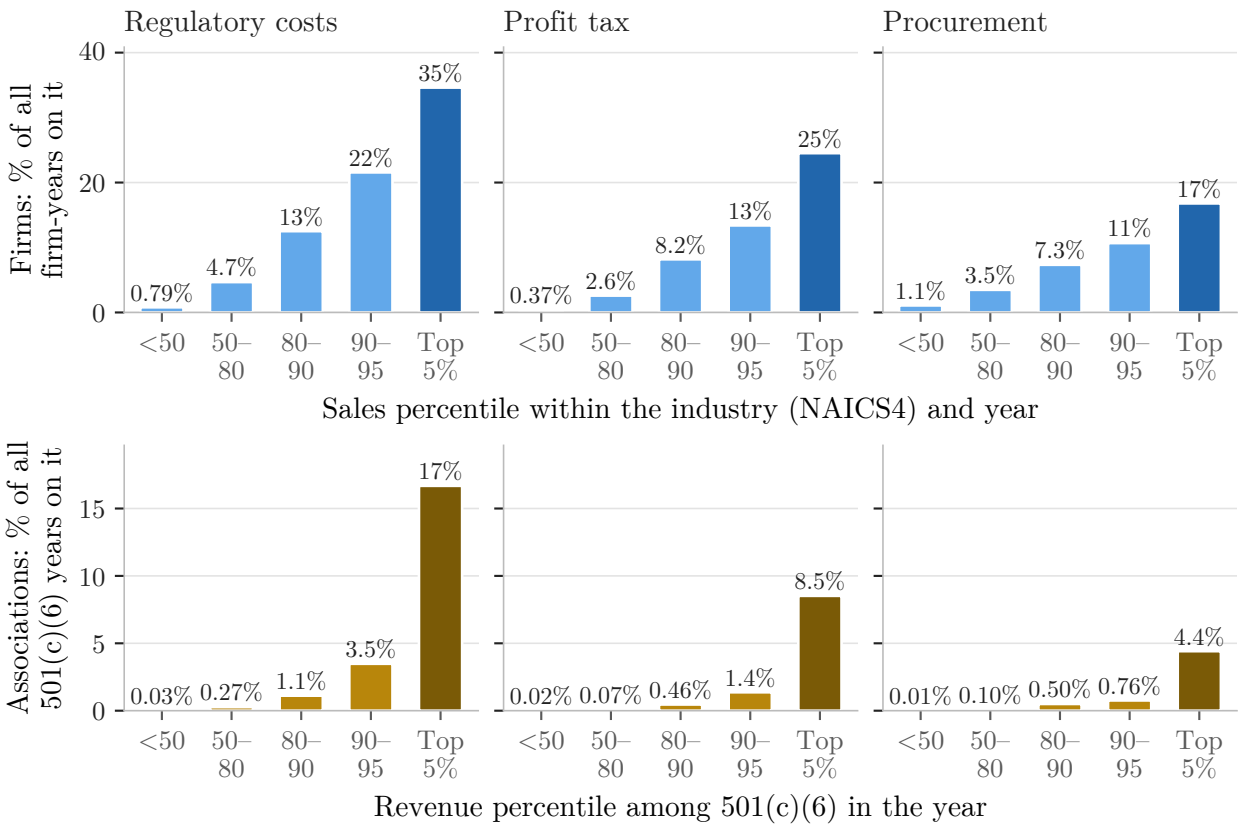


Figure 6. Participation in Lobbying by Firm Size

Notes: Share of all firm-years in which the firm lobbies on the instrument, by the firm's sales percentile within its industry (four-digit NAICS) and year, 75,957 firm-years of Compustat firms headquartered in the United States; for associations, the share of the years of 501(c)(6) organizations filing an IRS Form 990 or 990-EZ with positive revenue (30,112 a year on average) in which the association lobbies on it, by its revenue percentile among these organizations in the year. 2008–2022. Appendix Figure 14 reports the shares among the years with a lobbying report.

3.4 Fact 4: Lobbying is persistent across instruments

Nine in ten firms that lobby on an instrument in one quarter lobby on it in the next: 91% on regulatory costs, 89% on the profit tax, and 89% on procurement among all Compustat firms, and 93, 91, and 91% among the largest 5%. Few firms join in a given quarter: 0.4 to 0.6% of all firms and 1.6 to 3.4% of the largest. Associations are as persistent (92% on regulatory costs, among consecutive quarters in which the association has a report). Lobbying thus looks like a costly organizational commitment, made at infrequent times and maintained, rather than a flow chosen anew each quarter.

3.5 Associations and the largest firms

Industry associations and the largest firms lobby on regulatory costs side by side. We take all 268 four-digit NAICS industries and every quarter of 2008–2022, 16,080 industry-quarters; the association is active in a quarter when any industry association of the industry lobbies on regulatory costs, and the largest firms when any of the industry's four largest firms does. The association lobbies without these firms in 25% of industry-quarters, these firms lobby without the association in 9%, both lobby in 20%, and neither in 45%; the association is active in 46% of quarters, and in 83% of the quarters in which anyone in the industry lobbies on regulatory costs.

3.6 From facts to the model

The facts shape the model in four ways. Because tax and procurement lobbying is one-sided and firm-specific, the model treats it as each firm's own instrument, with an outcome observed in the data: the firm's tax rate and its contracts. Because lobbying over regulatory costs is two-sided and the rule applies to every firm in an industry, the model determines regulatory cost as a contest between firms with opposing stakes, and each firm's stake, and hence its side, is an equilibrium object. Because lobbying is concentrated among the largest firms, and associations side with the smaller firms more often than with the largest, each industry has a few strategic superstars that lobby individually and a competitive fringe that lobbies only through its association. Because participation is persistent and rises with size, the decision to lobby carries a fixed cost that is revised only infrequently. The model computes the same moments as the data: participation by rank, persistence, side shares, association activity, and the outcomes of tax and procurement lobbying.

4 The model

We build a continuous-time economy with multiple industries and a representative household that expands on Cavenaile et al. (2026). Each industry consists of a few large, strategic firms (superstars) and a competitive fringe. Superstars can increase their productivity through incremental innovation, fringe firms can become superstars through breakthrough innovation, and entrepreneurs create fringe firms. Firms can lobby on three policy instruments: the firm-specific profit tax, firm-specific procurement contracts, and the industry-wide regulatory fixed cost. We introduce the key new margin of selection in the fringe: a higher regulatory cost forces the least efficient fringe firms to stop producing, which raises superstars' profits and lowers the rate at which fringe firms break into the superstar tier.

4.1 Environment

Household. The lifetime utility of the representative household is a function of consumption of the final good and leisure

$$U = \int_0^\infty e^{-\rho t} \left[\ln C_t \; - \; \xi \, \frac{L_t^{1+1/\epsilon_L}}{1+1/\epsilon_L} \right] dt, \qquad C_t \; = \; C_t^p + G_t, \tag{1}$$

C_t is total consumption: private consumption C_t^p, chosen by the household, plus government purchases of the final good G_t, which the household takes as given. L_t is labor supply and ϵ_L the Frisch elasticity of labor supply.

Final good. The final good Y_t is produced competitively by aggregating over the continuum of industries $j \in [0,1]$:

$$\ln Y_t \; = \; \int_0^1 \ln(y_{jt}) \, dj. \tag{2}$$

We normalize the price of the final good to one. With Cobb–Douglas aggregation, the expenditure share of the final good is constant across industries.

Industry structure. At each instant t, each industry j is composed of $N_{jt} \leq \bar{N}$ *superstar* firms, who produce differentiated varieties and compete in quantities, and a continuum of competitive fringe firms with total mass m_t. The output of industry j at time t is a CES aggregate of superstar varieties and the fringe variety:

$$y_{jt} \; = \; \left(\sum_{i=1}^{N_{jt}} y_{ijt}^{\frac{\eta-1}{\eta}} + \tilde{y}_{fjt}^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}, \qquad \eta > 1, \tag{3}$$

where $\tilde{y}_{fjt} = \int_0^{m_t} y_{fkjt} \, dk$ is aggregate fringe output.

Technology. At each instant t, each firm, superstar or fringe, produces output with a decreasing returns to scale technology in labor:

$$y_{ijt} \; = \; q_{ijt} \, l_{ijt}^{\alpha}, \qquad \alpha \in (0,1). \tag{4}$$

q_{ijt} is the quality of firm i, α parametrizes the decreasing returns, and l_{ijt} is labor inputs.

Each superstar's quality follows a quality ladder. Denote by $n_{ijt} \in \{0, 1, \ldots, \bar{n}\}$ the quality gap of superstar i relative to the leader in industry j (the leader's gap is zero), with $\bar{n}$ the maximum quality gap to stay an oligopoly. The ladder depth $\bar{n}$ is not a free parameter: it is the largest gap at which the smallest superstar, in every configuration of its rivals' gaps, sells at least 10% of the leader's sales in the static equilibrium with $\bar{m}^{\mathrm{a}}$ active fringe firms,

the data's definition of a superstar. Superstar i's quality is

$$q_{ijt} \;=\; q_{jt}^{\text{leader}} \cdot (1+\lambda)^{-n_{ijt}}, \qquad \lambda > 0, \tag{5}$$

and q_{jt}^{leader} is the quality of the leader in industry j at time t.

Quality is constant across firms in the competitive fringe of industry j. The quality level in the fringe is inherited from j's superstars through technology diffusion given by

$$q_{fjt} \;=\; \zeta\Big(\sum_{i=1}^{N_{jt}} q_{ijt}^{\gamma}\Big)^{1/\gamma}, \qquad \gamma > 0, \tag{6}$$

The elasticity of fringe quality with respect to the quality of superstar i is $q_{ijt}^{\gamma}/\sum_{k=1}^{N_{jt}} q_{kjt}^{\gamma}$: the fringe learns the most from the most productive superstars. Each superstar i internalizes that its innovation spills over to the fringe in proportion to i's quality, eroding part of the innovator's advantage over its fringe competitors. The curvature γ governs how concentrated these spillovers are. As $\gamma \to \infty$, fringe productivity tracks the leader's productivity, as in Cavenaile et al. (2026); a lower γ spreads the spillovers more evenly across the superstars. Together with γ, ζ sets the level of fringe productivity and the fringe's share of industry sales.

Every active firm incurs a fixed cost of regulatory compliance, with R_{jt} set by lobbying: a fringe firm pays $R_{jt}Y_t$ and superstar i pays $\kappa_{ijt}R_{jt}Y_t$, where $\kappa_{ijt} = \bar{\kappa}\,(1+\lambda)^{-\gamma_R n_{ijt}}$, so that a superstar's compliance cost rises with its size. In addition, each *active* fringe firm k incurs an idiosyncratic operating cost $\varepsilon_k Y_t$ with ε_k drawn at the firm's birth according to

$$\varepsilon_k \;\sim\; \text{LogN}(\mu_\varepsilon, \sigma_\varepsilon^2), \qquad \mu_\varepsilon = \ln\bar{\varepsilon} - \sigma_\varepsilon^2/2, \tag{7}$$

so that $\mathbb{E}[\varepsilon_k] = \bar{\varepsilon}$. The cumulative distribution and density of ε are denoted $F(\cdot)$ and $f(\cdot)$. A fringe firm can stop and restart producing at no cost. Every instant it operates if and only if $\varepsilon_k \le \varepsilon_{jt}^*$. We call the operating firms *active*; their mass is $m_{jt}^{\text{a}} = m_{jt}\,F(\varepsilon_{jt}^*)$. The draw ε_k is the only source of heterogeneity among fringe firms, and it generates the *selection margin* that we exploit throughout: a higher R_{jt} lowers ε_{jt}^*, and the marginal fringe firms stop producing. Heterogeneous fringe fixed costs and a positive R_{jt} are the central structural deviation from Cavenaile et al. (2026): they create the margin through which regulation, and lobbying over it, operates.

Entry and exit. Entry into the fringe is governed by a unit mass of entrepreneurs who pay $\psi e_t^2 Y_t$ to generate a Poisson rate e_t of new fringe firms. New fringe firms are randomly allocated across industries, implying $m_{jt} = m_t$. Entrepreneurs sell each new firm at its

expected value before the draw of ε_k from (7) and remain in the entrepreneur population. New firms enter only the fringe; a new superstar arises only from a successful breakthrough by a fringe firm. Fringe firms die exogenously at Poisson rate δ; firms that are not producing stay in the pool until they die, and resume production whenever the industry state makes it profitable. A superstar exits when a rival's innovation would put it more than $\bar{n}$ steps behind the leader, or when it is displaced by an entrant.

Same as in Cavenaile et al. (2026), modeling entry into and exit from the fringe explicitly lets us map entry in the model to new business creation in the data. This contrasts with models in which entrants immediately become industry leaders or are otherwise large enough to interact strategically with incumbents. Our model instead generates the following life cycle: each industry has a large pool of fringe firms, only some of which produce at any time, and a much smaller number of breakthroughs create new superstars.

4.2 Innovation

Innovation occurs through two distinct technologies: incremental innovation by incumbent superstars on the quality ladder, and breakthrough innovation, through which active fringe firms become superstars.

Superstar incremental innovation. Each superstar i in industry j chooses an R&D intensity $z_{ijt} \geq 0$ that generates a Poisson arrival rate z_{ijt} of a one-step quality improvement, at cost

$$\text{cost}_i^{\text{R\&D}} \;=\; \chi z_{ijt}^{\phi} Y_t, \qquad \phi > 1. \tag{8}$$

Upon success, the firm's productivity rises by a factor $(1+\lambda)$. If a follower innovates, its gap falls by one. If the leader innovates, the frontier moves up by one step, and every other superstar's gap rises by one; a superstar whose gap would exceed $\bar{n}$ exits.

Fringe breakthrough innovation. Each active fringe firm k chooses the arrival rate $x_{fkjt} \geq 0$ of a breakthrough that would create a new superstar, at cost

$$\text{cost}_k^{\text{breakthrough}} \;=\; \chi_f \, x_{fkjt}^{\phi_f} Y_t, \qquad \phi_f > 1. \tag{9}$$

A breakthrough creates a new superstar owned by the innovating firm, whose fringe operation continues. The entrant lands $k \geq 0$ steps above the *entry margin*, the lowest position a superstar can hold, with probability

$$\Pr(k) \;=\; (1-p)\, p^k. \tag{10}$$

In an industry with the maximum number $\bar{N}$ of superstars, the margin is the position of the least productive incumbent, whom the entrant displaces; otherwise it is $\bar{n}$ steps behind the leader. An entrant that climbs past the leader becomes the new leader; it lands at most $\bar{n}$ steps above the former leader (landings beyond are placed at $\bar{n}$ steps), and incumbents pushed more than $\bar{n}$ steps behind exit. Unlike in Cavenaile et al. (2026), where entrants join $\bar{n}$ steps behind the leader and only industries with fewer than $\bar{N}$ superstars admit entry, entrants in our model, as in the data, also join full industries and can land anywhere above the margin, including at the top.

4.3 Lobbying

Firms can lobby to affect three policy instruments $I \in \{T, P, R\}$: their profit tax (T), their procurement contracts (P), and the industry-wide regulatory fixed cost (R). A superstar lobbying for instrument I chooses an intensity $b_{Ii} \geq 0$ at cost $\kappa_I \, b_{Ii}^{\phi_I} \, Y_t$, with $\phi_I > 1$, and those costs are not tax deductible.[1] $a_{Iijt} \in \{0, 1\}$ indicates that the firm lobbies on the instrument I.

Profit tax (T). Lobbying on the tax instrument reduces the firm's own effective profit tax rate from the statutory τ_0:

$$\tau_{ijt} \;=\; \frac{\tau_0}{1 + a_{Tijt} \, b_{Tijt}}. \tag{11}$$

Procurement contracts (P). Superstar i holds government contracts worth $g_{ijt} \, Y_t$, with

$$g_{ijt} \;=\; \left(g_0 + a_{Pijt} \, b_{Pijt}\right) (1 + \lambda)^{-\kappa_g n_{ijt}}, \tag{12}$$

where g_0 is the baseline contract, common to all superstars, b_{Pijt} is the increment won by lobbying, and $\kappa_g \geq 0$ governs how contract amounts relate to the firm's distance to the leader, which is a proxy for the firm's sales. Contracts are cost-plus: of the payment $g_{ijt} Y_t$, the share $1 - \theta_g$ covers the cost of the goods the firm delivers to the government, and the share θ_g is the firm's fee, which is taxed as profit.

Regulatory fixed cost (R): the contest. The industry-wide regulatory cost is set by an influence contest among the firms that lobby on it. Each participant l lobbies with intensity b_{Rljt} either for a higher cost (the pro side) or against it (the anti side). Individual intensities aggregate to the total influence of side s according to $B_{jt}^s = \sum_{l \in s} b_{Rljt}^{\theta}$ with $\theta \in (0, 1)$. The

[1] Internal Revenue Code §162(e) disallows the deduction of lobbying expenditures for federal income tax purposes.

lobbying outcome, regulatory cost R_{jt}, is determined by the ratio

$$R_{jt} \;=\; R_0 \,\frac{1+\beta^p\, B_{jt}^{pro}}{1+\beta^a\, B_{jt}^{anti}} \tag{13}$$

which is common knowledge to all lobbying participants. With $\theta < 1$, each participant faces diminishing marginal returns to its own effort, and β^p and β^a measure how effective lobbying is for and against the cost. Without lobbying, $R_{jt} = R_0$. The baseline R_0 is set by the compliance level c: $R_0\,(\bar{K} + \bar{m}^{\mathrm{a}}) = c$, where $\bar{K}$ is the average of $\sum_i \kappa_i$ over technology states with $\bar{N}$ superstars, so that compliance costs are a share c of output in a full industry with the observed number of active fringe firms. Each participant joins the side given by the sign of its marginal value of R, derived in Section 4.5. This contest resembles the legislative process, where firms and associations lobby for or against the bills that increase or decrease regulatory fixed costs.

This lobbying contest represents the distinction between lobbying by agents who are relatively large versus those who are relatively small (Olson, 1982, 1965). Superstars have an incentive to lobby individually, as each can affect regulatory fixed costs by itself. For the small fringe firms, however, lobbying is a public good: no fringe firm has an incentive to lobby individually, and would rather free ride on the lobbying efforts of the other fringe firms who share its interests. The role of a trade association, as in Olson (1982), is to solve the collective action problem. Our model is a reduced-form representation of the idea in Olson (1982) that associations sell private benefits—such as trade publications and conventions—to individual firms and produce lobbying insofar as it is a complement to those sales.

Participation in lobbying. Lobbying on an instrument requires a commitment, such as retaining lobbyists and maintaining relationships with legislators. A superstar that lobbies on instrument I pays a flow participation cost $\varphi_{Ii}\, Y_t$, where

$$\ln \varphi_{Ii} \;\sim\; N(\mu_I, \sigma_I^2), \qquad \text{i.i.d. across firms and instruments.} \tag{14}$$

Participation is sticky: the firm draws a new cost at Poisson rate λ_I and can switch its status $a_{Ii} \in \{0, 1\}$ on instrument I only when a new draw arrives; in between, a participating firm keeps paying its current draw. This timing makes lobbying status persistent, as in the data (Section 3.4), and, because the arrivals are independent across firms and instruments, firms revise their participation one at a time. Participation costs are partly shared across instruments. A firm that lobbies on instrument I pays $\varphi_{Ii} \prod_{J \neq I} (1 - \iota_{IJ})^{a_{Ji}}$ on it: each other instrument J on which it also lobbies lowers the cost of I by the share $\iota_{IJ} = \iota_{JI}$, which measures the economies of scope between the two instruments, such as lobbyists,

relationships, and information that serve both.

Fringe trade association. The association organizes with the same timing as a superstar participates: at Poisson rate λ_A it draws a flow organizing cost $\varphi_A Y_t$, with $\ln \varphi_A \sim N(\mu_A, \sigma_A^2)$, and if it organizes, $a_{Ajt} = 1$, it pays this cost until the next draw. An organized association enters the contest (13) as a single participant and chooses the side that maximizes the value of the industry's fringe firms. The cost of the intensity b_{Ajt} for the association is

$$C_A\big(b_{Ajt}\big) \;=\; \kappa_A \, b_{Ajt}^{\phi_A} \, Y_t, \qquad \phi_A > 1. \tag{15}$$

Members of the association, the fringe firms, pay the organizing and intensity costs through dues proportional to their surplus from producing: a fringe firm with cost ε pays $d_{jt}(\varepsilon)\, Y_t$ if it produces, $\varepsilon \le \varepsilon_{jt}^*$, and nothing otherwise, with

$$d_{jt}(\varepsilon) \;=\; \omega_{jt}(\varepsilon)\left[\kappa_A \, b_{Ajt}^{\phi_A} + \varphi_A\right], \qquad \omega_{jt}(\varepsilon) \;=\; \frac{\varepsilon_{jt}^* - \varepsilon}{m_t \int_0^{\varepsilon_{jt}^*} \left(\varepsilon_{jt}^* - \tilde{\varepsilon}\right) dF(\tilde{\varepsilon})}, \qquad \varepsilon \le \varepsilon_{jt}^*, \tag{16}$$

where ε_{jt}^* is the production cutoff (Section 4.5). A producing firm's surplus is $(1-\tau_0)(\varepsilon_{jt}^* - \varepsilon)\, Y_t$, so $\omega_{jt}(\varepsilon)$ is its share of the members' total surplus; the shares integrate to one, and dues cover the association's spending exactly. Every member pays the same fraction of its surplus, and the marginal producer pays nothing, so dues do not affect which firms produce as long as they are feasible: the spending must not exceed the members' after-tax surplus, $(1-\tau_0)\, m_t \int_0^{\varepsilon_{jt}^*} (\varepsilon_{jt}^* - \varepsilon)\, dF(\varepsilon)$. At the estimate, this holds in every organized state: spending stays below a third of the surplus.

4.4 Government problem

The government buys goods through the superstars' procurement contracts in the amount of $G_t^S = Y_t \int_0^1 \sum_i g_{ijt}\, dj$, so a contract that a superstar wins by lobbying is an additional purchase. Contracts held by other firms, including fringe firms, are outside the model.[2] For a superstar i it costs $(1-\theta_g) g_{it} Y_t$ to produce the good, and the firm keeps $\theta_g g_{it} Y_t$ as its fee. Thus, the consumer's value of government purchases is its fair value: $G_t = (1-\theta_g)\, G_t^S$. Corporate taxes are collected from every firm and returned to households: the government rebates lump-sum the revenue that the statutory rate τ_0 raises on every firm's tax base. It also pays households baseline lump-sum transfers $\bar{T} Y_t$, a constant share of output, set in the economy without lobbying so that the labor income tax equals 24.4% there (the average US labor income tax, OECD Taxing Wages) and held at that share in every economy we

[2]Federal contracts are about one to two percent of the sales of listed fringe firms.

compare. The labor income tax τ_w adjusts to finance the transfers, the contracts, and the revenue lost to tax lobbying, so that the government budget balances at each instant:

$$\tau_w \, w_t \, L_t \;=\; \bar{T} \, Y_t \;+\; G_t^S \;+\; \int_0^1 \sum_{i=1}^{N_{jt}} \left(\tau_0 - \tau_{ijt}\right) \Pi_{ijt} \, Y_t \, dj, \tag{17}$$

where Π_{ijt} is the tax base of superstar i in units of Y_t, defined below in (22). Fixed compliance costs are a resource cost, not government revenue.

4.5 Decisions in the decentralized economy

Households. The representative household owns all firms and chooses private consumption and labor supply, taking government purchases as given:

$$\max_{\{C_t^p, L_t\}} \int_0^\infty e^{-\rho t} \left[\ln\left(C_t^p + G_t\right) - \xi \, \frac{L_t^{1+1/\epsilon_L}}{1 + 1/\epsilon_L} \right] dt \quad \text{s.t.} \quad \dot{A}_t = r_t A_t + (1 - \tau_w) \, w_t L_t + T_t - C_t^p,$$

where A_t is the value of all firms, r_t the return on it, w_t the wage, and T_t the government's lump-sum payments: the rebate of corporate tax revenue at the statutory rate, τ_0 times every firm's tax base, and the transfers $\bar{T}Y_t$. Firms' profits, including the superstars' contract fees and net of their R&D, fixed and compliance costs and taxes, reach the household as the return on A_t. The Euler equation and the labor supply condition are:

$$\frac{\dot{C}_t}{C_t} \;=\; r_t - \rho, \qquad \xi \, L_t^{1/\epsilon_L} \;=\; \frac{(1 - \tau_w) \, w_t}{C_t}, \qquad C_t = C_t^p + G_t. \tag{18}$$

Private consumption and government purchases are both shares of output, so on the balanced growth path both grow at rate g and $r_t = \rho + g$. Combining the household's budget with the government's budget (17) and the firms' cash flows gives the resource constraint in Section 4.6, so the goods market clears. Because all firm payoffs are proportional to Y_t, firm decisions do not depend on the wage, labor supply, or the labor tax; they affect only the level of output and welfare.

Final good producer. A representative final good producer chooses the quantity of each variety in each industry competitively, which determines the inverse demand functions for superstars and the price of fringe output:

$$p_{ijt} = \frac{y_{ijt}^{-\frac{1}{\eta}} Y_t}{\sum_{k=1}^{N_{jt}} y_{kjt}^{\frac{\eta-1}{\eta}} + \tilde{y}_{fjt}^{\frac{\eta-1}{\eta}}}, \qquad p_{fjt} = \frac{\tilde{y}_{fjt}^{-\frac{1}{\eta}} Y_t}{\sum_{k=1}^{N_{jt}} y_{kjt}^{\frac{\eta-1}{\eta}} + \tilde{y}_{fjt}^{\frac{\eta-1}{\eta}}}. \tag{19}$$

Product market. Superstars compete à la Cournot. Each superstar chooses y_{ijt} to maximize operating profit $\pi_{ijt} = p_{ijt}\, y_{ijt} - w_t\, l_{ijt}$, taking rivals' quantities and the fringe aggregate $\tilde{y}_{fjt}$ as given, subject to (4), (5) and (19). Fringe firms take p_{fjt} as given. Appendix F derives the best responses and profits. A superstar's markup and operating profit are functions of its share of industry sales $\sigma_{ijt} = p_{ijt}y_{ijt}/Y_t$:

$$M_{ijt} \;=\; \frac{\eta}{(\eta-1)(1-\sigma_{ijt})}, \qquad \pi_{ijt} \;=\; \sigma_{ijt}\Big(1 - \frac{\alpha}{M_{ijt}}\Big)\, Y_t, \tag{20}$$

and each active fringe firm earns

$$\pi_{fjt} \;=\; \frac{(1-\alpha)\,\sigma_{fjt}\, Y_t}{m^{\mathrm{a}}_{jt}}, \tag{21}$$

where σ_{fjt} is the fringe's share of industry sales. Fringe firms price at marginal cost, but decreasing returns leave them with positive operating profits. Given productivities, all sales shares depend only on the industry's productivity gaps and the active fringe mass, and $\partial\pi_{ijt}/\partial m^{\mathrm{a}}_{jt} < 0$, $\partial\pi_{fjt}/\partial m^{\mathrm{a}}_{jt} < 0$.

Taxable income. A superstar's tax base, in units of Y_t, is its operating profit plus its contract fee, minus compliance and R&D costs:

$$\Pi_{ijt} \;=\; \frac{\pi_{ijt}}{Y_t} \;+\; \theta_g\, g_{ijt} \;-\; \kappa_{ijt} R_{jt} \;-\; \chi z^{\phi}_{ijt}. \tag{22}$$

The tax applies to the signed base: a firm whose base is negative receives $\tau_{ijt}|\Pi_{ijt}|Y_t$ (full loss offset), and the same holds for fringe firms at the rate τ_0. Fringe firms pay the statutory rate τ_0 on operating profit minus compliance, their fixed cost ε_k, and their breakthrough R&D. Lobbying intensities, participation costs, and association dues are not deductible.

States. The *technology state* of an industry is $\Theta = (N, \mathbf{n})$, where N is the number of superstars and $\mathbf{n} = (n_1, \ldots, n_{N-1})$, with $0 \le n_1 \le \cdots \le n_{N-1} \le \bar{n}$, is the vector of follower gaps behind the leader. The *industry state* adds every participation status: $S = (\Theta, \mathbf{a}_T, \mathbf{a}_P, \mathbf{a}_R, a_A)$, where $\mathbf{a}_I = (a_{I1}, \ldots, a_{IN})$ collects the superstars' statuses on instrument I and a_A is the association's status. A superstar's *own state* $s_i = (i, \boldsymbol{\varphi}_i)$ consists of its rank i in S, which gives its gap and its statuses, and the participation costs $\boldsymbol{\varphi}_i = (\varphi_{Ti}, \varphi_{Pi}, \varphi_{Ri})$ it has locked on the instruments it lobbies on. Locked costs are private, so a rival's policies enter a firm's problem through their expectation given S under the stationary distribution, which we mark with a bar. Values of the firms are normalized by output on the balanced growth path, $v \equiv V/Y_t$, and discounted at $r - g = \rho$; we drop time and industry subscripts where this creates no ambiguity.

Transitions. The industry state changes with the following events. After innovation by superstar i, the state is S^{+i}: a follower moves one step closer to the leader, while a superstar at the frontier advances it by one step, every other superstar falls one step further behind, and a superstar pushed more than $\bar{n}$ steps behind exits. After a breakthrough whose entrant lands k steps above the margin, the state is $S^{e,k}$. After a redraw of superstar i's cost on instrument I, the state is $S^{I,i,a}$, with its status on I set to $a \in \{0,1\}$; after a redraw of the association's organizing cost, it is $S^{A,a}$. Statuses and locked costs move with their firms when ranks change: tied superstars keep their order, an entrant ranks below the incumbents it ties and starts with every status off, and a superstar's own state after an event, such as s_i^{+i}, is its new rank with the same locked costs.

Superstar problem. A superstar chooses its R&D intensity and its lobbying intensities on the instruments it participates in at every instant, and its participation on instrument I whenever its cost on that instrument is redrawn. Its Hamilton–Jacobi–Bellman equation is

$$
\begin{aligned}
\rho\, v(S,s_i) \;=\; & \max_{z_i,\, b_{Ti},\, b_{Pi},\, b_{Ri}} \Big\{ (1-\tau_i)\,\Pi_i \;-\; \sum_I a_{Ii}\Big[\kappa_I\, b_{Ii}^{\phi_I} + \varphi_{Ii} \prod_{J\neq I} (1-\iota_{IJ})^{a_{Ji}}\Big] \\
& + z_i \left[v(S^{+i}, s_i^{+i}) - v(S,s_i) \right] \;+\; \sum_{i'\neq i} \bar{z}_{i'}(S) \left[v(S^{+i'}, s_i^{+i'}) - v(S,s_i) \right] \\
& + X(S) \sum_k P_k(S) \left[v(S^{e,k}, s_i^{e,k}) - v(S,s_i) \right] \\
& + \sum_{i'\neq i} \sum_I \lambda_I \Big[\bar{G}_{Ii'}(S)\, v(S^{I,i',1}, s_i) + \big(1-\bar{G}_{Ii'}(S)\big)\, v(S^{I,i',0}, s_i) - v(S,s_i) \Big] \\
& + \lambda_A \Big[G_A(S)\, v(S^{A,1}, s_i) + \big(1-G_A(S)\big)\, v(S^{A,0}, s_i) - v(S,s_i) \Big] \\
& + \sum_I \lambda_I \Big(\mathbb{E}_{\varphi'} \max \big\{ v(S^{I,i,1}, s_i^{I,\varphi'}),\; v(S^{I,i,0}, s_i^{I,0}) \big\} - v(S,s_i) \Big) \Big\},
\end{aligned}
\tag{23}
$$

subject to (13). The first line is the after-tax flow net of lobbying and locked participation costs. The second line contains innovation by the firm and by its rivals. The third contains fringe entry at rate $X(S)$, with landing probabilities $P_k(S)$ from (10), the probability beyond the cap placed at the cap, and the value averaged over the displaced incumbent when several are tied at the margin. The fourth and fifth lines contain the redraws of rivals, after which rival i' is on instrument I with probability $\bar{G}_{Ii'}(S)$, and of the association, after which it is organized with probability $G_A(S)$, given in the association problem below. The last line contains the firm's own redraws: it either joins instrument I, locking the new draw φ', which gives the own state $s_i^{I,\varphi'}$, or stays out, which gives $s_i^{I,0}$. A firm that exits has value zero. The

first-order conditions for innovation, procurement, and tax lobbying are

$$(1-\tau_i)\,\chi\phi\,z_i^{\phi-1} \;=\; v(S^{+i},s_i^{+i})-v(S,s_i), \tag{24}$$

$$(1-\tau_i)\,\theta_g\,(1+\lambda)^{-\kappa_g n_i} \;=\; \kappa_P\,\phi_P\,b_{Pi}^{\phi_P-1}, \tag{25}$$

$$\frac{\tau_0\,\Pi_i}{(1+b_{Ti})^2} \;=\; \kappa_T\,\phi_T\,b_{Ti}^{\phi_T-1}, \tag{26}$$

where the tax intensity is zero when the base Π_i is not positive. A higher locked cost lowers the firm's value at every date until its next redraw on the instrument, so the value after joining falls with the new draw. At a redraw, the firm therefore joins instrument I if and only if its draw is below a cutoff:

$$a'_{Ii}=1 \iff \varphi' \;\le\; \varphi_I^*(S,s_i), \qquad v\big(S^{I,i,1},s_i^{I,\varphi_I^*}\big) \;=\; v\big(S^{I,i,0},s_i^{I,0}\big), \tag{27}$$

and a rival's probability of being on the instrument after a redraw is $\bar{G}_{Ii'}(S)=\mathbb{E}\big[\Phi\big((\ln\varphi_I^*(S,s_{i'})-\mu_I)/\sigma_I\big)\mid S\big]$, where Φ is the standard normal distribution function.

The stake in regulation. A superstar participating in the R instrument chooses b_{Ri} to satisfy

$$\Delta_i(S,s_i)\,\frac{\partial R(S)}{\partial b_{Ri}} \;=\; \kappa_R\,\phi_R\,b_{Ri}^{\phi_R-1}, \tag{28}$$

where Δ_i is the marginal value of a higher R to the firm:

$$\Delta_i \;=\; \underbrace{-(1-\tau_i)\,\kappa_i}_{\text{(i) own compliance}} \;+\; \underbrace{(1-\tau_i)\,\frac{\partial(\pi_i/Y)}{\partial m^{\mathrm{a}}}\,\frac{dm^{\mathrm{a}}}{dR}}_{\text{(ii) fringe exit: Cournot gain}} \;+\; \underbrace{\frac{dX}{dR}\sum_k P_k\left[v(S^{e,k},s_i^{e,k})-v(S,s_i)\right]}_{\text{(iii) fewer entrants: displacement risk}}. \tag{29}$$

Equation (29) is the central object of the paper: it is the shadow value of regulation to an incumbent, and its sign determines which side of the contest the firm joins. Channel (i) is the firm's own compliance cost, κ_i per unit of R; it is negative and largest for the leader. Channel (ii) is the static Cournot gain as marginal fringe firms stop producing; it is positive, since $dm^{\mathrm{a}}/dR<0$ and $\partial\pi_i/\partial m^{\mathrm{a}}<0$. Channel (iii) is the lower rate of fringe breakthroughs, which would push the firm down in the rank or displace it; it is positive, since $dX/dR<0$ and, at the estimate, an entry lowers the value of every incumbent at every state. In a Markov-perfect equilibrium, each agent takes rivals' policies, the fringe's breakthrough intensity and continuation values as given, but internalizes the effect of its lobbying on R and, through R, on which fringe firms produce. The only fringe firm whose

status responds to R is the marginal one at ε^*, so

$$\frac{dm^{\mathrm{a}}}{dR} \;=\; m\, f(\varepsilon^*)\,\frac{d\varepsilon^*}{dR}, \qquad \frac{d\varepsilon^*}{dR} \;=\; \frac{-1}{1 - m\, f(\varepsilon^*)\,\dfrac{\partial(\pi_f/Y)}{\partial m^{\mathrm{a}}}}, \qquad \frac{dX}{dR} \;=\; x_f\,\frac{dm^{\mathrm{a}}}{dR}. \tag{30}$$

The denominator exceeds one, since $\partial\pi_f/\partial m^{\mathrm{a}} < 0$: as R removes marginal fringe firms, the survivors earn higher profits, which partially offsets the direct compliance burden. The firm lobbies for a higher cost if $\Delta_i > 0$ and against it if $\Delta_i < 0$:

$$\mathcal{P}(S) = \{i : a_{Ri} = 1,\ \Delta_i > 0\}, \qquad \mathcal{A}(S) = \{i : a_{Ri} = 1,\ \Delta_i < 0\}. \tag{31}$$

Given the influence function (13), the marginal influence of b_{Ri} is

$$\frac{\partial R(S)}{\partial b_{Ri}} \;=\; \begin{cases} \dfrac{R_0\,\beta^p\,\theta\,b_{Ri}^{\theta-1}}{1+\beta^a B^{\mathrm{anti}}} & \text{if } i \in \mathcal{P}(S), \\[2ex] -\dfrac{R\,\beta^a\,\theta\,b_{Ri}^{\theta-1}}{1+\beta^a B^{\mathrm{anti}}} & \text{if } i \in \mathcal{A}(S), \end{cases} \tag{32}$$

so that (28) gives $b_{Ri} = \big[\Delta_i R_0 \beta^p \theta/(\kappa_R \phi_R (1+\beta^a B^{\mathrm{anti}}))\big]^{1/(\phi_R-\theta)}$ on the pro side and $b_{Ri} = \big[|\Delta_i| R \beta^a \theta/(\kappa_R \phi_R (1+\beta^a B^{\mathrm{anti}}))\big]^{1/(\phi_R-\theta)}$ on the anti side, solved as a fixed point.

Fringe problem. Within an industry, all fringe firms share the same productivity and earn the same operating profit $\pi_f(S)$; they differ only in ε. A firm's dues (16) also depend on the organizing cost φ_A that an organized association has locked. The HJB equation of a fringe firm with cost draw ε is

$$\begin{aligned}(\rho+\delta)\,v^e(S,\varphi_A,\varepsilon) = \max_{a,\,x}\; a\Big\{(1-\tau_0)\Big[\frac{\pi_f(S)}{Y} - R(S) - \varepsilon - \chi_f x^{\phi_f}\Big] - a_A\, d(S,\varphi_A,\varepsilon) + x\, v^{\mathrm{entry}}(S)\Big\} \\ +\; \sum_{S'\neq S} \Gamma(S,S')\,\big[\mathbb{E}\,v^e(S',\varphi_A',\varepsilon) - v^e(S,\varphi_A,\varepsilon)\big],\end{aligned} \tag{33}$$

where $a \in \{0,1\}$ indicates that the firm produces, $x \geq 0$ is its breakthrough intensity, and $\Gamma(S,S')$ are the industry's transition rates, induced by superstars' innovation, breakthroughs, and the redraws of the superstars and the association. The organizing cost changes only when the association reorganizes at a redraw, and the expectation is over the new cost φ_A' in that case. A breakthrough creates a new superstar with all lobbying statuses off, owned by the innovating firm, whose expected value is

$$v^{\mathrm{entry}}(S) \;=\; \sum_k P_k(S)\; v\big(S^{e,k}, \text{entrant's rank}\big). \tag{34}$$

The first-order condition gives the same breakthrough intensity for every active firm in the industry,

$$(1-\tau_0)\,\chi_f\,\phi_f\,x_f^{\phi_f-1} \;=\; v^{\text{entry}}(S), \tag{35}$$

so breakthroughs arrive in the industry at rate $X(S) = m^{\text{a}}(S)\,x_f(S)$. Because switching is costless and ε is permanent, the decision to produce affects only the current flow, and because the marginal producer pays no dues, a fringe firm produces if and only if $\varepsilon \le \varepsilon^*(S)$, where

$$\varepsilon^*(S) \;=\; \frac{\pi_f(S)}{Y} - R(S) + \frac{\Omega(S)}{1-\tau_0}, \qquad \Omega(S) \;=\; \max_{x\ge 0}\left\{x\,v^{\text{entry}}(S) - (1-\tau_0)\,\chi_f x^{\phi_f}\right\}. \tag{36}$$

Because π_f depends on $m^{\text{a}} = mF(\varepsilon^*)$, (36) is a fixed point at each state; it has a unique solution, since π_f falls with m^{a}.

Let $\bar{v}^e(S,\varphi_A) \equiv \mathbb{E}_\varepsilon[v^e(S,\varphi_A,\varepsilon)]$ be the value of a firm in the industry's fringe pool. No decision depends on the organizing cost the association has locked, and each pool firm pays on average a share $1/m$ of it until the association's next redraw or the firm's death, so

$$\bar{v}^e(S,\varphi_A) \;=\; \bar{v}_0^e(S) \;-\; \frac{a_A\,\varphi_A}{m\,(\rho+\delta+\lambda_A)}, \tag{37}$$

where the pool value gross of the current organizing cost solves

$$\begin{aligned}(\rho+\delta)\,\bar{v}_0^e(S) \;=\; &(1-\tau_0)\,\mathbb{E}\big[(\varepsilon^*(S)-\varepsilon)^+\big] \;-\; \frac{a_A\,\kappa_A\,b_A(S)^{\phi_A}}{m} \;-\; \frac{\lambda_A\,\mathbb{E}\big[\varphi_A\,\mathbf{1}\{\varphi_A\le\varphi_A^*(S)\}\big]}{m\,(\rho+\delta+\lambda_A)} \\ &+\sum_{S'\neq S}\Gamma(S,S')\,\big[\bar{v}_0^e(S')-\bar{v}_0^e(S)\big].\end{aligned} \tag{38}$$

Its flow terms are the producers' after-tax surplus, the dues that pay for lobbying, and the members' share of any organizing cost committed at the next redraw. Entrepreneurs sell new firms at the pool value averaged over the stationary distribution μ; with $e = \delta m$ on the balanced growth path, free entry requires

$$2\psi\delta\,m \;=\; \sum_S \mu(S)\left[\bar{v}_0^e(S) - \frac{a_A\,\bar{\varphi}_A(S)}{m\,(\rho+\delta+\lambda_A)}\right], \tag{39}$$

where $\bar{\varphi}_A(S)$ is the expected cost, under μ, that an association organized at state S has locked at its last redraw.

Association problem. At each redraw, the industry's current fringe firms decide whether the association organizes. They organize if their gain covers their share of the new cost:

$$\varphi_A \;\le\; \varphi_A^*(S) \;=\; (\rho+\delta+\lambda_A)\, m\,\big[\bar v_0^e(S^{A,1}) - \bar v_0^e(S^{A,0})\big], \qquad G_A(S) \;=\; \Phi\Big(\frac{\ln\varphi_A^*(S)-\mu_A}{\sigma_A}\Big), \tag{40}$$

where $S^{A,1}$ and $S^{A,0}$ denote S with the association organized and not; firms that join the pool later pay the rest, which free entry prices in. While organized, the association chooses b_A at every instant to maximize the total value of the industry's fringe firms net of its lobbying cost, taking rivals' intensities as given:

$$m^{\mathrm{a}}\,\Delta_A\,\frac{\partial R}{\partial b_A} \;=\; \kappa_A\,\phi_A\,b_A^{\phi_A-1}, \tag{41}$$

where $\partial R/\partial b_A$ takes the form (32) and Δ_A is the marginal value of a higher R per producing member:

$$\Delta_A \;=\; \underbrace{-(1-\tau_0)}_{\text{own compliance}} \;+\; \underbrace{(1-\tau_0)\,\frac{\partial(\pi_f/Y)}{\partial m^{\mathrm{a}}}\,\frac{dm^{\mathrm{a}}}{dR}}_{\text{fringe exit: survivors' profit}} \;+\; \underbrace{\frac{m}{m^{\mathrm{a}}}\,\frac{dX}{dR}\sum_k P_k\,\big[\bar v_0^e(S^{e,k}) - \bar v_0^e(S)\big]}_{\text{breakthroughs: the pool's value}}. \tag{42}$$

By (36), the first two terms sum to $(1-\tau_0)\,d\varepsilon^*/dR < 0$: the exit of marginal rivals returns only part of a member's higher cost. Only the last term, which captures how rarer breakthroughs change the value of every firm in the pool, can make the association favor a higher cost; it lobbies on the side given by the sign of Δ_A.

Within-state political conflict. The contest at state S splits the industry whenever the participants' stakes differ in sign, for example when some superstars gain from a higher cost while the association loses. The two sides push R in opposite directions through (13), and $R(S)$ solves the joint fixed point of the superstar conditions (28) and the association condition (41).

4.6 Equilibrium

Growth. The frontier of an industry advances when its leader, or a firm tied with it, innovates, and when an entrant passes the leader. The aggregate growth rate is

$$g \;=\; \ln(1+\lambda)\,\mathbb{E}_\mu\Big[\sum_{i:\,n_i=0} z_i \;+\; X(S)\sum_k P_k(S)\,\ell_k(S)\Big], \tag{43}$$

where the expectation $\mathbb{E}_\mu$ is over industry states and over the locked costs, and $\ell_k(S) \geq 0$ is the number of steps by which an entrant that lands k steps above the margin passes the former leader.

Labor market and output. Within each industry, producers' labor demands follow from the static stage; each is proportional to Y/w. Labor market clearing requires

$$\int_0^1 \left(\sum_{i=1}^{N_{jt}} l_{ijt} + \int_0^{m_t} l_{fkjt}\, dk \right) dj \;=\; L_t, \tag{44}$$

with L_t given by the household's labor supply (18). Output then follows from (2).

Resource constraint. Final-good output is allocated between private consumption, government purchases, R&D, entry, fixed costs, compliance and lobbying. Dividing by Y_t and aggregating across industries with μ, the BGP resource constraint is

$$1 \;=\; \frac{C_t^p}{Y_t} \;+\; (1-\theta_g)\,\mathbb{E}_\mu\Big[\sum_i g_i\Big] \;+\; \mathcal{R}_{\text{R\&D}} \;+\; \psi\,(\delta m)^2 \;+\; \mathcal{R}_{\text{fixed}} \;+\; \mathcal{R}_{\text{comp}} \;+\; \mathcal{L}, \tag{45}$$

where

$$\begin{aligned}
\mathcal{R}_{\text{R\&D}} \;&=\; \mathbb{E}_\mu\Big[\sum_i \chi\, z_i^{\phi}\Big] \;+\; \sum_S \mu(S)\, m^{\text{a}}(S)\, \chi_f\, x_f(S)^{\phi_f}, \\
\mathcal{R}_{\text{fixed}} \;&=\; m \sum_S \mu(S)\, \mathbb{E}\big[\varepsilon\, \mathbf{1}\{\varepsilon \leq \varepsilon^*(S)\}\big], \\
\mathcal{R}_{\text{comp}} \;&=\; \sum_S \mu(S) \Big[\sum_{i=1}^{N(S)} \kappa_i(S) + m^{\text{a}}(S)\Big]\, R(S), \\
\mathcal{L} \;&=\; \mathbb{E}_\mu\Big[\sum_i \sum_I a_{Ii}\big(\kappa_I\, b_{Ii}^{\phi_I} + \varphi_{Ii} \prod_{J \neq I} (1-\iota_{IJ})^{a_{Ji}}\big) \;+\; a_A\Big(\frac{C_A}{Y} + \varphi_A\Big)\Big].
\end{aligned}$$

Compliance is a pure resource cost: paying it generates no output, consistent with the interpretation of regulation as administrative friction. Tax revenue, contract fees, transfers, and association dues are transfers and do not appear in (45).

Definition 1 (*Balanced-Growth-Path Equilibrium*)**.** *A stationary Markov-perfect balanced-growth-path (BGP) equilibrium, in which superstars' strategies depend on (S, s_i), locked costs are private, and beliefs about them are formed under μ, consists of*

- superstar values $v(S, s_i)$ and policies $\{z_i, b_{Ri}, b_{Ti}, b_{Pi}\}$ with participation cutoffs $\{\varphi_I^*\}$; fringe values $v^e(S, \varphi_A, \varepsilon)$, pool values $\bar{v}_0^e(S)$, breakthrough values $v^{\text{entry}}(S)$, cutoffs $\varepsilon^*(S)$ and breakthrough intensities $x_f(S)$; the association's cutoff $\varphi_A^*(S)$ and intensity $b_A(S)$;

- the regulatory cost $R(S)$ at every industry state;
- quantities, prices and sales shares at every industry state;
- a fringe pool m, an entry rate e, and active masses $m^{\mathrm{a}}(S) = mF(\varepsilon^*(S))$;
- a stationary distribution μ over industry states, superstars' locked costs, and the association's locked cost;
- a wage ratio w/Y, labor supply L, a labor tax τ_w, private consumption C^p/Y, lump-sum payments T/Y, an interest rate r, and a growth rate g;

such that:

1. **Product market.** Given $m^{\mathrm{a}}(S)$, quantities and prices satisfy the superstars' Cournot best responses and the fringe's price-taking conditions.
2. **Superstars.** Given the intensities of rivals and the association, the fringe's breakthrough intensity x_f and the organizing probability G_A, values and policies solve (23), where each firm internalizes through (13) the effect of its own intensity on $R(S)$ and, through $R(S)$, on $m^{\mathrm{a}}(S)$; participation follows (27).
3. **Fringe.** v^e solves (33), pool values satisfy (37) and (38), and v^{entry} is given by (34); $\varepsilon^*(S)$ satisfies (36), $x_f(S)$ satisfies (35), the association's decisions satisfy (40) and (41), and dues (16) are feasible.
4. **Contest.** At every state, intensities satisfy (28) for every superstar on the R instrument, with sides (31), and (41) for an organized association, and $R(S)$ is given by (13).
5. **Free entry.** m satisfies (39), with $e = \delta m$.
6. **Consistency.** μ is the invariant distribution of the process induced by the policies, and the rivals' expected policies $\bar{z}_{i'}$ and $\bar{G}_{Ii'}$ are computed under μ.
7. **Labor market and government.** The labor market clears (44) at the labor supply (18), the government budget (17) balances, and T is the rebate of corporate tax revenue plus the transfers.
8. **Growth and goods market.** g satisfies (43), $r = \rho + g$, and C^p/Y satisfies (45).

Computing the equilibrium. With $\bar{N} = 4$ and $\bar{n} = 5$, the full state has 481,040 industry states and 1.9 million superstar states. We compute the equilibrium with two simplifications. First, a firm forms beliefs about its rivals' tax and procurement statuses, equal to their stationary frequencies among superstars of the same rank in the same technology state; the state keeps the regulation statuses, the association's status, and each firm's own statuses, which leaves 2,180 industry states and 33,104 superstar states. Decisions use the beliefs, while the stationary distribution, growth, and every moment use the actual statuses. Second, a superstar's policies are evaluated at the expected cost of its current spell, $\hat{\varphi}_I = \mathbb{E}[\varphi \mid \varphi \leq \varphi_I^*]$, so that values are affine in the locked costs and the cutoff (27) has a closed form; spending uses the exact distribution of locked costs. Appendices G–I give the details.

4.7 Welfare and Counterfactual Analysis

We compare each equilibrium with a reference economy without lobbying. The reference economy is identical except that every participation cost is infinite, so $R = R_0$, every superstar pays τ_0, and contracts equal their baseline $g_0(1+\lambda)^{-\kappa_g n_i}$. It has the same statutory tax rate τ_0, baseline contracts, transfers $\bar{T}$, and entry cost ψ, and its fringe pool m is determined by free entry. A counterfactual ban on an instrument sets that instrument's participation costs to infinity. We compare balanced growth paths and report two kinds of results: the changes in output, resources, growth and hours, which do not depend on preferences, and a consumption-equivalent measure that aggregates them.

Changes without preferences. Consumption at date zero, $c_0 = C_0^p + G_0$, equals output times the share of output that remains after compliance, fixed costs, entry, R&D, and lobbying, and it grows at rate g. Every producer's technology is homogeneous of degree α in labor, so output splits exactly as

$$\ln Y_0 \;=\; \alpha \ln L + \ln A^* + \ln E, \quad \ln A^* \;=\; \sum_S \mu(S)\,\frac{1}{\upsilon} \ln \sum_k A_k(S)^{\upsilon}, \;\; \upsilon = \frac{\varsigma}{1-\alpha\varsigma}, \;\; \varsigma = \frac{\eta-1}{\eta}. \tag{46}$$

Here A^* is the productivity that an efficient allocation of the same labor would achieve given each industry's state and its operating fringe, with $A_k = q_k$ for a superstar and $A_f = (m^{\mathrm{a}})^{1-\alpha} q_f$ for the fringe taken as one producer, and $\ln E \leq 0$ is the loss from the allocation of labor across producers that markups distort. The share of output consumed splits by item of the resource constraint (45), and growth by source (43). None of these changes requires an assumption on preferences.

Consumption-equivalent welfare. On the balanced growth path, welfare is

$$W \;=\; \frac{\ln c_0}{\rho} \;+\; \frac{g}{\rho^2} \;-\; \frac{D}{\rho}, \qquad D \;=\; \xi\,\frac{L^{1+1/\epsilon_L}}{1+1/\epsilon_L}. \tag{47}$$

We summarize welfare by the consumption-equivalent variation of Lucas (1987), as in Jones and Klenow (2016): the permanent percentage increase in consumption, at every date of the equilibrium with lobbying, that would make the household as well off as in the reference economy. It solves

$$\ln(1+\mathrm{CEV}) \;=\; \rho\left(W^{\mathrm{ref}} - W\right) \;=\; \underbrace{\ln c_0^{\mathrm{ref}} - \ln c_0}_{\text{level}} \;+\; \underbrace{\frac{g^{\mathrm{ref}} - g}{\rho}}_{\text{growth}} \;+\; \underbrace{D - D^{\mathrm{ref}}}_{\text{hours}}, \tag{48}$$

so that CEV > 0 means that lobbying lowers welfare. As in Jones and Tonetti (2020), the measure splits additively into the level of consumption (which (46) and the consumption share decompose further), the growth rate, and hours. Preferences enter only through the weights on growth and on hours, and both weights are prices. For any balanced-growth preferences $u = [c\,e^{-D}]^{1-\sigma}/(1-\sigma)$, which reduce to ours at $\sigma = 1$, the growth term is $\frac{1}{1-\sigma}\ln\frac{r-g}{r^{\mathrm{ref}}-g^{\mathrm{ref}}}$, where $r - g = \rho - (1-\sigma)\,g$ is the gap between the interest rate and the growth rate. To first order, it is $(g^{\mathrm{ref}} - g)/(r^{\mathrm{ref}} - g^{\mathrm{ref}})$, the growth loss capitalized at the interest rate, so the curvature and the discount rate matter only through $r - g$, which equals ρ with log utility. By the labor supply condition (18), the hours term is, to first order, $[(1-\tau_w)\,wL/C]\,(\ln L - \ln L^{\mathrm{ref}})$: hours valued at the after-tax wage. Appendix M gives the details of the decomposition.

5 Estimation

We estimate the model by the simulated method of moments, in two steps. The first step estimates twelve technology parameters on moments of industry structure, firm dynamics, and aggregates, in the economy with compliance costs and without lobbying. The second step holds the technology fixed and estimates thirty lobbying parameters in the equilibrium with lobbying. On taxes and procurement we observe what lobbying buys. On regulation we do not, so we rely on who lobbies, what they ask for, and which bills pass. As in Cavenaile et al. (2026), the regressions below are not causal estimates: they are patterns the model must reproduce, computed the same way in the data and in the model.

5.1 Strategy

Let $\hat{m}_k$ be the data value of moment k and $m_k(\vartheta)$ its model counterpart. The estimator minimizes $\sum_k d_k(\vartheta)^2$ with equal weights. Technology moments use the relative deviation $d_k = m_k/\hat{m}_k - 1$. Lobbying moments use $d_k = \ln(m_k/\hat{m}_k)$ when the moment is positive by construction and $d_k = (m_k - \hat{m}_k)/\max(|\hat{m}_k|, \mathrm{se}_k)$ when it can take either sign. There are 30 technology moments and 38 lobbying moments. Every model moment has the definition of its data counterpart: moments of industry structure come from the stationary distribution over industry states, and moments measured on a panel of firms come from a simulated panel of superstars of the same length, with the same regressions and fixed effects. Growth must lie within 2.5 standard errors of the data, and compliance costs and their size elasticity must lie inside the bands of Section 5.4. Spending per ask on taxes and procurement carries a band, and two lobbying ceilings bind only when exceeded, because the data measure these objects with a wide margin.

An equilibrium takes about two hours to compute and the objective is not smooth in all parameters, so both steps start from a global search over a scrambled Sobol sequence. The technology step then refines the best points locally. The lobbying step solves 5,361 certified equilibria, of which 1,424 satisfy the constraints, and simulates every real bill at each of them (Section 5.5). The estimate is the point with the lowest objective among those that reproduce all eight statistics of the bill regression within 2.5 standard errors. Appendix J gives the identification argument for each block and every moment.

5.2 Parameters set outside the model

Table 5 lists the parameters set outside the model. The model describes the private business sector, so output maps to business value added, 75.6 percent of GDP. An industry has at most $\bar{N} = 4$ superstars, each selling at least 10 percent of the leader's sales, and the ladder depth $\bar{n}$ is the largest gap at which the smallest superstar meets this threshold, 5 at the estimate. The number of active fringe firms without lobbying or contracts is the Business Dynamics Statistics count in the median industry-year across 288 four-digit industries; free entry sets ψ to deliver it. The returns to scale follow from the cost data alone: the Economic Census payroll ratio gives the superstar markup, $1/0.764$, and the sales-to-cost ratio equals the markup over α, so $\alpha = 1.309/1.497 = 0.875$.

Table 5. Parameters Set Outside the Model

Parameter		Value	Source or target
Discount rate	ρ	0.04	Cavenaile et al. (2026)
Frisch elasticity of labor supply	ϵ_L	0.5	standard
Weight on the disutility of labor	ξ	0.519	hours equal one without lobbying, taxes, and contracts
Maximum number of superstars	$\bar{N}$	4	superstar definition
Smallest superstar, share of the leader's sales		0.10	superstar definition
Active fringe firms per industry, no lobbying or contracts	$\bar{m}^{\mathrm{a}}$	5,245	median industry-year (BDS)
Returns to scale in labor	α	0.875	payroll ratio and sales-to-cost ratio
Capital share (labor-share mapping)		0.195	2008–2022, as in Barkai (2020)
Contractor's margin	θ_g	0.101	Compustat, federal contractors
Effectiveness of pro-side lobbying	β^p	1	normalization

Notes: The entry cost ψ is set by free entry. BDS: Business Dynamics Statistics.

5.3 Technology

Table 6 reports the twelve technology estimates and the moments most informative about each. A larger step and a higher elasticity both widen the sales gaps between superstars; growth and the payroll ratio separate the two. Growth, R&D intensity, and the within-firm elasticity of patents to R&D pin down the cost of superstars' innovation. The rate at which new superstars appear, and how it varies with the sales gap to the leader, pin down the breakthrough cost. CR4 pins down fringe productivity, and fixed-cost dispersion, firm exit, and business applications pin down the fringe's fixed costs and turnover. Table 13 in the appendix reports all 30 moments.

5.4 Lobbying

Table 7 reports the thirty lobbying estimates. On taxes and procurement the outcome of lobbying is in the data. The effective tax rate identifies τ_0, contracts over sales and their elasticity to sales identify the baseline contract and its decline with the firm's gap, and what lobbying buys identifies the lobbying costs: a year of procurement lobbying is followed by contracts 0.22 log points higher within the firm, and among firms that lobby, a log point more spending per ask is followed by a tax rate on taxable income 0.42 percentage points lower over the next three years. On every instrument, participation by rank sets the mean of the participation cost, its slope in the firm's stake sets the dispersion, and persistence sets the redraw rate.

Regulation has no outcome in the data by industry and year. We bound compliance costs at the estimate of Trebbi et al. (2023): 0.96 percent of output is the upper bound and

Table 6. Technology Parameters

Parameter		Estimate	Most informative moments
Quality ladder and demand			
Quality step	λ	0.217	sales gaps among superstars; growth
Elasticity of substitution	η	3.97	sales gaps among superstars; payroll ratio
Superstars' innovation			
R&D cost: level	χ	655	growth; R&D intensity
R&D cost: convexity	ϕ	4.42	within-firm elasticity of patents to R&D
Breakthroughs			
Breakthrough cost: level, aggregate units	$\tilde{\chi}_f$	25.8	new superstars per industry-year
Breakthrough cost: convexity	ϕ_f	4.65	new superstars by #4/#1
Climb of new superstars	p	0.359	landing steps; new superstars' sales relative to the leader
Fringe productivity			
Fringe productivity, aggregate units	$\tilde{\zeta}$	0.765	CR4
Curvature of spillovers to the fringe	γ	1.23	CR4; leader's share of superstars' R&D
Fringe fixed costs and turnover			
Fixed cost: mean, per firm-year	$\bar{\varepsilon}$	2.04×10^{-4}	applications that become employers
Fixed cost: dispersion of logs	σ	1.67	dispersion of fixed costs
Death rate	δ	0.074	firm exit rate; applications per employer firm

Notes: Costs are in units of aggregate output Y_t.

the share serving our four regulatory instruments, 0.33 percent, is the lower bound. The size elasticity of superstars' compliance costs lies between 0.40, the low end of measured costs of specific rules, and 0.98, the occupation-based measures. Within this band, the share of positions asking for higher costs helps locate it: the faster the compliance bill rises with size, the more superstars ask for lower costs. How often the association is active, how long it stays active, and how often each side lobbies without the other identify its organizing cost, and ceilings discipline superstars' and associations' dollars. What regulatory force buys is identified on bills.

5.5 Bills

Enactment shows how lobbying force translates into regulation. A *proposal* is a set of bills of one Congress that are the same legislation, and a *regulation bill* is a proposal with a core regulatory direction and at least half of its provisions on regulation, excluding must-pass and appropriations bills. It raises or lowers fixed costs by the balance of its core instruments. Following Kang (2016), proposal k passes with probability

$$p_k = \frac{\pi_k + x_k}{1 + x_k + y_k}, \qquad x_k = \sum_{i \in \text{pushers}} \beta^{s_i} b_{Ri}^{\theta} \omega_{ik}, \qquad y_k = \sum_{i \in \text{blockers}} \beta^{s_i} b_{Ri}^{\theta} \omega_{ik}, \tag{49}$$

Table 7. Lobbying Parameters

Parameter		Estimate
Regulation		
Compliance without lobbying, share of output	c	0.0081
Size elasticity of superstars' compliance costs	e_R	0.82
Contest curvature	θ	0.15
Effectiveness of anti-side lobbying	β^a	7.63
Price of influence	κ_R	6.13×10^{-4}
Convexity of lobbying cost: superstars, association	ϕ_R, ϕ_A	2.11, 3.08
Participation cost of superstars: log mean, s.d.	μ_R, σ_R	−9.48, 2.81
Participation cost redraw rate (per year)	λ_R	0.10
Association organizing cost: log mean relative to superstars', s.d.	$\mu_A - \mu_R$, σ_A	1.96, 1.44
Association cost redraw rate (per year)	λ_A	0.10
Association lobbying cost: spending / industry sales	κ_A	2.22×10^{-4}
Profit tax		
Statutory tax rate	τ_0	0.244
Lobbying cost: leader's tax cut, share of τ_0	κ_T	4.2%
Lobbying cost convexity	ϕ_T	987
Participation cost: log mean, s.d.; redraw rate (per year)	μ_T, σ_T; λ_T	−6.56, 2.45; 1.45
Procurement		
Baseline contracts: contracts / sales	g_0	0.017
Decline of contracts with the gap	κ_g	2.34
Lobbying cost: leader's contracts due to lobbying	κ_P	1.5%
Lobbying cost convexity	ϕ_P	12.6
Participation cost: log mean, s.d.; redraw rate (per year)	μ_P, σ_P; λ_P	−6.62, 5.66; 0.63
Economies of scope in participation		
Regulation and tax; regulation and procurement; tax and procurement	ι_{RT}; ι_{RP}; ι_{TP}	0.03; 0.10; 0.09

Notes: The lobbying costs of the association and of the tax and procurement instruments are searched over, and reported, in the units of the moments they target.

where π_k is its probability of passing without model agents, $\beta^{s_i} b_{Ri}^{\theta}$ is firm i's force, and ω_{ik} is the share of that force the firm puts on proposal k. On a raising proposal pro firms push and anti firms block; on a lowering proposal the roles flip. The association is anti, so it blocks raising proposals and pushes lowering ones.

The data pin down π_k directly from bills that no superstar and no industry association lobbied, which we classify with the procedure of Section 2.2. A bill counts as passed if it became law or if its core regulatory provision is in a law of the same Congress. Without a model agent, 3.42 percent of raising bills pass and 7.67 percent of lowering bills. We then simulate every real proposal: each industry that lobbied a proposal in the data is drawn from the stationary distribution, forces add up across industries, and who lobbies each proposal is simulated, in the same way as its passage. Appendix J.4 gives the details.

We run one regression on real and simulated proposals (Table 8). The number of superstars on a bill disciplines the price of influence: cheaper influence means more force per firm, so passage rises faster with superstars. Association opposition disciplines the effectiveness of the anti side, because it measures how far blocking lowers passage. Passage on the 2,190 regulation bills in lobbying reports disciplines total force, given the baseline. The model reproduces all eight statistics within 2.5 standard errors, and it reproduces the ratio of compliance costs removed to costs added by the laws of the period, 0.050 in the data and 0.056 in the model.

Table 8. The Bill Regression: Data and Model

	Raising bills			Lowering bills		
	Data	s.e.	Model	Data	s.e.	Model
Number of superstars on the bill	0.404	0.111	0.275	0.368	0.181	0.535
Any association on the bill	0.160	0.247	0.190	0.220	0.355	0.426
Association supports the bill	0.659	0.232		−0.109	0.324	0.145
Association opposes the bill	−0.522	0.331	−0.292	−0.140	0.766	
Any other client on the bill	0.398	0.281	0.200	−0.064	0.369	0.171
Share of bills passed	0.0801	0.0067	0.0662	0.1458	0.0152	0.1450

Notes: Fractional logit of passing, one regression with coefficients by bill type. Outcome: law, or the core provision in a law of the same Congress. Superstars: log(1 + superstars listing the bill, weighted by the share of their lobbying on regulation; each counts one in the model). Any association, supports, opposes, and any other client are indicators. Controls: tax or contract content, sponsor in the majority, sponsor chairs a referral committee, bipartisan sponsorship, log cosponsors. Fixed effects: number of levers, Congress, lowering bill. Sample: all regulation bills, 2008–2022, 2,360 rows, 3,248 bills after sample weights, 255 passed rows. Standard errors clustered by bill family (2,015). Share passed: the 2,190 bills in lobbying reports. In the model the association never supports a raising bill or opposes a lowering bill, so these two cells are empty. Any other client is not targeted.

5.6 Fit

Table 9 compares the main moments with the data, and Figure 7 shows participation by rank; the appendix reports all of them. The model fits growth, the structure of sales among superstars, the arrival of new superstars, the tax rate, contracts over sales, and the payoff to tax lobbying. It reproduces the share of positions that ask for higher costs, for superstars and for associations, and both compliance moments lie inside their bands. The model overstates how often firms and associations lobby on regulation and how persistent that lobbying is, and it understates the payoff to procurement lobbying and the overlap between instruments.

Table 9. Main Moments: Data and Model

Moment	Data	Model
Technology		
Growth of business output per hour	0.0159	0.0172
CR4	0.213	0.255
Superstars per industry	3.64	3.71
New superstars per industry-year	0.132	0.144
Sales / cost of goods sold, superstars	1.497	1.686
Labor share	0.596	0.644
Regulation		
Compliance costs / output, band 0.0033 to 0.0096		0.0082
Size elasticity of compliance costs, band 0.40 to 0.98		0.81
Leader lobbying in a quarter	0.328	0.484
Still lobbying a year later, given the state	0.600	0.905
Superstars' positions asking for higher costs	0.533	0.671
Association active in an industry-quarter	0.558	0.897
Association's positions asking for higher costs	0.343	0.273
Tax and procurement		
Effective tax rate, all firms	0.222	0.242
Tax payoff	−0.0042	−0.0040
Contracts / sales	0.019	0.017
Procurement payoff	0.217	0.006

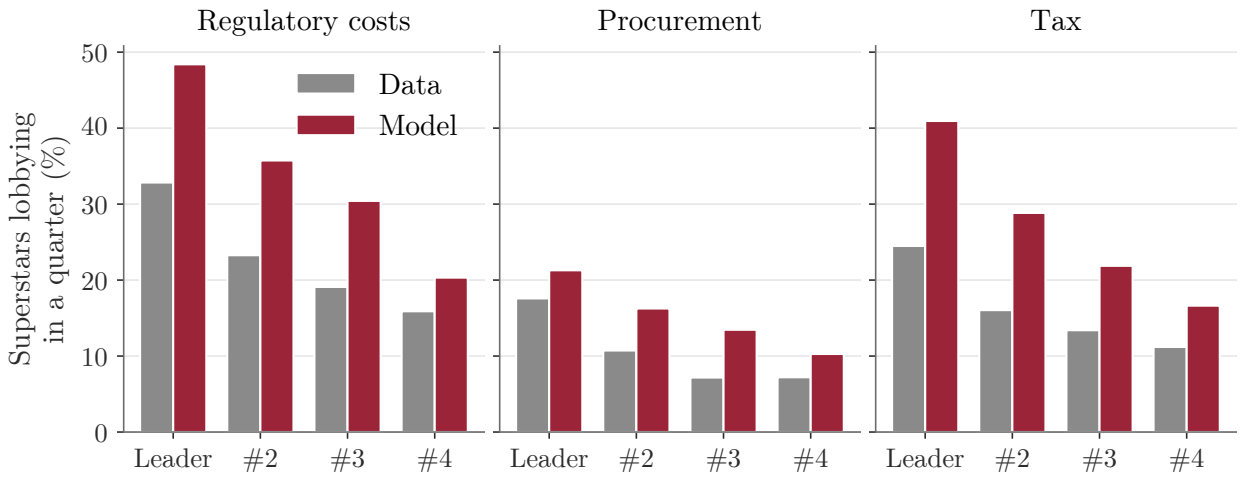


Figure 7. Who Lobbies: Superstars' Participation by Rank, Model and Data

Notes: Share of superstars lobbying in a quarter, by rank in the industry. Data: US-headquartered superstars, 2008–2022, weights equal across industries.

6 The Consequences of Lobbying

This section describes the effect of lobbying on regulation, industry structure, growth, and welfare, and which kind of lobbying causes the effect. We analyze the counterfactual of comparing the estimated economy with the same economy without lobbying by setting the participation cost in lobbying to infinity, so that the regulatory fixed cost is R_0 in every industry, and contracts and tax rates are at their baseline levels.

6.1 Who lobbies, on which side, and why

Superstars lobby for higher regulatory costs. For the estimated model, the marginal value of higher regulation (29) is positive for nearly every superstar, 99.9% of them across industry states, so superstars lobby on the side of a higher regulatory fixed cost, and the association lobbies against it. Figure 8 decomposes this marginal value by rank, per unit of R. One more unit of R costs the leader 173 after tax and the fourth superstar 70, because a superstar pays $\kappa_i R$ and κ_i rises with size. The gains are larger, because a higher R pushes marginal fringe firms out of production and the incidence of regulatory costs increases less than proportionately with firm size. All superstars gain on net, but the source of the gain varies with rank. The leader gains about as much from higher profits as the fringe firms stop producing (188) as from the lower rate of fringe breakthroughs, which would add a competitor or leapfrog it (215). The main gain of the fourth superstar is fewer breakthroughs (138 against 38 from profits), since a successful entrant could push it out of the top four.

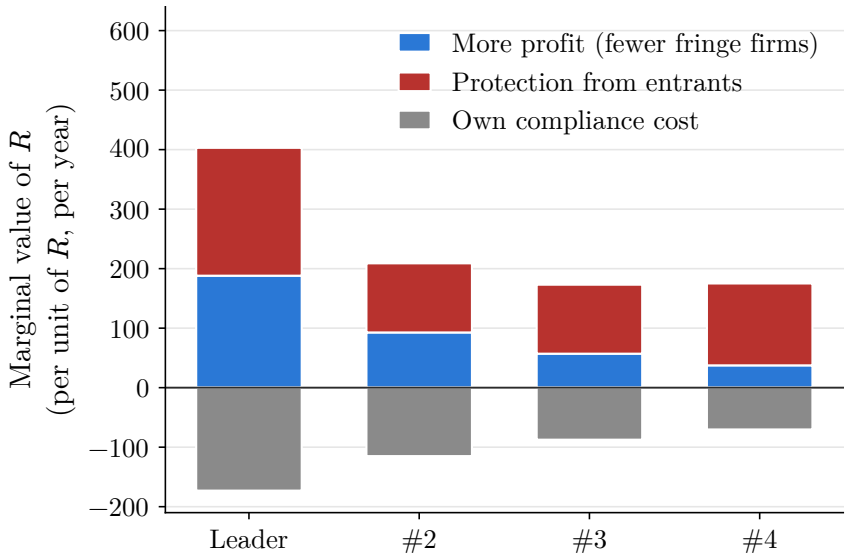


Figure 8. Why Superstars Lobby for Higher Regulatory Costs

Notes: The three components of a superstar's marginal value of a higher regulatory fixed cost (29), by rank, per unit of R and per year, averaged over the stationary distribution of industry states at the estimate. Blue: Cournot gain as marginal fringe firms stop producing. Red: value of fewer fringe breakthroughs. Grey: own compliance cost after tax.

The fringe opposes the regulatory costs. A fringe firm bears the compliance cost without much gain. Using the equation for the fixed costs cutoff to operate for fringe firms (36), the first two terms of the association's marginal value from increasing R (42) aggregate to the after-tax change in the operating cutoff, which is negative: the survivors' higher profits only partly offset the compliance cost. The third term, the value of a breakthrough to the industry's fringe pool, is too small to change the sign: the stake is negative in every industry state, so the association always lobbies against higher costs.

Lobbying moves regulatory costs in both directions. Superstars and the association lobby together in 72% of industry-quarters, and the balance of the two sides differs across industries. On average, regulatory costs are 1.03 times R_0, but they are above R_0 in 41% of industry states and below it in 57%; no one lobbies on regulation in the remaining 2%. At the 10th percentile across states the cost is 0.50 times R_0, at the median 0.86, and at the 90th percentile 1.53. The contest favors the fringe where few superstars compete: regulatory costs are on average 0.88 times R_0 in industries with one superstar, 0.99 with two, 1.06 with three, and 1.03 with four.

6.2 The welfare gain from removing lobbying

Removing lobbying raises the expected value of a firm in the fringe pool, so that the pool grows under free entry (by 0.9%) and the mass of operating fringe firms rises by 0.7%. More operating fringe firms mean more breakthroughs into the superstar set, and the stationary distribution of industries shifts toward states with more superstars (Figure 9). Compliance costs fall from 0.82 to 0.80% of output, and lobbying spending of 0.086% of output disappears. The full welfare on the Balanced Growth Path in this model is the following:

$$W = \frac{1}{\rho}\left[\underbrace{\ln Y_0}_{\text{output today}} + \underbrace{\ln C_0/Y_0}_{\text{share consumed}}\right] + \underbrace{\frac{g}{\rho^2}}_{\text{growth}} - \underbrace{\frac{D}{\rho}}_{\text{disutility of labor}}$$

The welfare change from removing lobbying is:

$$\ln(1+\text{CEV}) = \underbrace{\Delta \ln Y_0}_{+0.125\%} + \underbrace{\Delta \ln C_0/Y_0}_{+0.069\%} + \underbrace{\Delta g/\rho}_{+0.0013\text{pp}/\rho = 0.032\%} - \underbrace{\Delta D}_{+0.001\%} = 0.225\%$$

Removing lobbying raises consumption by 0.19% at every date, raises its growth rate by 0.0013 percentage points, from 1.7247 to 1.7259% a year, and leaves hours almost unchanged. We will summarize the effect of removing lobbying on growth, consumption share of output, output level, and disutility of labor.

Industry composition effect for innovation. Removing lobbying raises growth by 0.13 basis points, from 1.7247 to 1.7259%. The rise does not come from more innovation effort by a given firm. Superstars innovate *less* without lobbying: at every rank in industries with three or four superstars, and the follower in industries with two, a superstar's innovation rate is lower than in the same industry state with lobbying (Figure 10), consistent with the contest protecting the rents that fund innovation. Each fringe firm also attempts fewer breakthroughs, because becoming a superstar is worth 0.5% less when more fringe firms compete for it, but the larger number of fringe firms raises the total breakthrough rate. If the industry distribution stayed the same, growth would fall by 0.0003 percentage points: superstars lower growth by 0.0007 percentage points, and fringe breakthroughs raise it by 0.0004. However, growth rises overall because the industry composition changes, by 0.0016 percentage points. In Figure 10, we see that industries with 4 firms produce the most growth, as there is a higher chance that the top firms are tied in productivity and at least one will move the frontier and generate growth. Additionally, the reward for innovating is higher in such industries. Without lobbying, the share of industries with 4 superstars rises from 79.1 to 79.3% (Figure 9), so growth is higher.

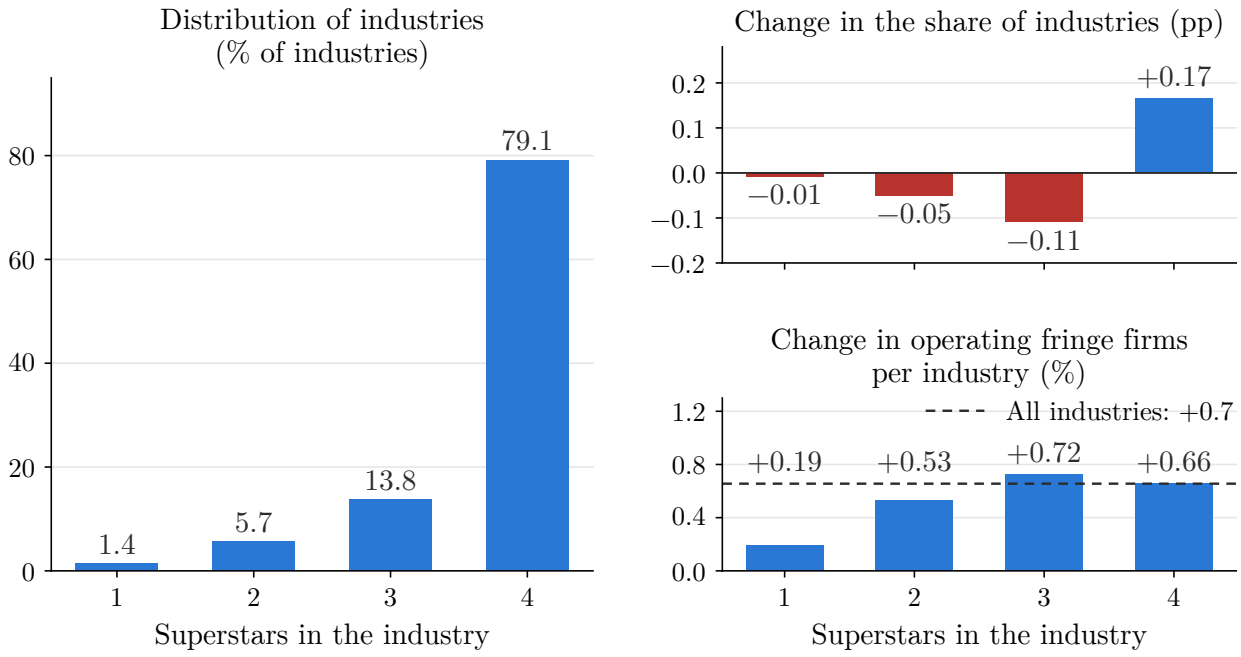


Figure 9. Industry Composition Without Lobbying

Notes: Left: share of industries by number of superstars with lobbying, percent. Right, top: change in the share of industries without lobbying, percentage points. Right, bottom: change in the average number of operating fringe firms per industry among industries with that number of superstars, percent; the dashed line is the change across all industries. At the estimate.

Output. Without lobbying, output rises by 0.125%. Output combines what the superstars produce, what the fringe produces, and the gain from buying from several producers, the love of variety. Figure 11 reports each part in two steps. In the same industry states, output rises by 0.048%. More operating fringe firms raise fringe output by 0.10%, adding 0.086% to total output, as the fringe draws sales and labor from the superstars, whose output falls by 0.10%, subtracting 0.015%. Love of variety lowers output by 0.024% because production shifts toward the fringe, already the largest producer. When industries adjust, output rises by a further 0.077%. More fringe firms produce more breakthroughs, so more industries have more superstars. Fringe output rises by 0.09%, adding 0.077% to total output, almost all of it productivity, because the fringe learns from a stronger set of superstars. Superstars' output and love of variety barely change.

Consumption to output. All the costs in the economy are in the units of output, which gives the following consumption-to-output ratio:

$$\underbrace{\frac{C_0}{Y_0}}_{\substack{\text{share of output}\\ \text{consumed}}} = 1 - \underbrace{\mathcal{R}}_{\text{regulatory costs}} - \underbrace{\mathcal{L}}_{\text{lobbying}} - \underbrace{\mathcal{F}_{\text{fringe}}}_{\substack{\text{fringe}\\ \text{fixed costs}}} - \underbrace{\psi\,(\delta m)^2}_{\substack{\text{entry}\\ \text{costs}}} - \underbrace{\mathcal{X}_{\text{R\&D}}}_{\text{R\&D}}$$

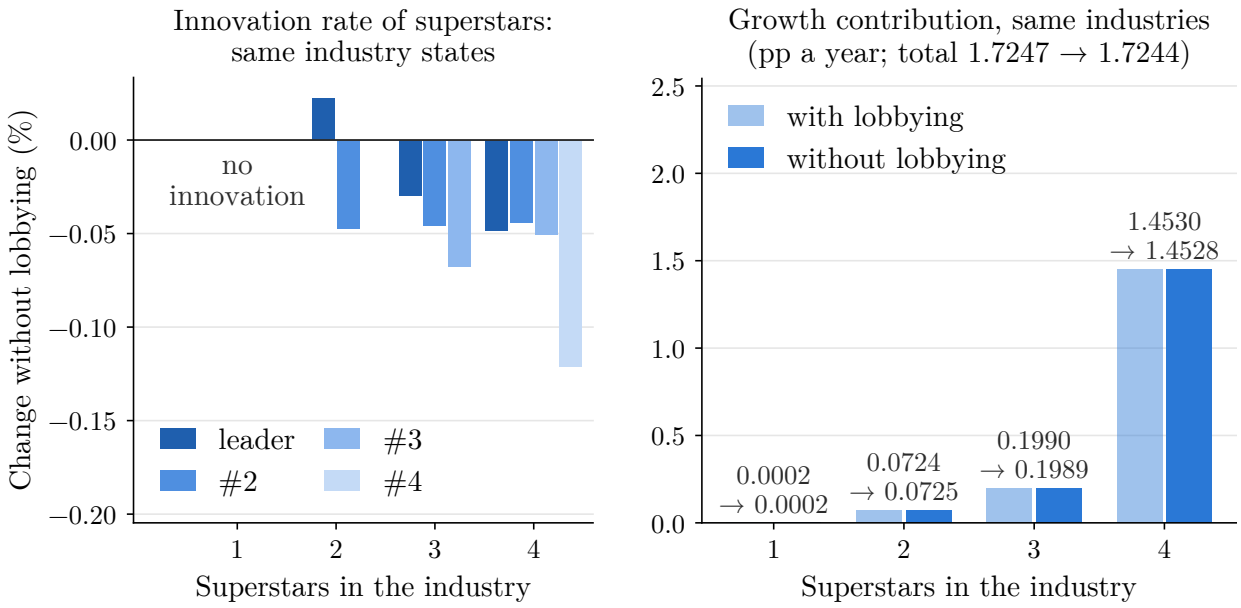


Figure 10. Innovation and Growth Within Industry States

Notes: Left: change in each superstar's innovation rate without lobbying, in the same industry state, by rank and number of superstars, percent. Right: contribution of the industries with each number of superstars to growth, percentage points per year: their share of industries with lobbying times their growth rate, with lobbying and without lobbying in the same industry states. At the estimate.

Figure 12 shows that the consumption-to-output ratio rises because the fall in lobbying spending and compliance costs exceeds the rise in the fixed and entry costs that come with more fringe firms. Lobbying spending alone is worth 0.094% of consumption; the association's organizing cost is 77% of it.

Labor hours. Without lobbying, labor hours rise by 0.001%, which raises the disutility of labor by 0.001%. By the household's FOC (18), hours depend only on the after-tax wage relative to consumption: a higher wage makes each hour pay more, while higher consumption makes each extra unit of income worth less. A 1% rise in this ratio raises hours by 0.5%, the Frisch elasticity ϵ_L. Removing lobbying raises output by 0.125%, and both the after-tax wage and consumption rise with it. Consumption rises by a further 0.07%, because a larger share of output is consumed, and the after-tax wage by a further 0.07%, because the labor share rises and the labor tax falls. Their ratio therefore rises by only 0.003%, and hours barely move.

To summarize, lobbying spending accounts for 0.094 percentage points of the 0.225% welfare gain and compliance costs for 0.016. The remaining 0.11 comes from the change in industry structure, net of the fixed and entry costs of the additional firms: with more entrants, industries have more superstars and the fringe learns from a stronger set of leaders, which raises output; and growth rises, because more industries have four superstars,

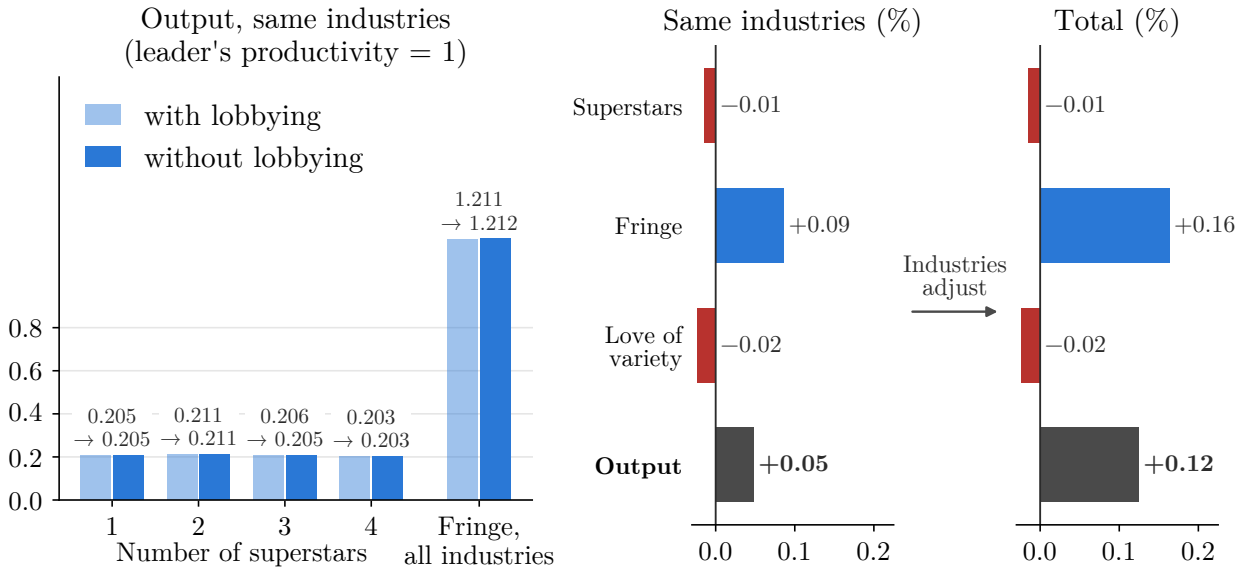


Figure 11. Output Without Lobbying

Notes: At the estimate. Left: output of all superstars in an industry, by number of superstars, and fringe output in all industries, in units of the leader's productivity: log average across industries, from with lobbying to without lobbying in the same industry states. Right: effect on the log of output without lobbying, percent. Same industries: every industry keeps its state, weighted by the distribution of states with lobbying. Total: including the shift in the distribution of states. Superstars and fringe: quantities added up, with log-mean quantity weights. Love of variety: the difference between the industry aggregate and that sum. The parts add up before rounding.

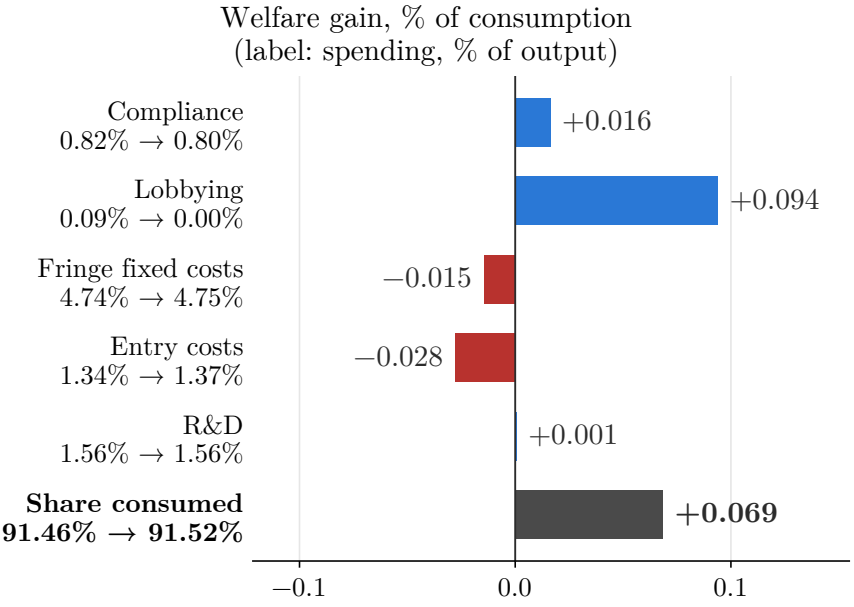


Figure 12. Resources With and Without Lobbying

Notes: Welfare gain from removing lobbying, by item, through the share of output consumed, percent of consumption. Labels show spending with and without lobbying, as a percent of output. At the estimate.

although superstars innovate less. We provide a welfare analysis of how lobbying-driven regulatory costs affect consumption and labor supply, and we abstract from any offsetting or complementary effects on the non-consumption utility effects of regulation. Some regulations affected by lobbying may nevertheless protect consumers, so we ask how valuable they would have to be for removing lobbying to bring no gain. Suppose each unit of output spent on compliance yields benefits worth v units of consumption to households, leaving firms' decisions unchanged. We see that removing lobbying raises welfare unless v exceeds 15.1. If each unit of compliance spending returned exactly one unit of benefits, compliance would no longer be a cost, yet removing lobbying would still raise welfare by 0.21% of lifetime consumption, because lobbying itself absorbs resources and it reduces the number of producing firms. Direct welfare evidence on the rules in R is scarce. Occupational licensing, the clearest case, raises prices and wages, and its welfare costs exceed its benefits (Kleiner and Soltas, 2023); the large benefit-cost ratios reported for federal regulation come mainly from air-quality rules (Office of Management and Budget, 2013), which lie outside of regulations we evaluate here.

6.3 Which lobbying to target?

Table 10 removes only the lobbying over regulation, by setting its participation cost to infinity and solving for the new equilibrium. This delivers almost the entire gain: 0.223% of lifetime consumption, against 0.225% when all lobbying is removed. Regulatory costs return to their levels without lobbying, and the remaining tax and contract lobbying costs only 0.0022% of consumption, almost all of it tax lobbying. Removing tax and contract lobbying while the contest over regulation stays delivers 0.012%, more than that amount, because regulatory costs then also fall, from 1.032 to 1.026 times their level without lobbying.

Table 10. The Welfare Cost of Each Kind of Lobbying

Lobbying allowed	$\mathbb{E}[R]/R_0$	Growth	Welfare loss (%)
All three (estimated economy)	1.032	1.7247%	0.2248
Regulation only	1.026	1.7245%	0.2124
Tax and procurement	1.000	1.7262%	0.0022
Tax only	1.000	1.7262%	0.0022
Procurement only	1.000	1.7260%	0.0000
None	**1.000**	**1.7259%**	**0**

Notes: Each row solves the equilibrium with the participation cost of the instruments not allowed set to infinity, at the estimated parameters. $\mathbb{E}[R]/R_0$: regulatory costs relative to their level without lobbying. Welfare loss: consumption-equivalent variation relative to the economy without lobbying.

Why removing tax and procurement lobbying produces a little gain. In the model, a tax cut due to lobbying is a transfer from the household as taxpayer to the household as owner of the firm, and the government finances it with a slightly higher labor tax. A contract obtained by lobbying is an additional government purchase: the household consumes the goods it delivers, which the government pays at their cost, and the contractor's fee is a transfer financed by the same labor tax. The only efficiency costs are lobbying resources (0.010% of output) and the distortion of labor supply from the labor tax, which rises by 0.036 percentage points. In percent of consumption, removing this lobbying raises output by 0.0006 and the share of output consumed by 0.0146, mostly by saving lobbying spending. However, growth falls, at a cost of 0.0059, because tax cuts and contracts won by lobbying made innovating slightly more valuable. Hours rise as the labor tax falls, at a cost of 0.0071 in leisure. The total gain is 0.0022%.

6.4 Policy implications

This counterfactual analysis highlights that the more salient lobbying on firm-specific tax breaks and contracts produces a much lower welfare loss than lobbying on regulatory costs. Our analysis yields two policy implications. First, lobbying reforms should target the contest over regulation rather than lobbying in general: banning regulatory-cost lobbying delivers 99% of the gain from removing all lobbying, while lobbying for tax cuts and contracts on its own costs almost nothing. Second, the welfare effect depends on whether lobbying raises or lowers regulatory costs, which depends on the balance between the superstars and the fringe's association. Measures that lower the cost of organizing small firms could therefore shift the contest without restricting political activity.

7 Conclusion

This paper studies what firms lobby for and how lobbying affects concentration, innovation, growth, and welfare. Using an LLM classification of federal lobbying reports, we construct a new dataset of the specific policy tools firms lobby for, from a taxonomy of 87 instruments. Almost all firms and associations ask for more appropriations and lower taxes, but they disagree on regulatory costs: 63% of the asks of the largest firms are for higher costs, against 36% for associations.

To quantify the consequences of lobbying, we build a dynamic general equilibrium model of oligopolistic competition and innovation in which firms lobby for lower taxes and procurement contracts and over industry-wide regulatory costs. In the estimated model, superstars

lobby for higher regulatory costs, while the trade association lobbies against them. The largest firms favor higher regulatory costs both for higher profits as marginal fringe firms stop producing and for protection against entrants; the smallest superstars favor them mainly for that protection. The two sides largely offset each other on average, while the contest moves regulatory costs widely across industries. Removing lobbying raises the number of producing fringe firms by 0.7% and increases the share of industries with four superstars. Superstars innovate less, but growth rises through an industry composition effect. Overall, removing lobbying raises consumption-equivalent welfare by 0.22%. Lobbying spending accounts for 0.09 percentage points of this gain, compliance costs for 0.02, and changes in industry structure for the rest.

Importantly, not all lobbying is the same. Lobbying over regulatory costs accounts for almost the entire welfare gain from removing lobbying, while tax and procurement lobbying, the most salient outcome of lobbying, mostly shifts money from the household as a taxpayer to the household as a firm owner. Thus, we need to think carefully about which types of lobbying to target. One limitation of our analysis is that we abstract from the non-consumption effects of regulation: removing lobbying raises welfare unless each unit of compliance spending yields more than 15.1 units of consumption.

The instrument-level classification of lobbying reports opens several directions for future research. First, one can evaluate reforms of individual instruments, for example, licensing requirements separately from reporting duties, and extend the model to the other wedges firms lobby on, such as government demand and administered prices. Second, one can match the rules that lobbying adds to evidence on their benefits, from which we abstract in this paper. Third, because the classification separates instruments by how their compliance costs scale with firm size, it can discipline heterogeneity in the stakes for large firms across regulation types, allowing a firm to take different sides on different types of regulation.

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Appendix

A Example of the lobbying report

Clerk of the House of Representatives	Secretary of the Senate
Legislative Resource Center	Office of Public Records
135 Cannon Building	232 Hart Building
Washington, DC 20515	Washington, DC 20510
http://lobbyingdisclosure.house.gov	http://www.senate.gov/lobby

LOBBYING REPORT

Lobbying Disclosure Act of 1995 (Section 5) - **All Filers Are Required to Complete This Page**

1. Registrant Name ☑ Organization/Lobbying Firm ☐ Self Employed Individual
APPLE INC.

2. Address
Address1 700 K Street NW Address2
City WASHINGTON State DC Zip Code 20001 Country USA

3. Principal place of business (if different than line 2)
City Cupertino State CA Zip Code 95014 Country USA

4a. Contact Name: Mr. Timothy Powderly | b. Telephone Number: 2027729500 | c. E-mail: gholger@apple.com | 5. Senate ID#: 4152-12

7. Client Name ☑ *Self* ☐ *Check if client is a state or local government or instrumentality*
APPLE INC. | 6. House ID#: 314500000

TYPE OF REPORT 8. Year 2026 Q1 (1/1 - 3/31) ☑ Q2 (4/1 - 6/30) ☐ Q3 (7/1 - 9/30) ☐ Q4 (10/1 - 12/31) ☐
9. Check if this filing amends a previously filed version of this report ☐
10. Check if this is a Termination Report ☐ Termination Date _____ 11. No Lobbying Issue Activity ☐

INCOME OR EXPENSES - YOU MUST complete either Line 12 or Line 13

12. Lobbying

INCOME relating to lobbying activities for this reporting period was:

Less than $5,000 ☐

$5,000 or more ☐ $ _____

Provide a good faith estimate, rounded to the nearest $10,000, of all lobbying related income for the client (including all payments to the registrant by any other entity for lobbying activities on behalf of the client).

13. Organizations

EXPENSE relating to lobbying activities for this reporting period were:

Less than $5,000 ☐

$5,000 or more ☑ $ 2,480,000.00

14. REPORTING Check box to indicate expense accounting method. See instructions for description of options.

☑ **Method A.** Reporting amounts using LDA definitions only

☐ **Method B.** Reporting amounts under section 6033(b)(8) of the Internal Revenue Code

☐ **Method C.** Reporting amounts under section 162(e) of the Internal Revenue Code

Signature Digitally Signed By: Timothy Powderly, Senior Director, Government Affairs **Date** 4/20/2026 10:26:49 PM

LOBBYING ACTIVITY. Select as many codes as necessary to reflect the general issue areas in which the registrant engaged in lobbying on behalf of the client during the reporting period. Using a separate page for each code, provide information as requested. Add additional page(s) as needed.

15. General issue area code CPT

16. Specific lobbying issues

General patent policy, including issues related to the Patent Trial and Appeal Board, S. 1553/H.R. 3160 the PREVAIL Act, and S. 708/H.R. 1574 the RESTORE Patent Rights Act
Issues related to standard essential patents
Issues related to anti-counterfeit policy
General trademark and copyright policy
Issues related to patent proceedings and the International Trade Commission, including the Advancing Americas Interests Act
General overview of music licensing legal framework

Figure 13. Sample lobbying report

B Issue-Text Splitting: Examples

Each example shows a raw LD-2 issue-text block and the subissues produced by the splitting stage. Deterministic pattern rules handle unambiguous structure (bill-led segments, enumerated lists, blank-line groups); blocks whose boundaries require semantic judgment are routed to the language-model stage. Original filers' typos are preserved.

Example 1: bill-led block, one subissue per bill.

> "S. 1816 and H.R. 3852: Chesapeake Reauthorization, H.R. 5088: America's Commitment to Clean Water Act, H.R. 5509: Chesapeake Bay Program Reauthorization and Improvement Act"

Split into three subissues: (1) S. 1816 / H.R. 3852, Chesapeake Reauthorization (a cross-chamber companion pair kept as one subissue); (2) H.R. 5088, America's Commitment to Clean Water Act; (3) H.R. 5509, Chesapeake Bay Program Reauthorization and Improvement Act.

Example 2: semicolon list mixing instruments.

> "Issues and incentives related to biodiesel, and the Renewable Fuel Standard; S. 1277/H.R. 2238, Biodiesel Tax Incentive Reform and Extension Act; S. 1294, Oil Independence for a Stronger America Act, Title V – Alternative Transportation Fuels; S. 1220, Fulfilling U.S. Energy Leadership Act, Title IIII – Alternative Fuel Deployment. Implementation of Budget Control Act. Extension of biodiesel tax credit. S. 3240, Agriculture Reform, Food and Jobs Act."

Split into seven subissues. The block mixes a regulatory quantity mandate (the Renewable Fuel Standard), a tax-credit companion pair (S. 1277 / H.R. 2238), title-restricted positions on two energy bills, statute implementation, a free-standing tax-extender request, and a farm bill. Keeping it whole would force a single label onto instruments from different wedges.

Example 3: language-model split required (no reliable delimiters).

> "Issues pertaining to intellectual property rights including, but not limited to, patent protection and enforcement, copyright protection and enforcement in international markets, and antitrust, S. 3197 – Platform Competition and Opportunity Act of 2021, S. 2992 – American Innovation and Choice Online Act"

Commas here separate both topics and bills, so pattern rules cannot place boundaries. The language model produces four subissues: (1) patent protection and enforcement; (2) copyright protection and enforcement in international markets; (3) antitrust, S. 3197; (4) antitrust, S. 2992. The intellectual-property topics and the antitrust bills belong to different wedges.

C Policy-Instrument Taxonomy and Model Crosswalk

The policy objects of the model, the industry-wide regulatory fixed cost $R(S)$, the firm's own profit tax τ_i, and its own procurement contracts g_i, must be measured from text that refers to none of them. A lobbying report names a statute, a rule, a bill, or a program; it does not say whether the policy at issue raises a fixed cost of operating or a cost that scales with production. Asking a language model to supply that judgment directly would place the central economic distinction of the paper inside a single step that can be neither inspected nor revised. We therefore separate the reading of the text from its economic interpretation. A closed codebook of policy instruments, each defined by the statutory or regulatory language that identifies it rather than by what it does to costs, is written in advance, and a crosswalk from instruments to the terms of the firm's problem is fixed before any text is classified. The language model assigns text to instruments; the assignment of instruments to economic channels is ours, and takes the form of a fixed table.

C.1 Design

Classification proceeds in three stages. In the first, a language model reads a single text segment and returns, for each distinct request the segment contains, the identifier of a codebook instrument, the conditions defining whom the policy covers, the direction and, where the text states one, the magnitude of the change requested relative to current law, and the position the filer takes. Each of these is a statement about what the text says. In the second stage the identifier is matched to the codebook, which supplies the economics: the wedge the instrument moves, the government account on which it falls, the payoff it typically delivers to a firm that qualifies for it, the base with which that payoff scales, and, for instruments delivered through the tax code, the Treasury provision numbers. The match is mechanical. The third stage derives the quantities the analysis requires: the breadth of the request, whether it favors the firms it covers, the side of the contest on which the filer stands, and the firm's portfolio across wedges.

Two consequences of this division matter. Because the economic content of the classification is held in a table written in advance, it can be examined, disputed, and replaced without reclassifying a single filing, and every result can be recomputed under an alternative crosswalk. Because the classifier's errors and the crosswalk's errors are of distinct kinds, they can also be measured separately. The unit of classification is the provision rather than the bill or the filing: an omnibus statute moves many wedges at once, so a bill enters the panel as a vector of classified provisions, and a subissue containing several distinct requests yields one observation for each. A segment consisting only of a bill reference is recorded as

such together with the resolved bill number, and its instruments arrive through the bill's provisions; a segment that names no policy object, or names a policy area too vaguely to place, is recorded as uninformative and excluded. Neither is forced onto an instrument.

C.2 The codebook and its sources

The codebook is intended to be exhaustive over the policy space on which firms lobby, and its completeness is borrowed rather than asserted: every instrument traces to an official or established inventory of its domain. We draw on the Treasury tax-expenditure inventory, whose provision numbers the codebook retains; the regulatory-forms taxonomy of Xie and Perez (2018); the UNCTAD International Classification of Non-Tariff Measures; the Harmonized Tariff Schedule of the United States and Title 19 of the U.S. Code; the federal Assistance Listings; the Federal Acquisition Regulation; the patent statute and the exclusivity provisions administered by the Food and Drug Administration; the Immigration and Nationality Act; and, as a check on the inventory as a whole, the tools-of-government catalog of Salamon (2002). Coverage failures are not random: they concentrate in the domains a researcher knows least well and bias the measured portfolio toward whatever the codebook covers. Deriving the list from external inventories makes omissions visible to any reader who consults the same sources.

Instruments are organized by what they do to the firm's problem, not by whom they reach. A research credit written for a single firm and one available to the whole economy are the same instrument, distinguished only by the eligibility conditions the classifier reads from the text. Breadth is thus a property of the observation rather than of the codebook, which keeps the observed breadth of lobbying falsifiable.

C.3 Wedges and the crosswalk to the model

A wedge names the term of the firm's problem an instrument moves. Table 11 lists the ten wedges, the margin each moves, and the object of the model to which each corresponds. Three objects of the model are measured from the classification: regulatory fixed costs are the industry-wide burden $R(S)$ of the contest (13); the profit-tax wedge is the object of tax lobbying (11); and procurement contracts, an instrument of the demand wedge, are the object of procurement lobbying (12). The other wedges are recorded rather than discarded, which allows the classification to be reused for extensions and for counterfactuals defined on individual instruments.

Instruments whose incidence is conditional on concentration, among them merger control, conduct rules for dominant platforms, and regulation triggered by size thresholds, are kept

Table 11. Wedges and the Crosswalk to the Model

Wedge	Margin moved	Model object
R_j	Industry-wide quasi-fixed cost of operating, entering, or introducing a product	$R(S)$: **regulatory costs lobbying**
b_T	Profit-tax level; scales after-tax profit with no first-order effect on marginal prices	τ_i: **tax lobbying**
s^d	Demand facing the firm: government purchases, buyer subsidies, trade protection	g_i (procurement contracts): **procurement lobbying**; other instruments recorded
s^y	Per-unit wedge on marginal cost or marginal revenue	recorded; extension
s^r	Price of R&D and the private return to innovation	R&D cost χz^{ϕ}; extension
s^k	User cost of capital, including financing	recorded; no capital in the model
s^w	Effective wage bill	labor cost; extension
s^p	Price the firm may charge	markup; extension
$A(\Theta)$	Concentration-conditional rules: act on the industry state or switch on above a size threshold	transitions of Θ; extension
s^i	Lump-sum transfer to, or fee from, the firm	profit level; extension

separate from R_j, since their burden depends on a firm's position in the industry rather than falling on every active firm alike; they are the natural object of an extension in which lobbying acts on the transitions of the industry state directly. A requirement enters R_j only if its compliance cost is quasi-fixed; any component of the same statute whose cost scales with output is assigned to the output wedge.

C.4 What the classifier records

Table 12 lists what the classifier records, one observation for each request it identifies.

Eligibility records only the conditions the text adds beyond the instrument's own definition, and because it is recorded in standard classification systems it also serves as the key that joins an observation to the firm panel. Conditions that bind on quantities the firm chooses are recorded against a closed list of variables that map into the model: output, revenue, price, employment, wages, capital, investment, research spending, profit, market share and concentration, merger events, and firm age. The direction of the requested change is one of a small set of operations, distinguishing the creation of an instrument from its expansion, reduction, extension, or repeal; magnitudes are recorded when the text states them and never imputed. Positions are recorded as the filer states them, and a filing that reports monitoring an issue is kept distinct from one that states no position.

Table 12. What the Classifier Records for Each Request

Field	Content
Instrument	The codebook instrument the text concerns, or a label marking the segment as a bare bill reference or as uninformative
Legislation	Bill and public-law numbers named in the text, resolved to the Congress in which they were introduced
Granularity	Whether the text is finer than the nearest instrument and, if so, a verbatim statement of the finer target
Eligibility	Whom the policy covers, over and above what the instrument itself implies: industry, geography, product, occupation, legal form, and certified status, recorded in standard classification systems
Conditions	Thresholds the policy places on quantities the firm chooses, expressed as comparisons against a closed list of model variables
Effect	The direction of the change requested relative to current law and, when the text states them, its magnitude and starting level
Position	The position the filer takes: support, opposition, either with an amendment, the pursuit of a specific change, monitoring, or none stated
Confidence	The classifier's confidence in the instrument assignment; low-confidence observations are dropped

C.5 From classified text to model objects

Besides the wedge, each instrument carries attributes that attach to every observation assigned to it: the government account (expenditure, receipt, or a mandate whose cost is borne privately), the sign of the payoff to a qualifying firm, and its shape, which records whether the payoff scales with a flow or is a lump independent of scale. The shape enforces the rule that only quasi-fixed burdens are regulatory costs.

The scope of a request is determined by the width of its eligibility conditions: a named entity yields a firm-specific request, an industry code with qualifiers a narrow one, an industry branch an industry-wide one, and unrestricted eligibility an economy-wide one. Whether a request favors the firms it covers follows from its direction and the instrument's sign. Let $\xi \in \{-1, 0, +1\}$ denote the requested change in the level of instrument k, with creation, increase, extension, reauthorization, and funding taking the value $+1$, reduction and repeal -1, and restatements and studies 0, and let $\text{sgn}_k \in \{-1, +1\}$ be the instrument's sign. Favorability is

$$\text{fav} \;=\; \xi \cdot \text{sgn}_k. \tag{50}$$

The same quantities place an observation on one side of the contest. For an observation on a regulatory costs instrument, let $\text{st} = +1$ when the filer supports the request and $\text{st} = -1$ when it opposes it. The filer would move the regulatory fixed cost in the direction $\text{st} \cdot \xi$:

supporting a bill that raises a compliance requirement and opposing the repeal of one are the same political act, and the classification records them on the same side without a further judgment. The share of a firm-year's regulatory costs requests with $\text{st} \cdot \xi = +1$ determines its side, and the share of positions that ask for higher regulatory costs is the moment that disciplines the elasticity of superstars' compliance costs to their size in Appendix J.3.

For a few instruments the sign is not a property of the instrument: a tariff protects the firms that produce the good and raises the costs of the firms that use it. Observations on such instruments are resolved filer by filer, first from the stated position, because supporting a duty reveals the protected side, and then from a match between the affected good and the firm's own products or inputs. Classified observations collapse to a panel of firm, quarter, instrument, wedge, and side, from which every moment of Sections 3 and 5 is computed.

C.6 The instruments

The list below groups the 87 instruments of the codebook by wedge.

Regulatory fixed costs *(wedge R_j, 18 instruments)*

- Licenses and charters required to operate; product certification and inspection; pre-market product approval; facility permitting and certificates of need.
- Recurring reporting, recordkeeping, and registration duties; labeling and warning mandates; securities disclosure and shareholder-rights rules; data privacy and cybersecurity rules; one-time outlays required by a standard; financial assurance (surety) bonds.
- Health, safety, and technical requirements on imports; services trade and data-flow restrictions; foreign investment screening; issuance of limited rights to public resources.
- Liability immunities and shields, limits on damages, expansion of liability exposure; conduct rules on how a transaction is made.

Profit tax *(wedge b_T, 7 instruments)*

- The general corporate income tax; deductions from taxable profit; reduced rates on classes of income; reduced tax on foreign and export income; exemption by legal form; deferral and basis rules; deductibility of losses and forgiven debt.

Demand *(wedge s^d, 14 instruments)*

- Government demand: program funding, appropriations, and authorizations; procurement contracts; public infrastructure and services used by firms; tax exemption for state and local government bonds.
- Buyer purchase subsidies and purchase mandates; domestic-content requirements for public purchases; markets reserved by law to designated sellers; advertising and marketing restrictions; government provision of competing services.

– Trade: import tariffs and tariff-rate quotas; limits on import quantities; antidumping and countervailing duties; duty-free trade under preference programs; export controls and sanctions.

Output *(wedge s^y, 17 instruments)*

– Per-unit production credits and support payments; percentage depletion; excise and sales taxes; taxes on emissions and discharges; regulatory levies on revenue; public-resource royalties.
– Technology and performance standards with per-unit costs; bank capital and liquidity requirements; limits on domestic production; tradable permits; portfolio and blending standards; prohibitions of products; obligations to serve; insurance coverage mandates.
– Duty suspensions and drawback; customs procedures and rules of origin; federal insurance backstops.

Innovation cost *(wedge s^r, 5 instruments)*

– R&D tax credits and expensing; R&D grants and technology transfer; patent, copyright, trademark, and trade-secret law; market exclusivity for approved products; enforcement of intellectual property against foreign infringement.

Capital cost *(wedge s^k, 6 instruments)*

– Investment tax credits and expensing; accelerated and bonus depreciation; tax-exempt private-activity bonds; federal lending and loan guarantees; creditor rights in bankruptcy; deductibility of business interest.

Labor cost *(wedge s^w, 8 instruments)*

– Hiring and wage credits; untaxed employee benefits; tax deferral on retirement contributions; payroll taxes; wage floors, hour rules, and benefit mandates; workplace safety and worker protections; employee status and collective bargaining; restrictions on hiring foreign workers.

Set prices *(wedge s^p, 5 instruments)*

– Government-set payment rates; mandatory discounts, rebates, and negotiated prices; price ceilings; price floors; allowed rates of return and revenue caps.

Competition *(wedge $A(\Theta)$, 4 instruments)*

– Merger control and conduct rules; platform gatekeeper regulation; antitrust exemptions; mandated access and interoperability.

Lump-sum transfers and fees *(wedge s^i, 2 instruments)*

– Grants and direct payments to firms; regulatory user fees.

Place-based bundles *(wedge 1 instrument)*

– Zone regimes that combine an employment credit, a financing subsidy, and a capital-gains preference; decomposed into their components, with the geographic restriction recorded as a condition of eligibility.

C.7 Validation

Since the crosswalk is deterministic, validation concerns the classifier's output alone. Section 2.2 reports the checks: agreement with an independent classification by a second language model, agreement with the positions recorded by MapLight, and the stability of repeated classification.

D Participation: All Observations and Lobbying Periods

The participation figures of Section 3 use two denominators. Figures 2 and 6 divide by all observations: all firm-years and, for associations, all years of 501(c)(6) organizations filing an IRS Form 990 or 990-EZ with positive revenue; they measure how often a client lobbies, the object of the model's participation targets. Figure 3 divides by the quarters with a lobbying report; it measures what lobbying clients lobby on. Figure 14 reports the shares from Figure 6 for the firm-years with a lobbying report, and for associations one bar over the 16,397 association-years with a report.

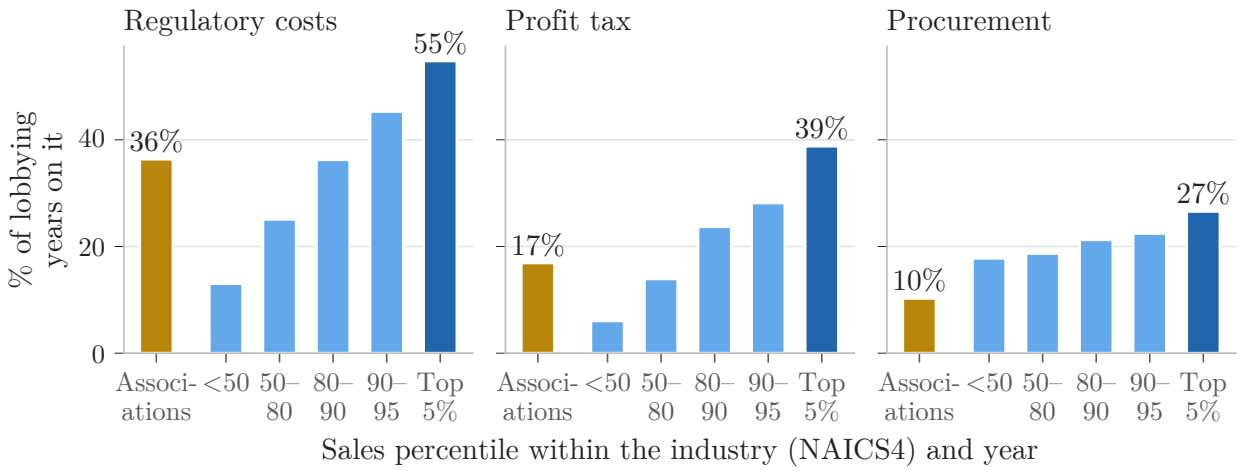


Figure 14. Participation in Lobbying by Firm Size, Among Lobbying Years

Notes: Share of the years with a lobbying report in which the client lobbies on the instrument: the 14,439 firm-years in which the firm lobbies and the 16,397 association-years with a report, by the firm's sales percentile within its industry (four-digit NAICS) and year. A year with a lobbying report has a report with a positive amount that names a classified instrument. 2008–2022.

E Robustness: Asks That Touch Several Instruments

The figures of Section 3 count an ask on an instrument only when the ask is about that instrument alone. An ask is one request in a report: a request in the filer's own words, a named section or title of a bill, or a bill or regulation that the filer cites. Each ask has total weight one. A request in the filer's own words counts for the instrument it names when the classification has high or medium confidence. A citation counts only when the provisions it refers to all concern a single instrument. The provisions a citation refers to are those its words point to (a named section or title, or the provisions that the readings of its words pick among the bill's provisions) and all the provisions of the bill otherwise. The bill's name plays no role. For example, a 2020 report that lists "H.R. 2, the Moving Forward Act, and the need for corrosion planning in federal bridge projects" refers to two of the bill's 41 provisions, both on public infrastructure, and counts on that instrument; a report of the same quarter that cites the bill by its number and title alone refers to all 41 provisions, on 18 instruments, and counts on none of them.

This appendix counts every ask instead. A citation divides its weight among the instruments of the provisions it refers to, in proportion to the number of provisions on each, as if each provision were a separate request. A 2019 citation of the America's Water Infrastructure Act (S. 3021, 115th Congress) refers to two provisions on procurement and thirteen on public infrastructure and gives 2/15 to procurement and 13/15 to public infrastructure. A 2020 report that asks to "Support H.R.748/S.3548 the CARES Act" refers to all 180 provisions of the Coronavirus Aid, Relief, and Economic Security Act; program funding and appropriations, the instrument of 44 of them, receives 0.24. A provision that the bill's summary states several times counts as many times, a request that names several sections of a bill divides its weight equally among them, and provisions classified at low confidence keep their share but do not count. A client lobbies on an instrument in a quarter with probability $1 - \prod_a (1 - w_a)$, where w_a is the weight of ask a on the instrument (an ask counted once within a report, and the weights of one request on a wedge or front summed up to one). The sides count every citation that states support or opposition and every request in the filer's own words; each provision's share goes to the side implied by the ask's stance and the provision's direction. Requests in the filer's own words keep weight one. The denominators do not change. The weights depend on how finely a bill's summary divides it into provisions. Giving each of the K instruments of a citation the same weight $1/K$ instead changes the participation shares of this appendix by at most 3.4 percentage points and the shares asking for higher regulatory costs by firm size by at most 1.3 points.

Counting every ask raises the shares and leaves the ranking of the fronts and of the

leading wedges in place. Lobbying firms lobby on regulatory costs in 40% of their quarters with a report instead of 33%, on the profit tax in 28% instead of 20%, and on procurement in 24% instead of 19%; associations in 39, 19 and 14% instead of 33, 15 and 8%. Government demand and regulatory costs remain the two most lobbied wedges for every group, and program funding and appropriations the most lobbied instrument (Figure 15). The mean share of asks for higher regulatory costs among filers taking a side is 34% below the 80th percentile of their industry's sales, 43% between the 80th and the 95th, and 62% among the largest 5%, against 27, 43 and 63% in the baseline; for associations it is 40% against 36%. Sides on single instruments rest on fewer filer-years and move more (Figure 16): the largest change is on licenses and charters among the largest 5% of firms, where the share is 63% against 45% in the baseline (23 expected filer-years with a side against 11).

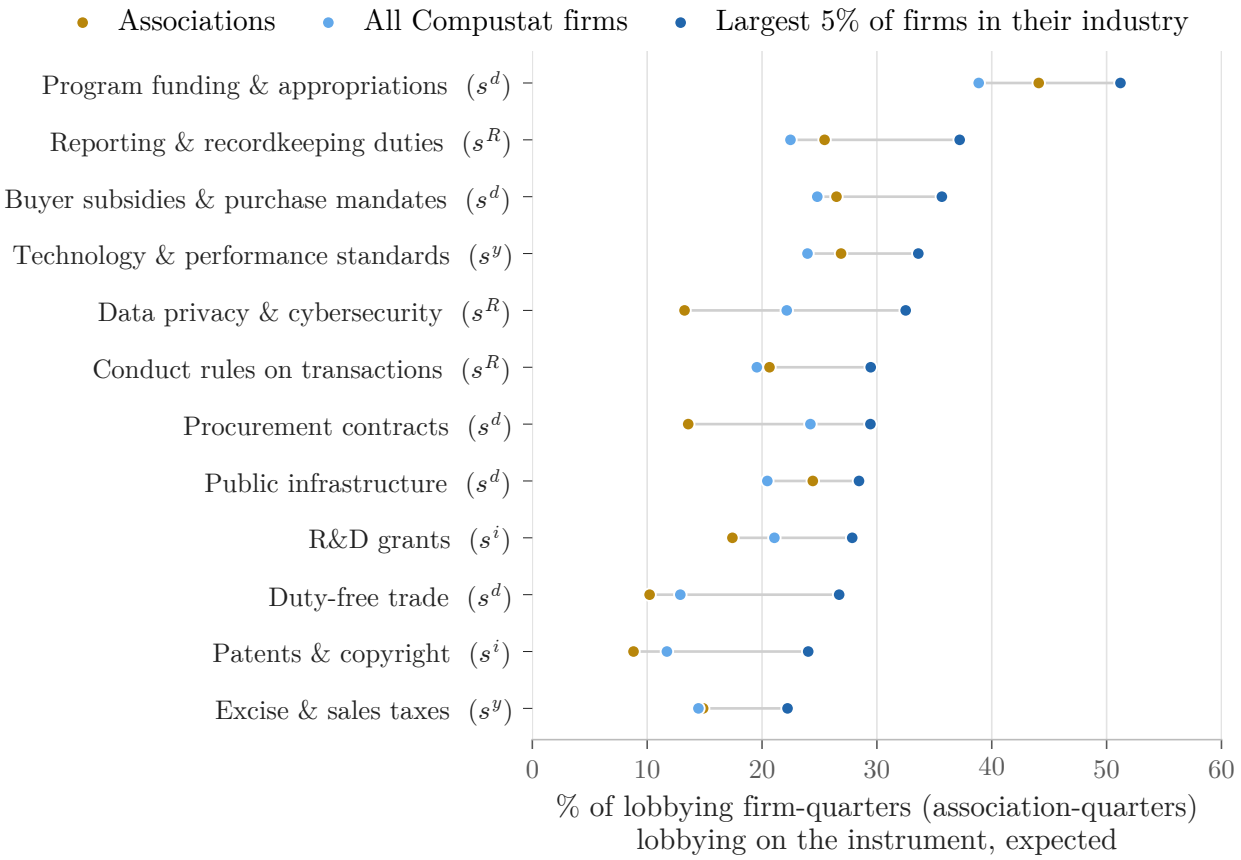


Figure 15. Lobbying by Instrument, Every Ask Weighted by Provision Share

Notes: Share of lobbying firm-quarters (association-quarters) in which the client lobbies on the instrument, counting every ask: a citation gives each instrument the share of its provisions on the instrument, and the share is the expected share of quarters with lobbying on the instrument. 52,079 Compustat firm-quarters, of which 14,970 of the largest 5% of firms in their industry by sales; 60,596 association-quarters; 2008–2022. The instruments are the 14 most lobbied by firms and the 10 most lobbied by associations, without government-set payment rates and investment tax credits.

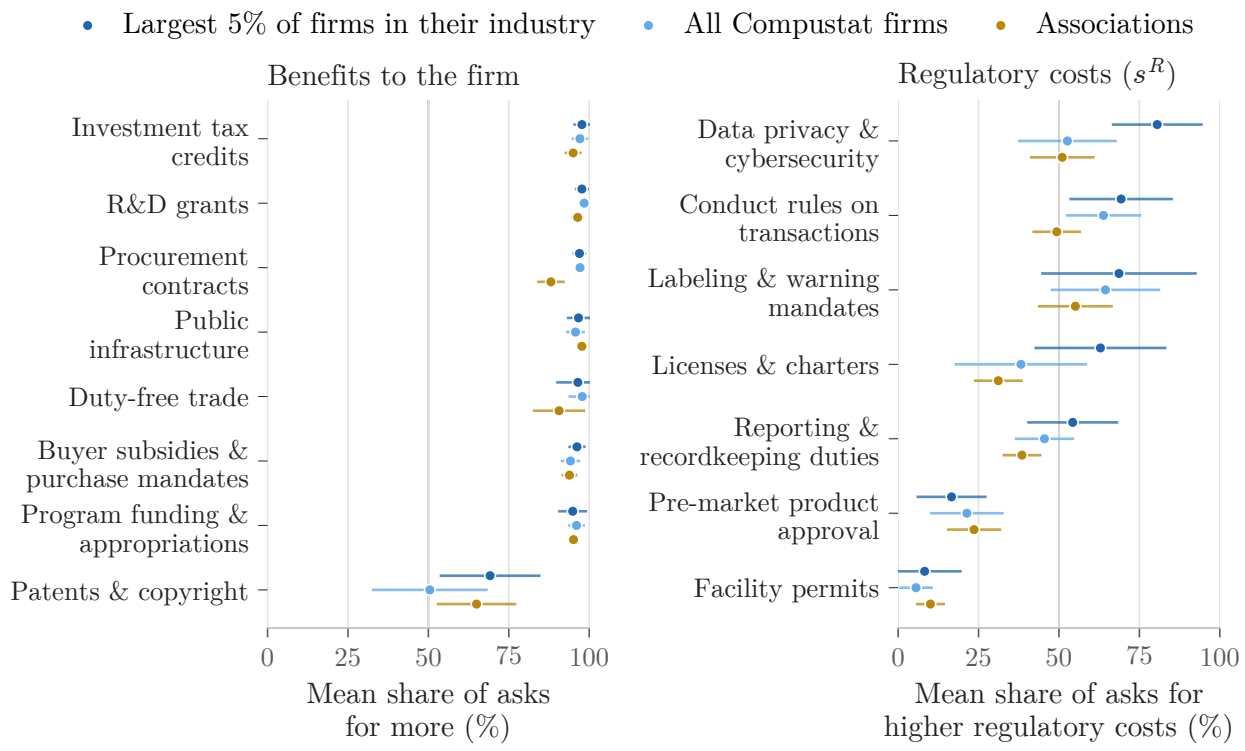


Figure 16. The Side Firms Take, Every Signed Ask Weighted by Provision Share

Notes: As Figure 4, counting every ask that states support or opposition, each with its provision shares.

F Static Cournot Equilibrium under Decreasing Returns to Scale

This appendix derives the static product-market objects used in Section 4: the superstar best responses, markups, and profits; the fringe output and profits; and the relative-output system solved at each industry state. The derivation follows Cavenaile et al. (2026), Appendix C.1, adapted to our notation. We fix an industry and a point in time and drop the j and t subscripts throughout. The industry contains N superstars with productivities $\{q_i\}_{i=1}^N$, an active fringe of mass m^{a} with common productivity q_f, the wage is w, and aggregate output is Y. Government contracts do not enter: government purchases are purchases of the final good, and a firm's contracts depend on its gap, not on its output choice.

Demand and notation. The final-good producer's problem yields the inverse demands

$$p_i \;=\; \frac{y_i^{-1/\eta}\,Y}{D}, \qquad p_f \;=\; \frac{\tilde{y}_f^{-1/\eta}\,Y}{D}, \qquad D \;\equiv\; \sum_{k=1}^{N} y_k^{\frac{\eta-1}{\eta}} + \tilde{y}_f^{\frac{\eta-1}{\eta}}. \tag{51}$$

It is convenient to define, for each superstar i and for the fringe aggregate,

$$\Sigma_i \;\equiv\; \sum_{k=1}^{N}\left(\frac{y_k}{y_i}\right)^{\frac{\eta-1}{\eta}} + \left(\frac{\tilde{y}_f}{y_i}\right)^{\frac{\eta-1}{\eta}} \;=\; \frac{D}{y_i^{\frac{\eta-1}{\eta}}}, \qquad \Sigma_{-i} \;\equiv\; \Sigma_i - 1, \qquad \Sigma_f \;\equiv\; \frac{D}{\tilde{y}_f^{\frac{\eta-1}{\eta}}}, \tag{52}$$

where Σ_i includes firm i's own term (equal to one). Revenues then satisfy $p_i y_i = Y/\Sigma_i$ and $p_f \tilde{y}_f = Y/\Sigma_f$: each producer's share of industry expenditure is the reciprocal of its own-normalized CES aggregator, $\sigma_i = 1/\Sigma_i$ and $\sigma_f = 1/\Sigma_f$, with $\sum_i \sigma_i + \sigma_f = 1$.

Superstar best response. Superstar i chooses y_i taking rivals' quantities $\{y_k\}_{k\neq i}$ and the fringe aggregate $\tilde{y}_f$ as given, internalizing the effect of y_i on D. Inverting the production function (4), $l_i = (y_i/q_i)^{1/\alpha}$, so operating profit is

$$\pi_i \;=\; \frac{y_i^{\frac{\eta-1}{\eta}}\,Y}{D} \;-\; w\left(\frac{y_i}{q_i}\right)^{1/\alpha}. \tag{53}$$

The first-order condition equates marginal revenue with marginal cost:

$$\frac{\eta-1}{\eta}\; y_i^{-1/\eta}\, Y\; \frac{D - y_i^{\frac{\eta-1}{\eta}}}{D^2} \;=\; \frac{w}{\alpha}\; y_i^{\frac{1-\alpha}{\alpha}}\; q_i^{-1/\alpha}. \tag{54}$$

The factor $(D - y_i^{(\eta-1)/\eta})/D^2$ on the left is the Cournot internalization: expanding output depresses the industry price index in proportion to the firm's own weight in D. Multiplying

both sides of (54) by $(\alpha/w)\, q_i^{1/\alpha}\, y_i^{(\eta-1)/\eta}$ collects all powers of y_i into $y_i^{1/\alpha}$; raising to the power α and substituting (52) gives the best response

$$y_i \;=\; \alpha^{\alpha} \left(\frac{Y}{w}\right)^{\alpha} \left(\frac{\eta-1}{\eta}\right)^{\alpha} q_i \left[\frac{\Sigma_{-i}}{\Sigma_i^2}\right]^{\alpha}. \tag{55}$$

Markups. From (51) and (52), $p_i = Y/(y_i\Sigma_i)$. Substituting this into the right side of (54) shows that marginal cost equals $\frac{\eta-1}{\eta}\, p_i\, \Sigma_{-i}/\Sigma_i$, so the markup is

$$M_i \;=\; \frac{p_i}{\mathrm{MC}_i} \;=\; \frac{\eta}{\eta-1} \cdot \frac{\Sigma_i}{\Sigma_{-i}} \;=\; \frac{\eta}{(\eta-1)(1-\sigma_i)}. \tag{56}$$

The markup formula as a function of relative outputs is identical to the constant-returns baseline of Cavenaile et al. (2026); decreasing returns alter the level of quantities but not the mapping from market shares to markups.

Superstar profits. The wage bill satisfies $wl_i = w\, y_i^{1/\alpha} q_i^{-1/\alpha} = \alpha\, \mathrm{MC}_i\, y_i$, so using (56),

$$\pi_i \;=\; p_i y_i - wl_i \;=\; \frac{Y}{\Sigma_i}\left[1 - \alpha\,\frac{\eta-1}{\eta}\,\frac{\Sigma_{-i}}{\Sigma_i}\right] \;=\; \frac{\big(\eta + [(1-\alpha)\eta+\alpha]\,\Sigma_{-i}\big)\,Y}{\eta\,\Sigma_i^2} \;=\; \sigma_i\Big(1 - \frac{\alpha}{M_i}\Big)Y,$$

and the wage bill is $wl_i = \alpha\,\sigma_i Y/M_i$. The term $(1-\alpha)\eta + \alpha$ multiplying Σ_{-i} captures the additional rent that decreasing returns generate on inframarginal units.

Relative outputs. Taking the ratio of the pre-exponentiation form of (54) for two superstars i and k and raising to the power α yields

$$\left(\frac{y_i}{y_k}\right)^{\frac{\eta(1-\alpha)+\alpha}{\eta}} \;=\; \frac{q_i}{q_k}\left[\frac{\Sigma_i - 1}{\Sigma_i - \left(\frac{y_k}{y_i}\right)^{\frac{\eta-1}{\eta}}}\right]^{\alpha}, \tag{57}$$

where both sums are normalized by y_i. Combining the superstar best response (55) with the fringe aggregate output (59) below gives the superstar-to-fringe condition

$$\left(\frac{y_i}{\tilde{y}_f}\right)^{\frac{\eta(1-\alpha)+\alpha}{\eta}} \;=\; \left(\frac{\eta-1}{\eta}\right)^{\alpha} \frac{q_i}{q_f}\,(m^{\mathrm{a}})^{\alpha-1}\left[\frac{\Sigma_{-i}}{\Sigma_i}\right]^{\alpha}. \tag{58}$$

Fringe output and profits. Each fringe firm is a price taker and equates price to marginal cost, $p_f = \frac{w}{\alpha}\, y_{fk}^{\frac{1-\alpha}{\alpha}}\, q_f^{-1/\alpha}$. Using $p_f = Y/(\tilde{y}_f\Sigma_f)$ and $\tilde{y}_f = m^{\mathrm{a}}\, y_{fk}$, all powers of y_{fk} again collect

into $y_{fk}^{1/\alpha}$, giving per-firm and aggregate fringe output

$$y_{fk} \;=\; \frac{\alpha^{\alpha}\, Y^{\alpha}\, q_f}{w^{\alpha}\,(m^{\mathrm{a}})^{\alpha}\,\Sigma_f^{\alpha}}, \qquad \tilde{y}_f \;=\; m^{\mathrm{a}}\, y_{fk} \;=\; \frac{(m^{\mathrm{a}})^{1-\alpha}\,\alpha^{\alpha}\, Y^{\alpha}\, q_f}{w^{\alpha}\,\Sigma_f^{\alpha}}. \tag{59}$$

Since $wl_{fk} = \alpha\, p_f\, y_{fk}$, per-firm fringe profit is

$$\pi_{fk} \;=\; (1-\alpha)\, p_f\, y_{fk} \;=\; \frac{(1-\alpha)\, Y}{m^{\mathrm{a}}\,\Sigma_f}. \tag{60}$$

Fringe firms charge no markup, yet decreasing returns ($\alpha < 1$) leave them positive operating profits. This is essential for the lobbying extension: it is these profits, net of the regulatory cost $R(S)$ and the idiosyncratic fixed cost ε, that give the cutoff $\varepsilon^*(S)$ its content.

Solving the static block. Productivities enter only through ratios. By (5) and (6),

$$\frac{q_i}{q_f} \;=\; \frac{(1+\lambda)^{-n_i}}{\zeta\left(\sum_{k=1}^{N}(1+\lambda)^{-\gamma n_k}\right)^{1/\gamma}},$$

where n_k is the gap of superstar k behind the leader. Given the technology state Θ and the active mass m^{a}, equations (57)–(58) form a system in the N relative outputs $\{y_i/\tilde{y}_f\}_{i=1}^{N}$, which we solve by Newton's method with the analytic Jacobian. Its solution determines the aggregators $\{\Sigma_i, \Sigma_f\}$, hence sales shares, normalized profits π_i/Y and π_f/Y, and markups (56); these depend only on (Θ, m^{a}), not on the wage, which the dynamic solver exploits. The derivatives $\partial(\pi_i/Y)/\partial m^{\mathrm{a}}$ and $\partial(\pi_f/Y)/\partial m^{\mathrm{a}}$ in the stakes (29) and (42) follow from the implicit function theorem on the same system. Labor demands $l_i = (y_i/q_i)^{1/\alpha}$ and $l_{fk} = (y_{fk}/q_f)^{1/\alpha}$ are each proportional to Y/w, and enter only labor-market clearing (44).

G Participation in Lobbying

This appendix gives the closed form of the participation cutoff (27) in the computed model of Section 4.6, in which a superstar's policies are evaluated at the expected costs of its current spells. Fix a superstar and write $e = (S, s_i)$ for its state without its locked costs; $e^{I,\mathrm{on}}$ and $e^{I,\mathrm{off}}$ denote its state after joining and after staying out of instrument I at a redraw.

Durations. For each instrument I on which the firm is active, $D_I(e)$ is the expected discounted time over which its current draw is paid, weighted by the scope discount. It

solves, on states where $a_{Ii} = 1$,

$$\Big(\rho + \sum_{I'} \lambda_{I'} + q(e)\Big)\, D_I(e) \;=\; \prod_{J \neq I} \big(1 - \iota_{IJ}\big)^{a_{Ji}(e)} \;+\; \sum_{e'} q(e,e')\, D_I(e') \\ + \sum_{I' \neq I} \lambda_{I'} \Big[\big(1 - G_{I'}(e)\big)\, D_I\big(e^{I',\mathrm{off}}\big) + G_{I'}(e)\, D_I\big(e^{I',\mathrm{on}}\big)\Big], \tag{61}$$

where $q(e,e')$ are the rates of the events in (23) other than the firm's own redraws (innovation, entry, and the redraws of rivals and of the association), $q(e) = \sum_{e'} q(e,e')$, $G_{I'}(e) = \Phi\big((\ln \varphi^*_{I'}(e) - \mu_{I'})/\sigma_{I'}\big)$ is the probability of being on instrument I' after its next redraw, and $D_I = 0$ after the firm exits. The firm's own redraw on I ends the spell, so it appears in the discount rate with no continuation term.

Values. When the policies do not depend on the locked costs, the value is affine in them, $v(e, \boldsymbol{\varphi}_i) = v^0(e) - \sum_I a_{Ii}\, \varphi_{Ii}\, D_I(e)$, where v^0 solves (23) without the locked costs in the flow and with the own-redraw terms

$$\lambda_I \Big[\big(1 - G_I(e)\big)\, v^0\big(e^{I,\mathrm{off}}\big) + G_I(e)\, v^0\big(e^{I,\mathrm{on}}\big) - D_I\big(e^{I,\mathrm{on}}\big)\, \mathbb{E}\big[\varphi\, \mathbf{1}\{\varphi \leq \varphi^*_I(e)\}\big] - v^0(e)\Big]. \tag{62}$$

The expectation is the partial mean of the lognormal distribution (14),

$$\mathbb{E}\big[\varphi\, \mathbf{1}\{\varphi \leq \varphi^*\}\big] \;=\; e^{\mu_I + \sigma_I^2/2}\, \Phi\Big(\frac{\ln \varphi^* - \mu_I - \sigma_I^2}{\sigma_I}\Big).$$

Cutoffs and policies. Comparing the values after joining and after staying out, with the new draw subtracted, gives the cutoff

$$\varphi^*_I(e) \;=\; \frac{v^0(e^{I,\mathrm{on}}) - v^0(e^{I,\mathrm{off}}) - \sum_{I' \neq I} a_{I'i}\, \varphi_{I'i}\, \big[D_{I'}(e^{I,\mathrm{on}}) - D_{I'}(e^{I,\mathrm{off}})\big]}{D_I(e^{I,\mathrm{on}})}. \tag{63}$$

The policies (z_i, b_{Ii}), the cutoffs, and the stake (29) are evaluated at the firm's expected locked costs, $\hat{\varphi}_I(e) = \mathbb{E}[\varphi \mid \varphi \leq \varphi^*_I(e)]$, so that the locked costs need not be tracked as a continuous state. Spending and resource costs use the exact distribution of locked costs: on each instrument, the stationary mass of locked costs at every state solves a linear balance in which a spell starts with a draw accepted at the cutoff of the state where it starts and moves with the firm until its next redraw or its exit.

H Beliefs about Rivals' Tax and Procurement Statuses

This appendix describes the beliefs that replace rivals' tax and procurement statuses in the computed model of Section 4.6. The reduced industry state $\tilde{S} = (\Theta, \mathbf{a}_R, a_A)$ keeps the technology state, the superstars' statuses on regulation, and the association's status, and each superstar's own state keeps its own tax and procurement statuses $b = (a_T, a_P)$. Let $\nu(\tilde{S}, j, b)$ be the stationary measure of superstars at rank j of reduced state $\tilde{S}$ with own statuses b. The belief about the statuses of the superstar at rank j in technology state Θ is

$$\pi(b \mid \Theta, j) \;=\; \frac{\sum_{\tilde{S}\in\Theta} \nu(\tilde{S}, j, b)}{\sum_{\tilde{S}\in\Theta} \sum_{b'} \nu(\tilde{S}, j, b')}, \tag{64}$$

where $\tilde{S} \in \Theta$ runs over every configuration of the regulation statuses and the association's status in technology state Θ. Decision objects use the beliefs: rivals' innovation rates and redraw probabilities in (23) are averages of rivals' policies at the actual reduced state over (64), and so are the fringe's pool value, the association's cutoff, and the contest stakes. A superstar's own stake is averaged over its own tax and procurement statuses with the same weights, so that its intensity on regulation, and the regulatory cost, depend on $\tilde{S}$ alone. The process uses the actual conditional mix, $w(\tilde{S}, j, b) = \nu(\tilde{S}, j, b) / \sum_{b'} \nu(\tilde{S}, j, b')$: the stationary distribution, growth, the locked-cost balances, and every moment. Because no participation status enters the conditioning, a firm's participation choice cannot change what it predicts about its rivals.

I Computing the Equilibrium

At the estimated ladder ($\bar{n} = 5$), the computed model has 84 technology states, 2,180 reduced industry states once the regulation statuses and the association's status are included, and 33,104 superstar states once each firm's rank and own tax and procurement statuses are included. The equilibrium is computed by three nested fixed points.

1. **Inner**, given the pool m and the contest outcome at every state: superstars' values, durations, policies and participation cutoffs; the fringe cutoffs, active masses and breakthrough intensities; the association's cutoffs; the beliefs; and the stationary distribution.

2. **Middle**, given m: the stakes from the inner solution, the contest at every state (the regulatory cost, dues and lobbying costs), iterated to a fixed point with Anderson acceleration (Walker and Ni, 2011).

3. **Outer**: the pool m that satisfies free entry (39), by a secant method.

An equilibrium is accepted only if one further evaluation at full precision certifies it: the contest residual is below 10^{-6}, the free-entry residual below 10^{-7} (relative), and the inner fixed point is solved to 10^{-9}. At the estimate, the lobbying equilibrium takes about two hours on one core, and the certificate gives a contest residual of 8.4×10^{-7} and a free-entry residual of 7.7×10^{-10}. The economy without lobbying is computed with the same code, with every participation cost set to infinity. The global search of Section 5.1 solves each point in a fresh process with the same certificate; points whose fixed points do not certify within the time limit are discarded.

J Estimation: Identification and Moments

This appendix gives the identification argument of each block and reports every moment.

J.1 Technology

We estimate twelve parameters on the 30 moments of Table 13. Every parameter affects every moment, but some moments are especially informative about each parameter.

1. *Quality step and demand (λ, η).* A larger step and a higher elasticity both widen the sales gaps between superstars, so the distribution of sales gaps pins down a combination of the two. Growth equals $\ln(1+\lambda)$ times the rate at which the frontier advances (43), and the payroll ratio is the inverse of the superstar markup, which falls with η. Together they separate the two parameters.

2. *Superstars' innovation (χ, ϕ).* Growth and R&D intensity measure the level of innovation and its cost. By (24), a superstar's innovation rises with the value of an innovation with elasticity $1/(\phi - 1)$. Patents are proportional to innovation, so the within-firm elasticity of patents to R&D equals $1/\phi$ and measures the convexity directly. The leader's share of superstars' R&D shows how the incentive to innovate differs between leaders and followers, which separates the cost level from the fringe's parameters.

3. *Breakthroughs ($\tilde{\chi}_f$, ϕ_f, p).* The rate at which new superstars appear pins down a combination of the level and convexity of the breakthrough cost. The convexity also sets how breakthroughs respond to the value of entry (35), and this value differs across industries; entry by the previous year's #4/#1 ratio carries that information. A share $(1-p)p^k$ of entrants lands k steps above the entry margin (10), so the landing shares and the entrants' sales relative to the leader identify p.

4. *Fringe productivity ($\tilde{\zeta}$, γ).* A more productive fringe takes a larger share of industry sales, so CR4 pins down $\tilde{\zeta}$. The curvature γ determines whether the fringe learns mostly from the leader or from all superstars (6). It changes how much of an innovation's value superstars keep, which the leader's R&D share reflects.

5. *Fringe costs and turnover ($\bar{\varepsilon}$, σ, δ).* The dispersion of fixed costs measured from selling, general, and administrative expenses pins down σ. Given σ, the mean $\bar{\varepsilon}$ sets the share of new firms that produce, measured by the share of business applications that become employers. Applications per employer firm equal the death rate divided by the producing share of the pool, and the exit rate adds the firms that stop producing; together they pin down δ. The exit premium of loss-making firms measures how close producers are to the cutoff in (36).

We hold out moments the model is not built to match. The gap dynamics of pairs of superstars are checks, because small changes in sales ratios in the data are permanent drift that a ladder of discrete steps cannot produce. The share of firms with losses is a check, because the fringe's gross margin is $1 - \alpha$ by construction.

J.2 Tax and procurement

The block has sixteen parameters: the statutory rate τ_0; the baseline contract g_0 and its decline with the firm's gap κ_g; for each instrument, the lobbying cost κ_I, its convexity ϕ_I, and the mean, dispersion, and redraw rate of the participation cost $(\mu_I, \sigma_I, \lambda_I)$; and the three economies of scope ι_{IJ}. Table 14 lists the moments, which answer six questions about each instrument.

1. *What happens without lobbying?* The effective tax rate of all US-headquartered firms identifies τ_0. Contracts over sales identify g_0, and the elasticity of contracts to sales across firms identifies κ_g.

2. *What does lobbying buy?* This identifies the lobbying cost κ_I. A year of procurement lobbying is followed by contracts 0.22 log points higher within the firm. On taxes, among firms that lobby, a log point more spending per ask is followed by a tax rate on taxable income 0.42 percentage points lower over the next three years, before the 2017 tax reform.

3. *Who lobbies?* Participation by rank sets the mean of the participation cost. Its slope in the firm's stake within an industry-year sets the dispersion: when the cost varies little across firms, the stake decides who lobbies and the slope is steep.

Table 13. Technology Moments: Data and Model

Moment	Data	s.e.	Model	Source
Sales among superstars				
Industries with fewer than four superstars	0.229	0.028	0.209	Compustat
Superstars per industry	3.64	0.05	3.71	Compustat
Leader's share of superstars' sales, median	0.458	0.010	0.469	Compustat
Leader's share of superstars' sales, 90th percentile	0.735	0.035	0.670	Compustat
#2 / #1 sales, 10th percentile	0.235	0.017	0.250	Compustat
#2 / #1 sales, median	0.613	0.019	0.527	Compustat
Industries with #2 / #1 below 0.3	0.162	0.021	0.127	Compustat
Industries with #2 / #1 above 0.8	0.252	0.021	0.188	Compustat
#4 / #1 sales, median	0.288	0.014	0.223	Compustat
Costs and R&D of the largest firms				
Sales / cost of goods sold, superstars	1.497	0.029	1.686	Compustat
Payroll per dollar of sales, four largest firms relative to the rest	0.764	0.029	0.662	Economic Census
Leader's share of superstars' R&D	0.484	0.034	0.455	Compustat
Within-firm elasticity of patents to R&D	0.262	0.104	0.226	Compustat, patents
New superstars				
New superstars per industry-year	0.132	0.013	0.144	Compustat
lowest tercile of the previous year's #4 / #1	0.105	0.013	0.136	Compustat
highest tercile of the previous year's #4 / #1	0.171	0.025	0.181	Compustat
Landing at the entry bar	0.636	0.032	0.480	Compustat
Landing one step above the bar	0.247	0.024	0.273	Compustat
Landing two steps above the bar	0.069	0.016	0.075	Compustat
Log sales relative to the leader, mean	−1.255	0.069	−1.223	Compustat
Log sales relative to the leader, standard deviation	0.639	0.029	0.545	Compustat
Fringe				
CR4	0.213	0.012	0.255	Economic Census
Exit premium of loss-making fringe firms	0.035	0.003	0.029	Compustat
Dispersion of log fixed costs, active fringe firms	0.648	0.030	0.608	Compustat
Firm exit rate	0.079	0.003	0.075	BDS
Business applications per employer firm	0.242	0.008	0.258	BFS, BDS
Applications that become employers within two years	0.273	0.006	0.288	BFS
Aggregates				
Growth of business output per hour	0.0159	0.0040	0.0172	Fernald (2014), 2008–2022
Labor share	0.596	0.004	0.644	Karabarbounis and Neiman (2014)
R&D funded by business / business value added	0.0261	0.0015	0.0125	NSF, BEA

Notes: Superstars are the four largest Compustat firms of a four-digit NAICS industry-year with at least 10 percent of the leader's sales, in industry-years with at least six Compustat firms, 2008–2022. CR4 is the median industry-year in the Economic Census. In the data, a step of the ladder is a sales ratio of 1.27. Standard errors from a bootstrap; growth: Newey-West. BDS: Business Dynamics Statistics; BFS: Business Formation Statistics. Appendix K defines each moment. The model's labor share and R&D are in the units of the data, with a capital share of 0.195.

4. *How long do they stay?* Persistence, net of the industry state, identifies the redraw rate: given the state, a firm stays in or out because its participation cost persists.

5. *How much do they spend?* At an interior optimum, spending on an instrument equals the gain from lobbying divided by ϕ_I, so spending relative to the stake identifies the convexity. The data bound this ratio rather than measure it, because a report's amount covers all of its asks; the bounds come from two ways of splitting the amount.

6. *Which instruments go together?* A discount on the participation cost of a second instrument makes a firm that lobbies on one more likely to lobby on another, net of rank, industry, and year. These links identify the economies of scope.

Table 14. Tax and Procurement Moments: Data and Model

	Tax			Procurement			
Moment	Data	s.e.	Model	Data	s.e.	Model	Identifies
Effective tax rate, all firms	0.222	0.006	0.242				τ_0
Contracts / sales				0.019	0.006	0.017	g_0
Elasticity of contracts to sales				0.705	0.234	1.242	κ_g
Payoff: outcome after lobbying	−0.0042	0.0028	−0.0040	0.217	0.098	0.006	κ_I
Lobbying in a quarter, leader	0.245	0.025	0.409	0.176	0.021	0.213	μ_I
Lobbying in a quarter, ranks 2–4	0.135	0.016	0.225	0.084	0.012	0.133	μ_I
Lobbying on log stake, within industry-year	0.062	0.009	0.145	0.036	0.005	0.054	σ_I
Still lobbying a year later, given the state	0.591	0.024	0.283	0.626	0.029	0.547	λ_I
Spending per ask / stake, lower bound ($\times 10^{-5}$)	0.88		0.99	3.9		8.8	ϕ_I
Spending per ask / stake, upper bound ($\times 10^{-5}$)	7.4		0.99	38		8.8	ϕ_I
Across instruments, in a quarter							
Tax on regulation	0.252	0.026	0.019				ι_{RT}
Procurement on regulation				0.119	0.020	0.029	ι_{RP}
Tax on procurement	0.119	0.028	0.028				ι_{TP}

Notes: Superstars headquartered in the United States, fiscal years 2008–2022, weights equal across industries, unless stated. Effective rate: total tax over pretax income, all US-headquartered firms with positive income (46,235 firm-years). Tax payoff: three-year cash taxes over sales on taxable income over sales, interacted with lobbying last year and log dollars per ask; windows ending by December 2017. Procurement payoff: log prime contracts on last year's share of quarters lobbied, with firm and year fixed effects. Stake: pretax income (tax), last year's contracts (procurement). Persistence: lobbying in the fiscal fourth quarter of next year on this year's, with industry, rank, and year fixed effects and controls for the industry state. Across instruments: linear probability model of one instrument on the other within a quarter, with rank, industry, and year fixed effects. Appendix K gives the definitions.

J.3 Regulation

The block has fourteen parameters: the compliance level c, which sets R_0; the elasticity e_R of superstars' compliance costs to their size; the contest curvature θ and the effectiveness of

the anti side β^a; the price of influence κ_R and the convexities ϕ_R and ϕ_A; the participation cost of superstars $(\mu_R, \sigma_R, \lambda_R)$; and the association's organizing cost $(\mu_A, \sigma_A, \lambda_A)$ and lobbying cost κ_A. Table 15 lists the moments.

1. *Compliance costs (c, e_R).* We bound the level from Trebbi et al. (2023). Their measure of compliance labor in private establishments, as a share of output, is 0.96 percent, and we use it as an upper bound; the share of that cost serving our four regulatory instruments, 0.33 percent, is the lower bound. No study measures how the cost of these instruments rises with firm size, so e_R lies in the band from 0.40 to 0.98. The lower end reflects measured costs of specific rules, 0.33 to 0.84; the upper end reflects occupation-based measures across all compliance work, 0.95 to 0.98, which likely overstate it.

2. *Sides (e_R).* Within this band, the share of positions asking for higher costs helps locate e_R. A superstar gains from higher costs through rivals' costs and deterred entry, and pays its own compliance bill. The faster that bill rises with size, the more its own cost weighs against those gains, and the more superstars ask for lower costs. In the data 53 percent of superstars' positions and 34 percent of associations' positions ask for higher costs. In the model, a position's share is the part of the stake coming from channels that favor higher costs.

3. *Who lobbies (μ_R, σ_R, λ_R).* As on the other instruments, participation by rank and its slope in the firm's share of industry sales identify the mean and dispersion of the participation cost, and persistence identifies the redraw rate.

4. *Force and spending (θ, ϕ_R, κ_R).* How fast spending rises with size within an industry disciplines the curvature θ and the convexity ϕ_R. Superstars' spending per lobbying quarter is held below a ceiling, and what spending buys is identified on bills (Section 5.5).

5. *The association (μ_A, σ_A, λ_A, κ_A).* How often the association is active, how long it stays active, and how often it lobbies without superstars and superstars lobby without it identify its organizing cost. A ceiling on its spending disciplines κ_A and ϕ_A. The difference in superstars' participation between quarters with and without an active association measures how the two sides respond to each other.

J.4 Bills

Bills without model agents. The data pin down π_k directly from bills that no superstar and no industry association lobbied. We classify these bills with the procedure of Section 2.2:

Table 15. Regulation Moments: Data and Model

Moment	Data	s.e.	Model	Identifies
Compliance costs				
Compliance costs / output, band	0.0033 to 0.0096		0.0082	c
Size elasticity of compliance costs, band	0.40 to 0.98		0.81	e_R
Superstars				
Lobbying in a quarter, leader	0.328	0.030	0.484	μ_R
Lobbying in a quarter, ranks 2–4	0.194	0.022	0.288	μ_R
Lobbying on log share of industry sales	0.149	0.013	0.163	σ_R
Still lobbying a year later, given the state	0.600	0.023	0.905	λ_R
Positions asking for higher costs	0.533	0.089	0.671	e_R
Log dollars on log sales, relative to the leader	0.545	0.115	1.268	θ, ϕ_R
Leader's dollars per lobbying quarter / value added, ceiling		2.0×10^{-4}	6.5×10^{-5}	κ_R
Associations				
Active in an industry-quarter	0.558	0.031	0.897	μ_A
Active next quarter, given the state	0.658	0.021	0.975	λ_A
Association lobbies, no superstar	0.244	0.024	0.175	μ_A, σ_A
Superstar lobbies, no association	0.134	0.018	0.083	μ_A, σ_A
Superstars lobbying: association active − inactive	0.042	0.027	−0.004	κ_A
Positions asking for higher costs	0.343	0.029	0.273	e_R
Dollars / industry value added, ceiling		4.3×10^{-5}	6.8×10^{-5}	κ_A, ϕ_A

Notes: Superstars headquartered in the United States and associations in the census of industry associations, 2008–2022, on industry-quarters with at least six Compustat firms, weights equal across industries. Compliance band: Trebbi et al. (2023), $103 billion in 2014, over business value added and one minus the capital share; the lower bound multiplies it by 0.345, the share of their compliance cost serving our four instruments. Elasticity band: rule-level studies, including Dahl et al. (2018) and Ewens et al. (2024), and occupation-based measures from Trebbi et al. (2023). Positions: each position counts once, by the share of its asks that raise costs. Appendix K gives the definitions.

all that became law, every bill whose text passed inside another law, and a random sample of 4,000 others, weighted to the population. A bill counts as passed if it became law or if its core regulatory provision is in a law of the same Congress. Table 16 shows the result. Without a model agent, 3.42 percent of raising bills pass (95 percent interval 2.21 to 4.95) and 7.67 percent of lowering bills (4.07 to 13.61). Lobbying by other clients changes these rates little. With a model agent alone the rates are 4.4 and 8.3 percent; with model agents and other clients together, 9.0 and 16.0 percent. Big bills attract every kind of lobbyist, so these differences are not the effect of lobbying. They show why the model must fit the level of passage as well as its slope in lobbying.

Table 16. Passage of Regulation Bills by Who Lobbied

	Raising			Lowering		
Who lobbied	Rows	Bills	Passed	Rows	Bills	Passed
Nobody	64	441	2.7%	13	61	9.8%
Other clients only	149	507	4.0%	55	159	6.9%
Superstars or associations only	228	228	4.4%	84	84	8.3%
Superstars or associations, and other clients	1,341	1,341	9.0%	426	426	16.0%

Notes: Regulation bills, 2008–2022; rows read, bills after sample weights. Other clients: any client other than a superstar or an industry association that listed the bill before its outcome. Passed: law, or the core provision in a law of the same Congress.

Baseline. We estimate how bill features move the baseline on the 39,835 proposals without model agents, by a probit of passing on the sponsor's majority status, whether the sponsor chairs a referral committee, bipartisan sponsorship, original cosponsors, whether other clients lobbied, and Congress. Other clients raise the index by 0.076 (standard error 0.034). This gives each proposal an index I_k, and

$$\pi_k \;=\; \Phi\Big(\sqrt{1+\sigma_\xi^2}\,\big(I_k+\varsigma_{\text{type}}\big)+\sigma_\xi\,\xi_k\Big), \qquad \xi_k\sim N(0,1), \tag{65}$$

where the shift ς_{type} sets the average of π_k on the classified bills to the rates above, for every value of the unobserved component σ_ξ. Other clients are not agents in the model, so they enter only here, as a feature of the bill.

Simulation. We simulate every real proposal. Each industry that lobbied a proposal in the data is drawn from the stationary distribution, conditional on one of its firms being active, and forces add up across industries. A firm appears on the proposal if its dollars on it exceed a threshold, and it reveals its side with the probability observed in the data. Who lobbies each proposal is therefore simulated, in the same way as the passage of the proposal. On proposals without model agents $p_k=\pi_k$. At each parameter point, the repeal intensity is

set by the ratio of compliance costs removed to costs added by the laws of the period, 0.050 (standard error 0.068), the listing threshold by the number of superstars and the presence of associations on each industry's row of a proposal, and σ_ξ by the fit to the regression below and to passage.

J.5 Untargeted moments

Table 17 compares moments that the estimation does not target.

Table 17. Untargeted Moments

	Data	Model
Who lobbies on regulatory costs in an industry-quarter		
Both superstars and the association	0.314	0.722
Neither	0.308	0.020
Instruments a superstar lobbies on (share of quarters)		
None	0.661	0.466
Regulatory costs only	0.096	0.213
Tax only	0.040	0.114
Procurement only	0.038	0.062
All three	0.042	0.016
Industry structure and markups		
Industries with at most two superstars	0.111	0.071
Cost-weighted superstar markup	1.27	1.47

K Target Moments

This appendix defines each targeted moment and its model counterpart. Table 13 reports the technology moments and Tables 14 and 15 the lobbying moments. All moments receive equal weights (Section 5.1).

K.1 Technology moments

Sample. The firm sample is the Compustat panel of firms headquartered in the United States, 2008–2022. An industry is a four-digit NAICS code, and an industry-year enters if it has at least six Compustat firms: 2,131 industry-years in 170 industries. The superstars of an industry-year are the firms among its four largest by sales with at least 10% of the leader's sales (7,752 firm-years of 1,082 firms); all other firms form the fringe. For moments that follow firms across years, a change in a firm's reported industry code is treated as a re-coding, not a move, when its segment data show the same business in both years. A firm-year

is affected by a material acquisition if cash acquisitions or the sales of acquired firms exceed 10% of the firm's sales, or goodwill rises by more than 5% of sales. In the model, statistics of industries are computed over the stationary distribution of industry states, and statistics of firms over time from the equilibrium transition rates or a simulated panel. Table 18 lists the sources.

Table 18. Technology Moments: Sources and Samples

Moments	Source	Years	Observations
Sales among superstars	Compustat	2008–2022	2,131 industry-years
Sales / cost of goods sold	Compustat	2008–2022	superstar firm-years
Payroll ratio	Economic Census	2017, 2022	132 industries
New superstars	Compustat	2008–2022	2,089 industry-years; 275 new superstars
CR4	Economic Census	2012, 2017, 2022	132 industries
Exit premium of loss-making firms	Compustat	2008–2022	61,266 fringe firm-years
Dispersion of fixed costs	Compustat	2008–2022	5,801 fringe firms
Firm exit rate	Business Dynamics Statistics	2008–2022	all four-digit industries
Business applications	Business Formation Statistics; Business Dynamics Statistics	2008–2021	all sectors
Growth	Fernald (2014), business output per hour	2008–2022	annual
Labor share	Karabarbounis and Neiman (2014); OECD sector accounts	2008–2022	annual
R&D intensity	National Science Foundation	2008–2022	annual

Sales among superstars. The share of industries with fewer than four superstars is the share of industry-years in which the fourth-largest firm sells less than 10% of the leader's sales. The leader's share is its sales over the superstars' total sales (median and 90th percentile). The ratio #2/#1 is measured in industry-years with at least two superstars (10th percentile, median, and shares below 0.3 and above 0.8), and #4/#1 in industry-years with four superstars (median). The model computes each statistic with the same conditioning over the stationary distribution.

Cost structure of the largest firms. The sales-to-cost ratio is the superstars' total sales over their total cost of goods sold, with each firm's dollars divided by the Compustat sales of its industry-year so that industries have equal size, as in the model; in the model, it is superstars' sales over their variable cost, which equals their cost-weighted markup divided by α. The payroll ratio is the payroll per dollar of receipts of the four largest firms of an industry over that of all its other firms in the Economic Census, averaged over 2017 and 2022 and across the 132 industries of our panel in the Census, weighted by industry-years; in the model, it is the superstars' labor cost per dollar of sales over the fringe's, which equals

the inverse of the superstars' markup.

New superstars. A new superstar in year t is a superstar that was in the same industry in $t-1$ without being a superstar, has no material acquisition in t, and would remain a superstar if the firms that left the industry's four largest between $t-1$ and t were still present, with their sales grown at the industry's median rate. The entry rate is the number of new superstars per industry-year, overall and in the lowest and highest terciles of the previous year's #4/#1 (2,089 industry-years, 275 new superstars). A new superstar's landing is its real sales in t over the real entry bar of $t-1$, the larger of the fourth firm's sales and 10% of the leader's; its number of steps above the bar is the log landing divided by $\ln 1.27$, rounded, with landings below the bar set to zero. The step of 1.27 is the median sales ratio of the third to the fourth superstar in industry-years with four superstars. We target the shares landing zero, one, and two steps above the bar and the mean and standard deviation of log sales relative to the leader in t, excluding the two new superstars that become leaders. The model computes the same statistics exactly over one year of the continuous-time process, with sales grown by one year of aggregate growth so that landings are in real terms.

Gap dynamics. For each pair of superstars in year t, the sales ratio is the smaller firm's sales over the larger firm's. Pairs are tied (ratio at least 0.9), close (0.7 to 0.9), separated (0.4 to 0.7), or far (below 0.4). A pair moves *apart* if its ratio is below its value in t in both $t+1$ and $t+2$, and *closer* if it is above in both years. The sample keeps pairs whose firms remain superstars of the industry in $t+1$ and $t+2$ without material acquisitions: 3,825 of the 9,521 pairs with base years 2008–2020. The model computes these probabilities exactly from the transition rates, for every pair of superstars in every state, weighted by the stationary distribution and conditional on both firms remaining superstars.

Concentration. CR4 is the receipts share of the four largest firms in the Economic Census, averaged over 2012, 2017, and 2022; we take its median across the 132 industries of our panel present in all three Censuses, weighted by industry-years. In the model, it is the superstars' share of industry sales, averaged over the stationary distribution.

Fringe firms. The loss share is the share of corporate returns without net income among corporations with assets of \$1 to \$500 million, by minor industry of the IRS Corporation Source Book, 2008–2013, each industry-year weighted equally; in the model, it is the share of producing fringe firms whose operating profit net of compliance, breakthrough spending, and fixed cost is negative. The exit premium is the coefficient from a regression of an indicator that a fringe firm's last Compustat year is t and it was deleted for bankruptcy, liquidation, or other reasons (deletion codes 2, 3, and 10) on an indicator of negative operating income before depreciation in t, with industry-year fixed effects and industry-years weighted equally; the model runs the same regression on a simulated panel in which a firm exits when it dies

or stops producing. The dispersion of fixed costs is the standard deviation across 5,801 fringe firms of the firm average, over at least three years, of the residual from regressing log selling, general, and administrative expenses net of R&D on log sales with industry-year fixed effects (slope 0.667); in the model, it is the same statistic for the log of the fixed cost plus compliance of producing fringe firms, net of its industry-year mean. The exit rate is firm deaths over firms in the Business Dynamics Statistics, summed over industries and averaged over 2008–2022; in the model, it is the share of producing fringe firms that die or stop producing within a year. Applications per employer firm are high-propensity applications in the Business Formation Statistics over employer firms in the Business Dynamics Statistics, averaged over 2008–2021, and the share that become employers is business formations within eight quarters over high-propensity applications. In the model, applications are entries into the pool, so the first moment is the entry flow per producing fringe firm and the second the share of new firms that produce within two years before dying.

Aggregates. Growth is the mean annual log growth of business output per hour of Fernald (2014) over 2008–2022; in the model, it is the growth rate (43). The labor share is the corporate labor share of Karabarbounis and Neiman (2014), extended with the OECD accounts of nonfinancial and financial corporations, averaged over 2008–2022; in the model, it is the wage bill over output times one minus the capital share. R&D intensity is business-funded R&D over GDP from the National Science Foundation, averaged over 2008–2022; in the model, it is superstars' R&D plus fringe breakthrough spending, over output.

Untargeted moments. The superstar markup is the production-function markup computed as in De Loecker et al. (2020), weighted by cost of goods sold relative to industry-year sales; in the model, superstars' markups are weighted by their variable cost. The patent elasticity is the coefficient from a Poisson regression of a superstar's granted patents, by filing year, on the log of its R&D over industry-year sales, with firm and year fixed effects, on superstar firm-years with positive R&D, 2008–2020 (2,356 firm-years of 380 firms). Patents are matched to firms with the match of Dyèvre and Seager, extended to recent filing years with that of Kogan et al. (2017). In the model, innovation is $z = (\text{R\&D}/\chi)^{1/\phi}$, so the elasticity is $1/\phi$.

K.2 Lobbying moments

Tables 14 and 15 give the lobbying moments and their samples. Compustat superstars are headquartered in the United States; regulation and association moments cover 2008–2022, and tax and procurement participation is measured at the last quarter of the fiscal year, fiscal years 2008–2022. Every lobbying moment weights industries equally.

L From Bills to the Contest

This appendix describes how the enactment of bills maps into the regulatory fixed cost of the contest (13), and how the bill moments are simulated.

Rules add up. The regulatory fixed cost of an industry is the stock of rules in force. Proposals that raise it arrive at rate Λ_U; if enacted, proposal k adds r_k to the stock. Proposals that lower it arrive at rate Λ_L; if enacted, proposal k removes a share s_k of the stock. Rules lapse at rate δ_R. Proposal k is enacted with probability p_k, given by (49), and with probability π_k without lobbying. In the steady state, $\Lambda_U \mathbb{E}_U[rp] = \big(\delta_R + \Lambda_L \mathbb{E}_L[sp]\big) R$, so that

$$\frac{R}{R_0} \;=\; \frac{\mathbb{E}_U[r\,p]}{\mathbb{E}_U[r\,\pi]} \cdot \frac{1+\ell}{1+\ell\,\mathbb{E}_L[s\,p]/\mathbb{E}_L[s\,\pi]}, \qquad \ell \;=\; \frac{\Lambda_L\,\mathbb{E}_L[s\,\pi]}{\delta_R}, \tag{66}$$

where ℓ measures how fast lowering proposals erode the stock relative to lapsing; we fit $\ell = 0.035$. Without lobbying, $p = \pi$ and $R = R_0$. The result is exact when proposals arrive independently of the rules in force and the industry's lobbying is constant while a rule lasts.

How a firm splits its force. Firm i puts a share h_{ik} of its lobbying dollars on proposal k, with mean one across proposals. With the lobbying cost of convexity ϕ_R, its force on proposal k grows as h_{ik}^{θ/ϕ_R}. To first order in the forces, the value of force on proposal k for R/R_0 is

$$v_k^U \;=\; \frac{r_k}{\lambda_U\,\mathbb{E}_U[r\pi]} \times \begin{cases} 1-\pi_k & \text{push} \\ \pi_k & \text{block} \end{cases}, \qquad v_k^L \;=\; \frac{\ell}{1+\ell}\,\frac{s_k}{\lambda_L\,\mathbb{E}_L[s\pi]} \times \begin{cases} 1-\pi_k & \text{push} \\ \pi_k & \text{block} \end{cases},$$

where λ_U and λ_L are the shares of each type of proposal. Let e_k be the firm's efficiency on the proposal, one for pushing and β^a for blocking. The optimal split is $h_{ik} \propto (v_k e_k)^{1/(1-\theta/\phi_R)}$, and we scale the force so that the firm's force on proposal k is $f_{ik} = F_i\, e_k h_{ik}^{\theta/\phi_R} / \mathbb{E}[v\, e\, h^{\theta/\phi_R}]$, with $F_i = \beta^p b_{Ri}^{\theta}$ on the pro side and $\beta^a b_{Ri}^{\theta}$ on the anti side; the association splits its force in the same way with its own convexity ϕ_A. Then $\sum_k v_k f_{ik} = F_i$, and to first order the bills give $R/R_0 - 1 = \beta^p B^{\text{pro}} - \beta^a B^{\text{anti}}$, which is the first-order expansion of the contest (13).

Pushing and blocking. A firm with force f alone on a raising proposal changes its enactment probability by $f(1-\pi)/(1+f)$ by pushing and by $-\pi\beta^a f/(1+\beta^a f)$ by blocking. Since π is small, blocking matters only if β^a is large: to first order, lobbying superstars raise enactment on average only if $\beta^a < s(1-\pi)/((1-s)\pi)$, where s is the pro share. The association's dues fix its blocking force rather than β^a: a larger β^a with less effort gives the same dues and the same blocking. Information on β^a comes from association opposition on raising proposals and from lowering proposals, on which the roles flip.

M Welfare Decomposition

Exact decomposition. Superstar k of an industry produces $y_k = q_k l_k^{\alpha}$, and the fringe of m^{a} identical firms, efficient internally, produces as one producer $Y_f = (m^{\mathrm{a}})^{1-\alpha} q_f L_f^{\alpha}$. Industry output is $y_j = (\sum_k y_k^{\varsigma} + Y_f^{\varsigma})^{1/\varsigma}$ with $\varsigma = (\eta - 1)/\eta$, and $\ln Y_0 = \sum_S \mu(S) \ln y(S)$. Since every producer is of degree α in labor, $\ln Y_0 = \alpha \ln L + \ln A^* + \ln E$ exactly, with

$$\ln A^* \;=\; \sum_S \mu(S)\, \frac{1}{\upsilon} \ln \sum_k A_k(S)^{\upsilon}, \qquad \upsilon = \frac{\varsigma}{1 - \alpha\varsigma},$$

the productivity of a planner who allocates the same labor equally across industries (all have the same expenditure share and the same α) and within each industry in proportion to A_k^{υ}, given the industry states and the operating fringe. The term $\ln E = \ln Y_0 - \alpha \ln L - \ln A^* \leq 0$ splits into a within-industry and an across-industry part. Welfare (47) is $W = \ln c_0/\rho + g/\rho^2 - D/\rho$ with $\ln c_0 = \ln Y_0 + \ln(c_0/Y_0)$, so that

$$100 \ln(1 + \mathrm{CEV}) \;=\; 100 \left[\alpha\, \Delta \ln L + \Delta \ln A^* + \Delta \ln E + \Delta \ln(c_0/Y_0) + \frac{\Delta g}{\rho} - \Delta D \right],$$

where Δ is the economy without lobbying minus the lobbying economy. The change in $\ln A^*$ is split into the change within industry states, at the stationary distribution of the economy with lobbying, and the change due to the shift in the industry states, which itself splits into the superstars' configuration and the fringe's technology. The change in c_0/Y_0 is split by item of the resource constraint (45), and the change in growth by source (43).